

Unit 5

Module 5: Complex Variables (Integration)

6

Complex Variables (Integration)

Syllabus

Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series, Residues, Cauchy residue theorem (without proof), Rouché's theorem.

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6.1 Introduction

- In case of a real variable, the path of integration of $\int_a^b f(x) dx$ is always along the real axis from $x = a$ to $x = b$. But in the case of a complex variables, the path of the definite integrals $\int_a^b f(z) dz$ may be any curve joining from $z = a$ to $z = b$ and the value of this integral depends upon the path.

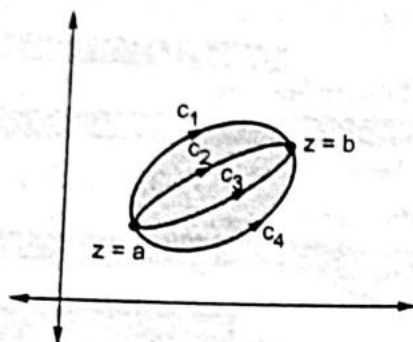


Fig. 6.1.1

6.2 Definitions

- Continuous arc** : The set of points (x, y) defined by parametric equation $x = f(t)$, $y = g(t)$ with parameter t in the interval (a, b) define a continuous arc if f and g are continuous functions.

- Smooth arc** : If parametric equation $f(t)$ and $g(t)$ are differentiable then arc is said to be smooth.

- Simple curve** : The curve is said to be simple if for two distinct values of z the images are distinct.

e.g. If $z_1 \neq z_2$ then $f(z_1) \neq f(z_2)$

- Closed curve** : The curve having same values of end points is said to be closed curve.

i.e. $f(a) = f(b)$ if a and b are end points of interval.

- Simple closed curve** : The curve does not intersect and coincide at end points is called simple closed curve.

- Contour** : A continuous chain of a finite number of smooth arcs is said to be contour.

- Closed contour** : The piecewise smooth closed curve without points of self intersection is called closed contour.

- Positively oriented** : When it is in the counterclockwise direction the curve is positively oriented.

• Note :

If the direction is clockwise then the integral is negative.

$$\int_C f(z) dz = - \int_C f(z) dz$$

6.3 Line Integral

- Suppose that the equation $z = z(t)$ represents a contour C from point $z = a$ to $z = b$, also $f(z)$ is piecewise continuous on C then line integral or contour integral of f along C in terms of parameter t is

$$\int_C f(z) dz = \int_a^b f(z(t)) \cdot z'(t) dt$$

provided $z'(t)$ is piecewise continuous.

• Note :

- $\int_C z_0 f(z) dz = z_0 \int_C f(z) dz$ (z_0 is constant)

- $\int_C [f(z) + g(z)] dz = \int_C f(z) dz + \int_C g(z) dz$

- If $C = C_1 + C_2$, then

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

- $\int_C dz = \text{Length of arc}$

- If C is closed contour then integral can be written as

$$\oint_C f(z) dz$$

- The parametric representation of the circle of radius r with centre z_0 is given by

$$z(t) = z_0 + r e^{i\theta}, \quad 0 \leq \theta \leq 2\pi$$

Ex. 6.3.1 : Find the parametric representation for the following

- The line segment from 0 to $4 + 3i$
- The line segment from $1 - i$ to $2 + i$
- The right half of $|z - 2 + i| = 2$
- The curve $y = x^3$ from $(1, 1)$ to $(2, 8)$

Sol. : i) Here C is from 0 to $4 + 3i$

C is from (0, 0) to (4, 3)

The equation of line joining two points (x_0, y_0) and (x_1, y_1) is

$$\frac{y-y_0}{y_1-y_0} = \frac{x-x_0}{x_1-x_0}$$

$$\Rightarrow \frac{y-0}{3-0} = \frac{x-0}{4-0}$$

$$\Rightarrow 4y = 3x$$

$$\Rightarrow 3x - 4y = 0$$

Let $x(t) = t$

$$y(t) = \frac{3}{4}t$$

∴ The parametric representation is

$$z(t) = t + i\frac{3}{4}t; \quad 0 \leq t \leq 4$$

ii) Here C is from $1 - i$ to $2 + i$

⇒ C is from (1, -1) to (2, 1)

∴ The equation of line is

$$\frac{y+1}{1+1} = \frac{x-1}{2-1}$$

$$\Rightarrow y + 1 = 2x - 2$$

$$\Rightarrow 2x - y = 3$$

Let $x(t) = t$

$$y(t) = 2t - 3$$

∴ The parametric representation is

$$z(t) = t + i(2t - 3); \quad 1 \leq t \leq 2$$

iii) Here C is circle $|z - 2 + i| = 2$

$$\therefore |z - (2 - i)| = 2$$

which represents the circle with centre (2, -1) and radius 2.

∴ The parametric representation is

$$z(t) = 2 - i + 2e^{i\theta}; \quad -\pi/2 \leq \theta \leq \pi/2$$

... ∴ (Right half of the circle)

iv) Here C is $y = x^3$ from (1, 1) to (2, 8).

Let $x(t) = t$

$$y(t) = t^3$$

∴ The parametric representation is

$$z(t) = t + it^3; \quad 1 \leq t \leq 2$$

Ex. 6.3.2 : Evaluate $\int_0^{4+2i} \bar{z} dz$ along the curve $z = t^2 + it$

Sol. : Here $z(t) = t^2 + it; \quad 0 \leq t \leq 2$

$$z'(t) = 2t + i$$

$$\int_0^{4+2i} \bar{z} dz = \int_0^2 (t^2 - it)(2t + i) dt$$

$$= \int_0^2 [2t^3 + it^2 - i2t^2 + t] dt$$

$$= \int_0^2 [2t^3 - it^2 + t] dt$$

$$= \left[\frac{t^4}{2} - \frac{it^3}{3} + \frac{t^2}{2} \right]_0^2 = 8 - \frac{i8}{3} + 2$$

$$= 10 - \frac{8i}{3}$$

Ex. 6.3.3 : Evaluate

$\int_C (x^2 - iy^2) dz$ where C is the parabola $y = 2x^2$ from (1, 2) to (2, 8).

Sol. : Here C is $y = 2x^2$ from (1, 2) to (2, 8)

Let $x(t) = t$

$$y(t) = 2t^2$$

$$z(t) = t + i2t^2; \quad 1 \leq t \leq 2$$

$$z'(t) = 1 + i4t$$

$$\int_C (x^2 - iy^2) dz = \int_1^2 (t^2 - i4t^4)(1 + i4t) dt$$

$$= \int_1^2 [t^2 + i4t^3 - i4t^4 + 16t^5] dt$$

$$\begin{aligned}
 &= \left[\frac{t^3}{3} + it^4 - i \frac{4t^5}{5} + \frac{8t^6}{3} \right]_1 \\
 &= \left[\frac{8}{3} + i16 - i \frac{128}{5} + \frac{512}{3} - \left(\frac{1}{3} + i - i \frac{4}{5} + \frac{8}{3} \right) \right] \\
 &= \left[\frac{8}{3} + 16i - \frac{128}{5}i + \frac{512}{3} - \frac{1}{3} - i + \frac{4}{5}i - \frac{8}{3} \right] \\
 &= \left[\frac{511}{3} + 15i - \frac{128}{5}i + \frac{4}{5}i \right] \\
 &= \frac{511}{3} - \frac{49}{5}i
 \end{aligned}$$

Ex. 6.3.4 : Evaluate

$\int_C \operatorname{Re}(z^2) dz$ where C is the boundary of the square with vertices $0, i, 1+i, 1$ in the clockwise direction.

Sol. : $z = x + iy \Rightarrow z^2 = x^2 - y^2 + 2xyi$

$$\therefore \operatorname{Re}(z^2) = x^2 - y^2$$

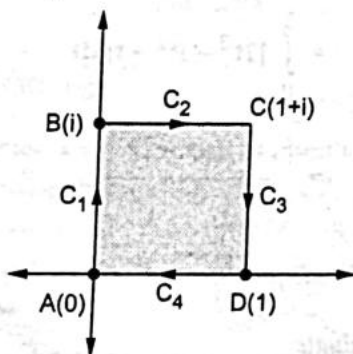


Fig. 6.3.1

$$\begin{aligned}
 \text{So, } \int_C f(z) dz &= \int_{C_1} f(z) dz + \int_{C_2} f(z) dz \\
 &\quad + \int_{C_3} f(z) dz + \int_{C_4} f(z) dz
 \end{aligned}$$

Along C_1 : $x = 0, y = t, 0 \leq t \leq 1$

$$z(t) = it, \quad 0 \leq t \leq 1$$

$$z'(t) = i$$

$$\int_{C_1} f(z) dz = \int_0^1 -t^2 i dt = \left[-\frac{it^3}{3} \right]_0^1 = -\frac{i}{3}$$

Along C_2 : $y = 1, x = t, 0 \leq t \leq 1$

$$z(t) = t + i$$

$$z'(t) = 1$$

$$\begin{aligned}
 \int_{C_2} f(z) dz &= \int_0^1 (t^2 - 1) \cdot 1 \cdot dt \\
 &= \left[\frac{t^3}{3} - t \right]_0^1 \\
 &= \frac{1}{3} - 1 = -\frac{2}{3}
 \end{aligned}$$

Along C_3 : $x = 1, y = t, 0 \leq t \leq 1$

$$z(t) = 1 + it$$

$$z'(t) = i$$

$$\int_{C_3} f(z) dz = \int_1^0 (1 - t^2) i dt$$

$$= - \int_0^1 (1 - t^2) i dt$$

$$= -i \left[t - \frac{t^3}{3} \right]_0^1$$

$$= -i \left[1 - \frac{1}{3} \right] = -\frac{2}{3} i$$

Along C_4 : $y = 0, x = t, 0 \leq t \leq 1$

$$z(t) = t$$

$$z'(t) = 1$$

$$\int_{C_4} f(z) dz = \int_1^0 t^2 \cdot 1 \cdot dt = - \int_0^1 t^2 dt$$

$$= - \left[\frac{t^3}{3} \right]_0^1 = -\frac{1}{3}$$

$$\therefore \int_C \operatorname{Re}(z^2) dz = -\frac{i}{3} - \frac{2}{3} - \frac{2}{3}i - \frac{1}{3} = -1 - i$$

Ex. 6.3.5 : Evaluate

$\int_C \bar{z} dz$, where C is the right half

of the circle $|z| = 2$. Hence show that $\int_C \frac{dz}{z} = \pi i$

Sol. : Here C is right half of the circle $|z| = 2$.

$$z(t) = 2e^{it}; \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

$$z'(t) = 2ie^{it}$$

$$\int_C \bar{z} dz = \int_{-\pi/2}^{\pi/2} 2e^{-it} \cdot 2ie^{it} dt$$

$$= \int_{-\pi/2}^{\pi/2} 4i dt$$

$$= 4i [t]_{-\pi/2}^{\pi/2}$$

$$= 4i \left[\frac{\pi}{2} + \frac{\pi}{2} \right]$$

$$= 4\pi i$$

Also, we know that $Z\bar{z} = |z|^2$

$$\bar{z} = \frac{|z|^2}{z} = \frac{4}{z}$$

$$\therefore \int_C \bar{z} dz = 4\pi i$$

$$\Rightarrow \int_C \frac{4}{z} dz = 4\pi i$$

$$\Rightarrow 4 \int_C \frac{dz}{z} = 4\pi i$$

$$\Rightarrow \int_C \frac{dz}{z} = \pi i$$

Ex. 6.3.6 : Evaluate

$\int_C (z - z^2) dz$, where C is the

upper half of the circle $|z| = 1$

Sol. : Here, C is upper half of $|z| = 1$

$$z(t) = e^{it}, \quad 0 \leq t \leq \pi$$

$$z'(t) = ie^{it}$$

$$\int_C (z - z^2) dz = \int_0^\pi (e^{it} - e^{i2t}) ie^{it} dt$$

$$= i \int_0^\pi (e^{i2t} - e^{i3t}) dt$$

$$= i \left[\frac{e^{i2t}}{2i} - \frac{e^{i3t}}{3i} \right]_0^\pi$$

$$= i \left[\frac{e^{i2\pi}}{2i} - \frac{e^{i3\pi}}{3i} - \frac{1}{2i} + \frac{1}{3i} \right]$$

$$= i \left[\frac{1}{2i} + \frac{1}{3i} - \frac{1}{2i} + \frac{1}{3i} \right] = i \frac{2}{3i} = \frac{2}{3}$$

Ex. 6.3.7 : Evaluate

$\int_C |z| dz$, where C is the left half of the unit circle $|z| = 1$ from $z = -i$ to $z = i$.

Sol. : Here C is left half of circle $|z| = 1$

$$z = e^{it}, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

$$z'(t) = ie^{it}$$

$$\int_C |z| dz = \int_{-\pi/2}^{\pi/2} ie^{it} dt$$

$$= \left[i \frac{e^{it}}{i} \right]_{-\pi/2}^{\pi/2}$$

$$= e^{i\pi/2} - e^{-i\pi/2}$$

$$= \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) - \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

$$= i + i = 2i$$

Ex. 6.3.8 : Prove that $\int_C \frac{1}{z} dz = -\pi i$ or πi ,

according as C is the semi-circular arc $|z| = 1$ from $z = -1$ to 1 above or below the real axis.

Sol. : Here $|z| = 1 \Rightarrow z = e^{it}$

$$z'(t) = ie^{it}$$

$$\therefore \int_C \frac{1}{z} dz = \int_C \frac{1}{e^{it}} ie^{it} dt = i \int_C dt$$

If C is the semi-circle $|z|=1$ from $z=-1$ to 1 above the real axis.

$$z(t) = e^{it}; t \text{ from } \pi \text{ to } 0$$

$$\therefore \int_C \frac{1}{z} dz = i \int_{\pi}^0 dt = i [t]_{\pi}^0 = -i\pi = -\pi i$$

If C is the semi-circle $|z|=1$ from $z=-1$ to 1 below the real axis.

$$z(t) = e^{it}, t \text{ from } -\pi \text{ to } 0$$

$$\therefore \int_C \frac{1}{z} dz = i \int_{-\pi}^0 dt = i [t]_{-\pi}^0 = \pi i$$

Ex. 6.3.9 : Evaluate $\int_C \ln z dz$, where C is circle $|z|=1$

Sol. : $|z|=1 \Rightarrow z = e^{it}, 0 \leq t \leq 2\pi$

$$z'(t) = ie^{it}$$

$$\begin{aligned} \therefore \int_C \ln z dz &= \int_0^{2\pi} \ln e^{it} ie^{it} dt \\ &= \int_0^{2\pi} it ie^{it} dt \\ &= - \int_0^{2\pi} te^{it} dt \\ &= - \left[t \frac{e^{it}}{i} - \frac{e^{it}}{-1} \right]_0^{2\pi} \\ &= - \left[\frac{2\pi e^{i2\pi}}{i} + e^{i2\pi} - 1 \right] \\ &= - [-2\pi i + 1 - 1] = 2\pi i \end{aligned}$$

6.4 Green's Theorem

Statement : If $M(x, y)$, $N(x, y)$ and its first order partial derivative $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ be continuous everywhere in a region R of xy -plane bounded by a closed curve C , then

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

6.5 Cauchy's Integral Theorem

Statement : If a function $f(z)$ is analytic and its derivative $f'(z)$ continuous at each points inside and on a simple closed curve C then,

$$\oint_C f(z) dz = 0$$

Proof : Let R be the region consisting of all points interior to and on the simple closed contour C .

Let $f(z) = u(x, y) + iv(x, y)$ be analytic on R and C .

$\therefore u(x, y)$ and $v(x, y)$ satisfies CR equation, so

$$u_x = v_y \text{ and } u_y = -v_x$$

Also, $f'(z) = u_x + iv_x$ is continuous on R and C .

$\therefore u_x, u_y, v_x, v_y$ are continuous on R and C .

And, $z = x + iy \Rightarrow dz = dx + idy$

$$\begin{aligned} \int_C f(z) dz &= \int_C (u + iv)(dx + idy) \\ &= \int_C [(udx - vdy) + i(vdx + udy)] \\ &= \int_C (udx - vdy) + i \int_C (vdx + udy) \\ &= \iint_R \left[-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right] dx dy + i \iint_R \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right] dx dy \\ &\quad \dots \text{[Applying Green's Theorem]} \\ &= \iint_R \left[-\frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} \right] dx dy + i \\ &\quad \iint_R \left[\frac{\partial u}{\partial x} - \frac{\partial u}{\partial x} \right] dx dy = 0 + i0 = 0 \end{aligned}$$

6.6 Cauchy-Goursat Theorem

GTU : Summer-21,22,23, Winter-23

Statement : If a function $f(z)$ is analytic at all points interior to and on a simple closed contour C , then

$$\oint_C f(z) dz = 0$$

Ex. 6.6.1 : Evaluate $\int_C \frac{z^2}{z-3} dz$ if C is the circle $|z|=1$

Sol. : Here, $f(z) = \frac{z^2}{z-3}$ is analytic everywhere except at $z=3$.

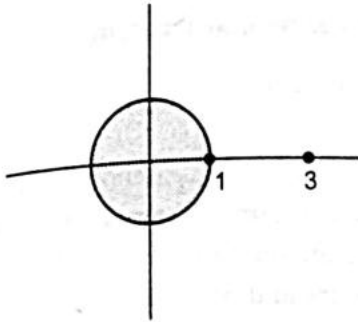


Fig. 6.6.1

But $z = 3$ is outside the contour $|z| = 1$.

$\therefore f(z)$ is analytic on and inside C .

$\therefore$ By Cauchy's integral theorem,

$$\int_C f(z) dz = 0$$

Ex. 6.6.2 : Evaluate $\int_C \tan z dz$ if C is the circle $|z| = 1$.

GTU : Summer-21, CVNM, Marks 4

Sol. : Here $f(z) = \tan z = \frac{\sin z}{\cos z}$ is analytic everywhere except at

$$z = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

But all that points are outside the contour $|z| = 1$.

$\therefore f(z)$ is analytic on and inside C .

$\therefore$ By Cauchy's integral theorem

$$\int_C f(z) dz = 0$$

Ex. 6.6.3 : Evaluate $\oint_C \frac{e^{-z}}{z+1} dz$ Where C is the circle

$$|z| = 1/2.$$

GTU : Summer-22, Maths-4, Marks 3

Sol. : Here, $f(z) = \frac{e^{-z}}{z+1}$ is not analytic

at $z = -1$

$C : |z| = 1/2$ is circle with center $(0, 0)$ and radius $1/2$.

But $z = -1$ outside the circle $|z| = 1/2$

$\therefore$ By Cauchy-Goursat theorem $\oint_C \frac{e^{-z}}{z+1} dz = 0$

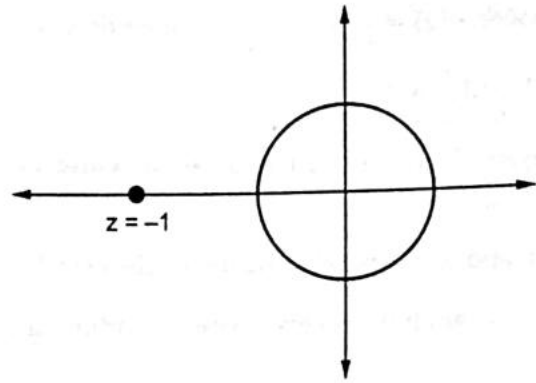


Fig. 6.6.2

Ex. 6.6.4 : Evaluate $\int_C \frac{e^{2z}}{z^2+1} dz$ Where C is

$$|z| = \frac{1}{2}.$$

GTU : Winter-23, CVPDE, Marks 4

Sol. : Here, $f(z) = \frac{e^{2z}}{z^2+1} = \frac{e^{2z}}{(z+i)(z-i)}$ is not analytic at $z = -i = (0, -1)$ and $z = i = (0, 1)$

Now $C : |z| = 1/2$ is the circle at center $(0, 0)$ and radius $1/2$

Here, $z = i$ and $z = -i$ outside of the circle $|z| = 1/2$.

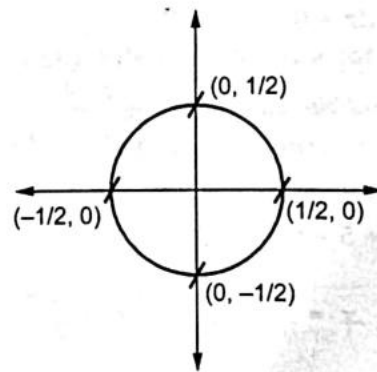


Fig. 6.6.3

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{e^{-z}}{z^2+1} dz = 0$$

Ex. 6.6.5 : Evaluate $\int_C \frac{z dz}{(z-1)(z-2)}$ Where C is the circle

$$|z| = 1/2.$$

GTU : Summer-23, CVPDE, Marks 3

Sol. : Here, $f(z) = \frac{z}{(z-1)(z-2)}$ is analytic every where at $z = 1$ and $z = 2$.

Our region is $|z| = 1/2$ (Circle with center $(0, 0)$ and radius $1/2$).

$\therefore z = 1$ and $z = 2$ outside of the circle $|z| = 1/2$

$\therefore f(z)$ is analytic every where within the circle $|z| = 1/2$

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{z \, dz}{(z-1)(z-2)} = 0$$

Ex. 6.6.6 : Evaluate $\int_C \frac{3z^2 + 7z + 1}{z+1} dz$ where C is

$$|z| = 1/2.$$

GTU : Summer-23, CVPDE, Marks 3

Sol. : Here, $f(z) = \frac{3z^2 + 7z + 1}{z+1}$ is not analytic at $z = -1$

But $z = -1$ is outside of the region $|z| = 1/2$.

$\therefore f(z)$ is analytic every where within the circle $|z| = 1/2$

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{3z^2 + 7z + 1}{z+1} dz = 0$$

Ex. 6.6.7 : Find the value of $\int_C \frac{z+4}{z^2 + 2z + 5} dz$ if C is the circle $|z+1|=1$

Sol. : $z^2 + 2z + 5 = 0$

$$\Rightarrow z = \frac{-2 \pm \sqrt{4-20}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

$\therefore f(z)$ is analytic everywhere except $z = -1 \pm 2i$

But $z = -1 \pm 2i$ is outside the contour $C : |z+1|=1$

$\therefore f(z)$ is analytic on and inside C .

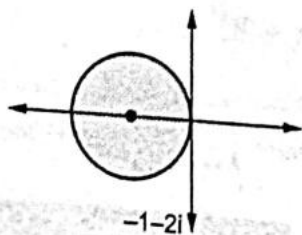


Fig. 6.6.4

$\therefore$ By Cauchy's integral theorem,

$$\int_C f(z) \, dz = 0$$

Corollary : Let C_1 and C_2 denote positively oriented simple closed contours, where C_1 is interior to C_2 . If a function f is analytic in the closed region consisting of those contours and all points between them, then

$$\int_{C_1} f(z) \, dz = \int_{C_2} f(z) \, dz$$

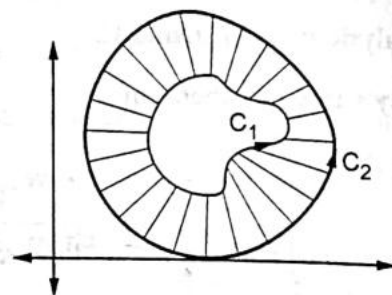


Fig. 6.6.5

Ex. 6.6.8 : Let C_1 denote the positively oriented boundary of the square whose sides lie among the lines $x = \pm 1$, $y = \pm 1$ and let C_2 be the positively oriented circle $|z| = 4$ point out why

$$\int_{C_1} f(z) \, dz = \int_{C_2} f(z) \, dz, \text{ for}$$

$$\text{i) } f(z) = \frac{1}{3z^2 + 1} \quad \text{ii) } f(z) = \frac{z+2}{\sin\left(\frac{z}{2}\right)} \quad \text{iii) } f(z) = \frac{z}{1-e^z}$$

Sol. :

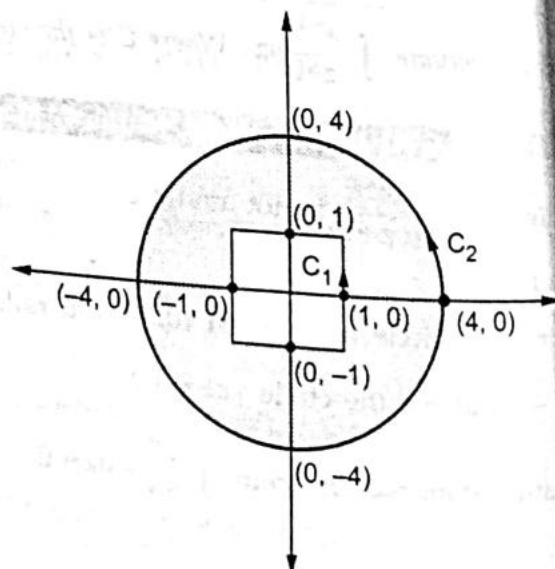


Fig. 6.6.6

Mathematics - 2

i) $f(z) = \frac{1}{3z^2 + 1}$ is analytic everywhere except

$$z = \pm \frac{i}{\sqrt{3}}$$

But $z = \pm \frac{i}{\sqrt{3}}$ is not on C_1 and C_2 and between C_1 and C_2 .

$$\therefore \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

ii) $f(z) = \frac{z+2}{\sin\left(\frac{z}{2}\right)}$ is analytic everywhere

except $z = 0, \pm 2\pi, \pm 4\pi, \dots$

But $z = 0, \pm 2\pi, \pm 4\pi, \dots$ is not on C_1 and C_2 and between C_1 and C_2 .

$$\therefore \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

iii) $f(z) = \frac{z}{1-e^z}$ is analytic everywhere except $z = 0$.

But $z = 0$ is not on C_1 and C_2 and between C_1 and C_2 .

$$\therefore \int_{C_1} f(z) dz = \int_{C_2} f(z) dz$$

6.7 Cauchy Integral Formula

GTU : Summer-19,21,22,23,24, Winter-21,22,23,24

Statement : If $f(z)$ is analytic within and on a closed curve C and if a is any point within C then

$$\oint_C \frac{f(z)}{z-a} dz = 2\pi i \cdot f(a)$$

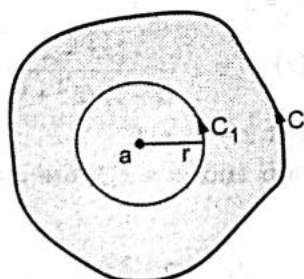


Fig. 6.7.1

Proof : The function $\frac{f(z)}{z-a}$ is analytic everywhere in C except $z = a$.

Consider a small circle C_1 with centre at a and radius r in C .

Thus, $\frac{f(z)}{z-a}$ is analytic in the region between C and C_1 and on C and C_1 .

Hence by Cauchy's integral corollary for multiply connected region, we get

$$\int_C \frac{f(z)}{z-a} dz = \int_{C_1} \frac{f(z)}{z-a} dz$$

Now C_1 is $|z-a| = r$

$$\Rightarrow z-a = re^{i\theta}, 0 \leq \theta \leq 2\pi$$

$$\Rightarrow dz = ire^{i\theta} d\theta$$

$$\therefore \int_C \frac{f(z)}{z-a} dz = \int_0^{2\pi} \frac{f(re^{i\theta} + a)}{re^{i\theta}} ire^{i\theta} d\theta$$

$$= i \int_0^{2\pi} f(re^{i\theta} + a) d\theta$$

$$= i \int_0^{2\pi} f(a) d\theta \quad \dots \{ \because \text{As } r \rightarrow 0$$

$$= if(a) \int_0^{2\pi} d\theta = 2\pi i f(a)$$

$$\therefore \int_C \frac{f(z)}{z-a} dz = 2\pi i f(a)$$

6.7.1 Cauchy Integral Formula for Derivative

Statement : If $f(z)$ is analytic within and on a closed curve C and if a is any point within C then

$$\oint_C \frac{f(z)}{(z-a)^n} dz = \frac{2\pi i}{(n-1)!} f^{(n-1)}(a)$$

Ex. 6.7.1 : Evaluate $\oint_C \frac{z^2+5}{z-3} dz$ where C is $|z|=4$

Sol. : Here, $f(z) = \frac{z^2+5}{z-3}$ is analytic everywhere except $z = 3$.

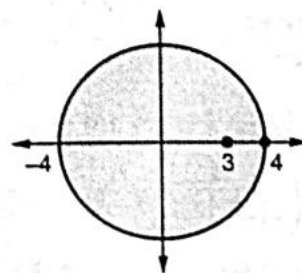


Fig. 6.7.2

But $z = 3$ is inside contour C .

Let $\phi(z) = z^2 + 5$

$\therefore$ By Cauchy's integral formula

$$\begin{aligned}\oint_C \frac{z^2 + 5}{z - 3} dz &= 2\pi i \phi(3) \\ &= 2\pi i (14) \\ &= 28\pi i\end{aligned}$$

Ex. 6.7.2 : Evaluate $\int_C \frac{dz}{z^2 + 1}$, where C is $|z + i| = 1$, counterclockwise.

**GTU : Summer-21 Maths-4, Marks 7,
Winter-24, CVNM, Marks 4
Winter-21 CVNM, Marks 3**

Sol. : Here, $f(z) = \frac{1}{z^2 + 1}$ is analytic everywhere except $z = \pm i$.

But $z = -i$ is inside and $z = i$ is outside the contour C .

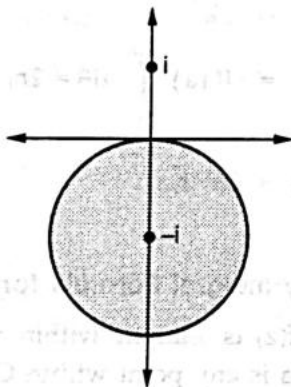


Fig. 6.7.3

Let $\phi(z) = \frac{1}{(z - i)}$

$\therefore$ By Cauchy's integral formula

$$\begin{aligned}\oint_C \frac{dz}{z^2 + 1} &= \oint_C \frac{1}{(z + i)(z - i)} dz \\ &= \oint_C \frac{1/z - i}{z + i} dz \\ &= 2\pi i \phi(-i) \\ &= 2\pi i \left(\frac{1}{-i - i} \right) \\ &= \frac{2\pi i}{-2i} = -\pi\end{aligned}$$

Ex. 6.7.3 : Evaluate $\int_C \frac{3z - 1}{z^3 - z} dz$, where C is $|z| = \frac{1}{2}$

Sol. : Here, $f(z) = \frac{3z - 1}{z(z^2 - 1)}$ is analytic everywhere except $z = 0, \pm 1$.

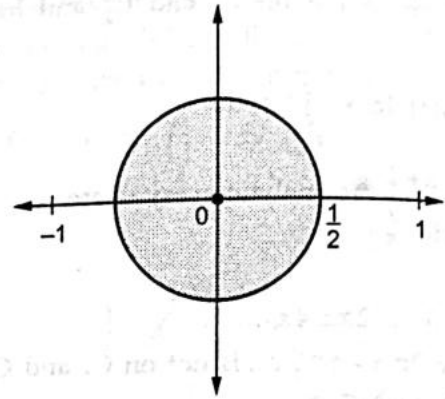


Fig. 6.7.4

But $z = 0$ is inside and $z = \pm 1$ are outside the contour $|z| = \frac{1}{2}$.

Let $\phi(z) = \frac{3z - 1}{z^2 - 1}$

$\therefore$ By Cauchy's integral formula,

$$\begin{aligned}\oint_C \frac{3z - 1}{z(z^2 - 1)} dz &= \oint_C \frac{(3z - 1) / (z^2 - 1)}{z} dz \\ &= 2\pi i \phi(0) = 2\pi i (1) = 2\pi i\end{aligned}$$

Ex. 6.7.4 : Find the value of the integral $\int_C \frac{3z^2 + 2}{(z - 1)(z^2 + 9)} dz$ taken counterclockwise around the circle $C : |z - 2| = 2$

Sol. : Here, $f(z) = \frac{3z^2 + 2}{(z - 1)(z^2 + 9)}$ is analytic

everywhere except $z = 1, \pm 3i$

But $z = 1$ is inside and $z = \pm 3i$ are outside the circle $|z - 2| = 2$.

Let $\phi(z) = \frac{3z^2 + 2}{z^2 + 9}$

∴ By Cauchy's integral formula

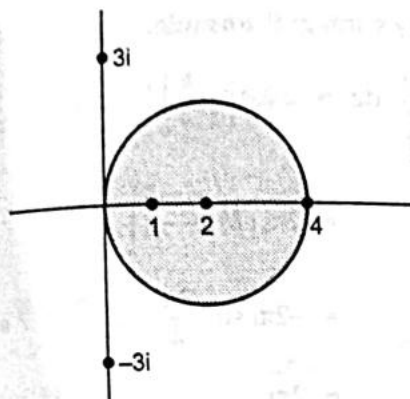


Fig. 6.7.5

$$\oint_C \frac{3z^2 + 2}{(z-1)(z^2 + 9)} dz = \oint_C \frac{3z^2 + 2}{z-1} \cdot \frac{1}{z^2 + 9} dz$$

$$= 2\pi i \phi(1) = 2\pi i \left(\frac{5}{10} \right) = \pi i$$

Ex. 6.7.5 : Find $\oint_C \frac{z+5}{z^2 + 3z - 4} dz$, where $C : |z - i| = 2$

Sol. :

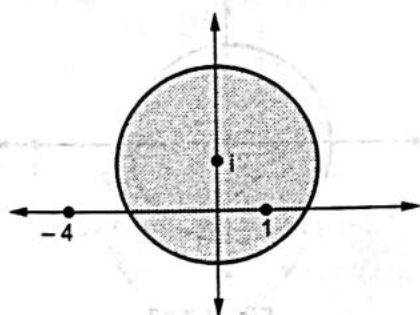


Fig. 6.7.6

Here $f(z) = \frac{z+5}{(z^2 + 3z - 4)} = \frac{z+5}{(z+4)(z-1)}$

∴ $f(z)$ is analytic everywhere except $z = -4, 1$.

But $z = 1$ is inside and $z = -4$ is outside the circle $|z - i| = 2$

Let $\phi(z) = \frac{z+5}{z+4}$

∴ By Cauchy's integral formula

$$\oint_C \frac{z+5}{z^2 + 3z - 4} dz = \oint_C \frac{z+5}{z-1} \cdot \frac{1}{z+4} dz$$

$$= 2\pi i \phi(1) = 2\pi i \cdot \frac{6}{5} = \frac{12\pi i}{5}$$

Ex. 6.7.6 : Evaluate $\oint_C \frac{z dz}{z^2 - 1}$, where $C : |z| = 2$.

Sol. : Here,

$$f(z) = \frac{z}{z^2 - 1} = \frac{z}{(z-1)(z+1)}$$

∴ $f(z)$ is analytic everywhere except $z = \pm 1$

But $z = \pm 1$ are inside the circle $|z| = 2$.

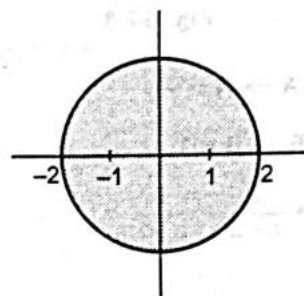


Fig 6.7.7

Now $\frac{1}{(z-1)(z+1)} = \frac{1}{2} \left[\frac{1}{z-1} - \frac{1}{z+1} \right]$

$$\oint_C \frac{z}{z^2 - 1} dz = \frac{1}{2} \oint_C \left(\frac{z}{z-1} - \frac{z}{z+1} \right) dz$$

$$= \frac{1}{2} \left[\oint_C \frac{z}{z-1} dz - \oint_C \frac{z}{z+1} dz \right]$$

$$= \frac{1}{2} [2\pi i \phi(1) - 2\pi i \phi(-1)]$$

... $\{\phi(z) = z\}$

$$= \frac{1}{2} [2\pi i (1) - 2\pi i (-1)] = \frac{4\pi i}{2} = 2\pi i$$

Ex. 6.7.7 : Evaluate $\oint_C \frac{\cos \pi z^2}{(z-1)(z-2)} dz$, where $C : |z| = 3$

GTU : Summer-24, CVPDE, Marks 7

Sol. : Here, $f(z) = \frac{\cos \pi z^2}{(z-1)(z-2)}$ is analytic everywhere except $z = 1, 2$.

But $z = 1, 2$ are inside the circle $|z| = 3$

now, $\frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$

$\Rightarrow 1 = A(z-2) + B(z-1)$

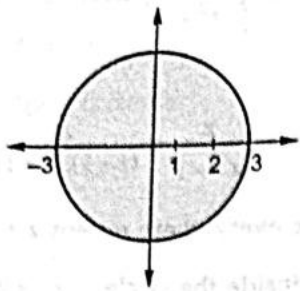


Fig. 6.7.8

$$\text{If } z = 1 \Rightarrow 1 = -A \Rightarrow A = -1$$

$$\text{If } z = 2 \Rightarrow 1 = B \Rightarrow B = 1$$

$$\therefore \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$$

So,

$$\begin{aligned} \oint_C \frac{\cos \pi z^2}{(z-1)(z-2)} dz &= \oint_C \left(\frac{\cos \pi z^2}{z-2} - \frac{\cos \pi z^2}{z-1} \right) dz \\ &= \oint_C \frac{\cos \pi z^2}{z-2} dz - \oint_C \frac{\cos \pi z^2}{z-1} dz \\ &= 2\pi i \phi(2) - 2\pi i \phi(1) \\ &\quad \dots \{ \phi(z) = \cos \pi z^2 \} \\ &= 2\pi i \cos 4\pi - 2\pi i \cos \pi \\ &= 2\pi i + 2\pi i = 4\pi i \end{aligned}$$

Ex. 6.7.8 : Evaluate $\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz$, where $C: |z| = 5$

Sol. : Here, $f(z) = \frac{\sin 3z}{z + \frac{\pi}{2}}$ is analytic everywhere except $z = -\frac{\pi}{2}$

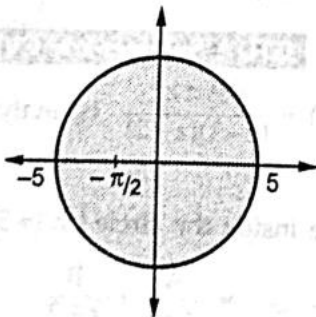


Fig. 6.7.9

But $z = -\frac{\pi}{2}$ is inside the circle $|z| = 5$

$$\phi(z) = \sin 3z$$

$\therefore$ By Cauchy's integral formula,

$$\begin{aligned} \oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz &= 2\pi i \phi\left(-\frac{\pi}{2}\right) \\ &= 2\pi i \sin\left(-\frac{3\pi}{2}\right) \\ &= -2\pi i \sin \frac{3\pi}{2} \quad \dots \{ \because \sin \frac{3\pi}{2} = -1 \} \\ &= 2\pi i \end{aligned}$$

Ex. 6.7.9 : Evaluate $\oint_C \frac{e^z}{z(1-z)^3} dz$, where C is

$$a) |z| = \frac{1}{2}, \quad b) |z-1| = \frac{1}{2}$$

GTU : Winter-22, CVNM, Marks 7

Sol. : Here, $f(z) = \frac{e^z}{z(1-z)^3}$ is analytic everywhere except $z = 0, 1$.

a) For $|z| = \frac{1}{2}$, $z = 0$ is inside and $z = 1$ is outside.

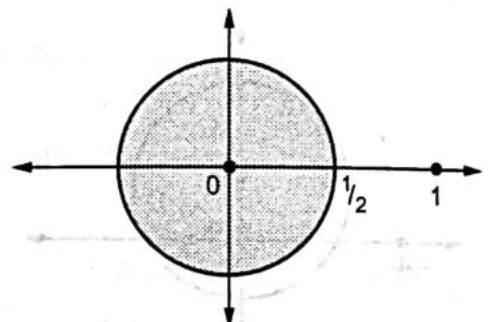


Fig. 6.7.10

$$\phi(z) = \frac{e^z}{(1-z)^3}$$

$$\begin{aligned} \oint_C \frac{e^z}{z(1-z)^3} dz &= \oint_C \frac{e^z / (1-z)^3}{z} dz \\ &= 2\pi i \phi(0) = 2\pi i (1) = 2\pi i \end{aligned}$$

b) For $|z-1| = \frac{1}{2}$, $z = 0$ is outside and $z = 1$ is inside.

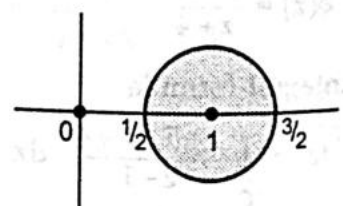


Fig. 6.7.11

$$\phi(z) = \frac{e^z}{z}$$

By Cauchy's integral formula,

$$\oint_C \frac{e^z}{z(1-z)^3} dz = \oint_C \frac{e^z/z}{(1-z)^3} dz$$

$$= -\frac{2\pi i}{2} \phi''(1)$$

$$\phi(z) = \frac{e^z}{z}$$

$$\phi'(z) = \frac{e^z \cdot z - e^z}{z^2} = \frac{e^z(z-1)}{z^2}$$

$$\phi''(z) = \frac{[e^z(z-1) + e^z]z^2 - e^z(z-1) \cdot 2z}{z^4}$$

$$\oint_C \frac{e^z}{z(1-z)^3} dz = \frac{2\pi i}{2!} \phi''(1) = -\frac{2\pi i}{2} \cdot e = -\pi i e$$

Ex 6.7.10 : Evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$, where C is the circle

$$|z|=3.$$

GTU : Winter-22, CVNM, Marks 7

Sol : Here $f(z) = \frac{e^{2z}}{(z+1)^4}$ is analytic everywhere except $z = -1$.

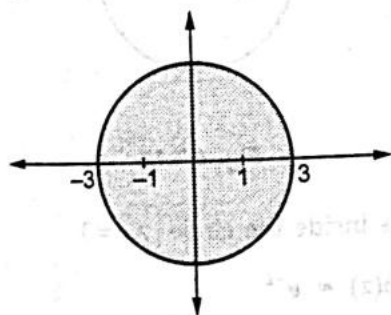


Fig. 6.7.12

But $z = -1$ is inside the circle $|z| = 3$.

$$\phi(z) = e^{2z}$$

By Cauchy's integral formula

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{3!} \phi'''(-1)$$

$$\phi(z) = e^{2z}$$

$$\phi'(z) = 2e^{2z}$$

$$\phi''(z) = 4e^{2z}$$

$$\phi'''(z) = 8e^{2z}$$

$$\phi'''(-1) = 8e^{-2}$$

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i}{6} 8e^{-2} = \frac{8}{3} \pi i e^{-2}$$

Ex. 6.7.11 : Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$ **GTU : Winter-23, CVPDE, Marks 7, Winter-23, Maths-4, Marks 7**

Sol : Here $f(z) = \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}$ is analytic everywhere except $z = 1, 2$.

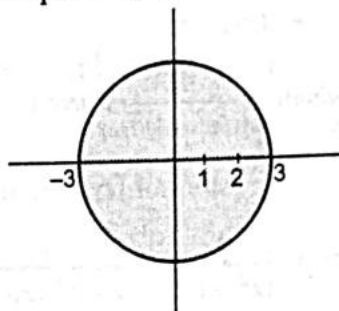


Fig. 6.7.13

But $z = 1, 2$ are inside the circle $|z| = 3$

$$\text{Now, } \frac{1}{(z-1)^2(z-2)} = \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{z-2}$$

$$\therefore 1 = A(z-1)(z-2) + B(z-2) + C(z-1)^2$$

$$\text{Take } z = 1 \Rightarrow 1 = -B \Rightarrow B = -1$$

$$\text{Take } z = 2 \Rightarrow 1 = C \Rightarrow C = 1$$

$$\text{Take } z = 0 \Rightarrow 1 = 2A - 2B + C \Rightarrow A = -1$$

$$\text{So, } \frac{1}{(z-1)^2(z-2)} = \frac{-1}{(z-1)} - \frac{1}{(z-1)^2} + \frac{1}{(z-2)}$$

$\therefore$ By Cauchy's integral formula,

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$$

$$= -\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)} dz$$

$$- \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2} dz$$

$$+ \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{z-2} dz$$

$$= -2\pi i \phi(1) - \frac{2\pi i}{1!} \phi'(1) + 2\pi i \phi(2)$$

$$\phi(z) = \sin \pi z^2 + \cos \pi z^2$$

$$\phi'(z) = \cos \pi z^2 \cdot 2\pi z - \sin \pi z^2 \cdot 2\pi z$$

$$\begin{aligned} \therefore \oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz &= -2\pi i (-1) - 2\pi i (-2\pi) + 2\pi i (1) \\ &= 2\pi i + 4\pi^2 i + 2\pi i \\ &= 4\pi i + 4\pi^2 i \\ &= 4\pi(1 + \pi)i \end{aligned}$$

Ex. 6.7.12 : Evaluate $\oint_C \frac{dz}{(z^2-1)^2}$ where $C: |z+i|=1$

Sol. : Here,

$$f(z) = \frac{1}{(z^2+1)^2} = \frac{1}{(z-i)^2(z+i)^2}$$

$\therefore f(z)$ is analytic everywhere except $z = \pm i$.

But $z = -i$ is inside and $z = i$ is outside the circle $|z+i|=1$.

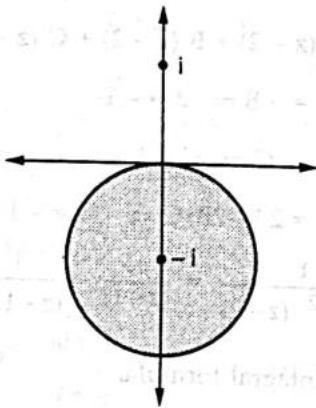


Fig. 6.7.14

Let, $\phi(z) = \frac{1}{(z-i)^2}$

$$\begin{aligned} \oint_C \frac{dz}{(z^2+1)^2} &= \oint_C \frac{1}{(z-i)^2(z+i)^2} dz \\ &= \oint_C \frac{1/(z-i)^2}{(z+i)^2} dz \end{aligned}$$

$\therefore$ By Cauchy's integral formula

$$\begin{aligned} \oint_C \frac{dz}{(z^2+1)^2} &= \oint_C \frac{1/(z-i)^2}{(z+i)^2} dz \\ &= \frac{2\pi i}{1!} \phi'(-i) \end{aligned}$$

$$\phi'(z) = \frac{-2}{(z-i)^3}$$

$$\begin{aligned} \oint_C \frac{dz}{(z^2+1)^2} &= \frac{2\pi i}{1!} \left(\frac{-2}{(-2i)^3} \right) \\ &= 2\pi i \left(\frac{-2}{-8i^3} \right) = \frac{4\pi i}{8i^3} = -\frac{\pi}{2} \end{aligned}$$

Ex. 6.7.13 : Evaluate $\oint_C \frac{e^{-z}}{(z+2)^5} dz$, where $C: |z|=3$

Sol. : Here, $f(z) = \frac{e^{-z}}{(z+2)^5}$ is analytic everywhere except $z = -2$.

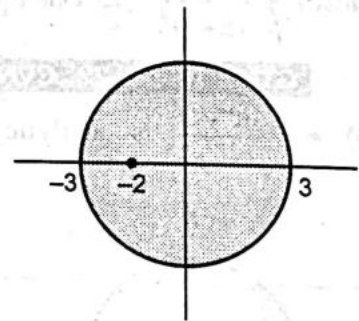


Fig. 6.7.15

But $z = -2$ is inside the circle $|z|=3$.

$$\phi(z) = e^{-z}$$

$\therefore$ By Cauchy's integral formula,

$$\oint_C \frac{e^{-z}}{(z+2)^5} dz = \frac{2\pi i}{4!} \phi^{iv}(-2)$$

$$\phi'(z) = -e^{-z}$$

$$\phi''(z) = e^{-z}$$

$$\phi'''(z) = -e^{-z}$$

$$\phi^{iv}(z) = e^{-z}$$

$$\oint_C \frac{e^{-z}}{(z+2)^5} dz = \frac{2\pi i}{24} e^2$$

$$= \frac{\pi i e^2}{12}$$

Ex. 6.7.14 : Evaluate

$$\oint_C \frac{\sin^2 z}{(z - \frac{\pi}{2})^3} dz, \text{ where } C : |z| = \pi$$

Sol. : Here, $f(z)$ is analytic everywhere except $z = \frac{\pi}{2}$.

But $z = \frac{\pi}{2}$ is inside the circle $|z| = \pi$

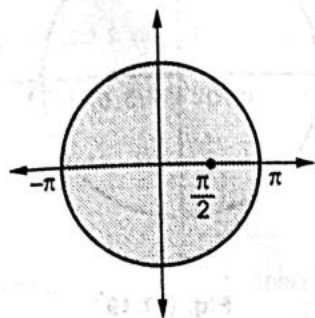


Fig. 6.7.16

$$\phi(z) = \sin^2 z$$

$\therefore$ By Cauchy's integral formula,

$$\oint_C \frac{\sin^2 z}{(z - \frac{\pi}{2})^3} dz = \frac{2\pi i}{2!} \phi''\left(\frac{\pi}{2}\right)$$

$$\phi(z) = \sin^2 2z$$

$$\phi'(z) = \sin 2z$$

$$\phi''(z) = 2 \cos 2z$$

$$\phi''\left(\frac{\pi}{2}\right) = -2$$

$$\oint_C \frac{\sin^2 z}{(z - \frac{\pi}{2})^3} dz = \frac{2\pi i}{2} (-2) = -2\pi i$$

Ex. 6.7.15 : Evaluate $\int_C \frac{5z+7}{z^2+2z-3} dz$ where C is the circle

$$|z-2| = 2.$$

GTU : Winter-24, Maths-4, Marks 7

Sol. : Here, $f(z) = \frac{5z+7}{z^2+2z-3} = \frac{5z+7}{(z-1)(z+3)}$ is analytic everywhere except $z = 1, z = -3$.

$C : |z-2| = 2$ is the circle with center $(2, 0)$ and radius is 2.

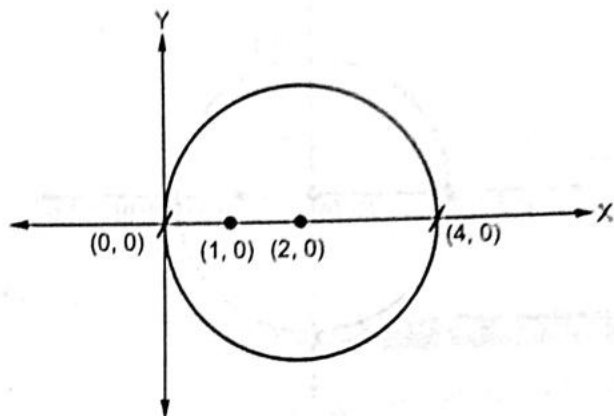


Fig. 6.7.17

Now $z = 1 = (1, 0)$ is inside and $z = -3 = (-3, 0)$ is outside of the circle $|z-2| = 2$.

$\therefore$ By Cauchy integral formula.

$$\oint_C \frac{(5z+7)/(z+3)}{(z-1)} dz = 2\pi i f(z_0)$$

$$\text{where, } f(z) = \frac{5z+7}{z+3} \text{ and } z_0 = 1$$

$$\therefore f(z_0) = f(1) = \frac{12}{4} = 3$$

$$\therefore \oint_C \frac{5z+7}{z^2+2z-3} dz = 2\pi i (3) = 6\pi i$$

Ex. 6.7.16 : State Cauchy's Integral formula. Also, Evaluate $\oint_C \frac{e^z}{z(z-1)} dz$ where C is the circle $|z| = 2$.

GTU : Winter-22, Maths-4, Marks 7

Sol. : Cauchy's integral formula :

If $f(z)$ is analytic within and on a closed curve C and if a is any point within C then $\oint_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$.

Now, $\oint_C \frac{e^z}{z(z-1)} dz, C : |z| = 2$

Here, $f(z) = \frac{e^z}{z(z-1)}$ is analytic every where except $z = 0$ and $z = 1$.

$C : |z| = 2 \rightarrow$ Circle with center $(0, 0)$ and radius 2

Here, $z = 0$ and $z = 1$ inside the circle $|z| = 2$.

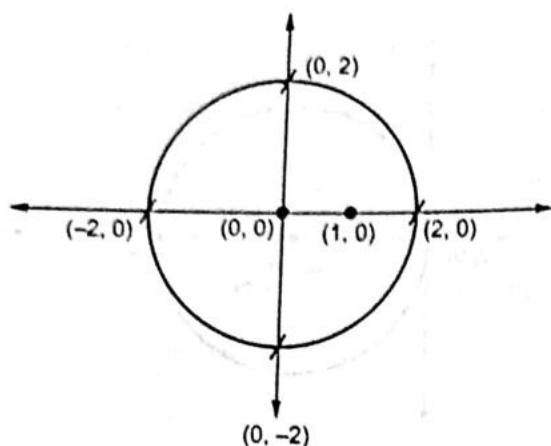


Fig. 6.7.18

∴ First we use partial fraction

$$\frac{1}{z(z-1)} = \frac{A}{z} + \frac{B}{z-1}$$

$$1 = A(z-1) + Bz$$

$$z=0 \Rightarrow 1 = -A \Rightarrow A = -1$$

$$z=1 \Rightarrow 1 = B \Rightarrow B = 1$$

$$\therefore \frac{1}{z(z-1)} = \frac{-1}{z} + \frac{1}{z-1}$$

$$\begin{aligned} \therefore \oint_C \frac{e^z}{z(z-1)} dz &= -\oint_C \frac{e^z}{z} dz + \oint_C \frac{e^z}{z-1} dz \\ &= -2\pi i f(z_0) + 2\pi i f(z_0) \end{aligned}$$

Where

$$f(z) = e^z \text{ and}$$

$$z_0 = 0$$

$$\therefore f(z_0) = f(0) = e^0 = 1$$

Where

$$f(z) = e^z \text{ and}$$

$$z_0 = 1$$

$$f(z_0) = f(1) = e^1$$

$$\begin{aligned} \therefore \oint_C \frac{e^z}{z(z-1)} dz &= -2\pi i(1) + 2\pi i(e) \\ &= 2\pi i(e-1) \end{aligned}$$

Ex. 6.7.17 : Evaluate $\int_C \frac{dz}{z^2 - 7z + 12}$ where C is the circle $|z| = 3.5$.

GTU : Winter-21, Maths-4, Marks 7, Summer-22, CVPDE, Marks 4

Sol. : Here $f(z) = \frac{1}{z^2 - 7z + 12} = \frac{1}{(z-3)(z-4)}$ is analytic everywhere except $z = 3$ and $z = 4$

Now C is the circle $|z| = 3.5$ whose center is $z = 0$ and radius is 3.5.

∴ $z = 3$ is inside but $z = 4$ is outside of the circle $|z| = 3.5$

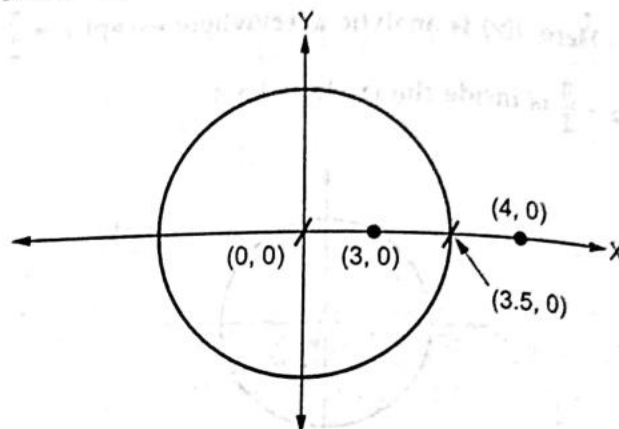


Fig. 6.7.19

∴ By Cauchy integral formula

$$\oint_C \frac{1/(z-4)}{z-3} dz = 2\pi i f(z_0)$$

$$\text{Where } f(z) = \frac{1}{z-4} \text{ and } z_0 = 3$$

$$\therefore f(z_0) = f(3) = \frac{1}{3-4} = -1$$

$$\begin{aligned} \therefore \oint_C \frac{1}{z^2 - 7z + 12} dz &= 2\pi i (-1) \\ &= -2\pi i \end{aligned}$$

Ex. 6.7.18 : Evaluate $\int_C \frac{\cos z}{z+\pi} dz$, where C is the circle $|z| = 4$.

GTU : Summer-19, Maths-4, Marks 3

Sol. : Here, $\frac{\cos z}{z+\pi}$ is not analytic at $z = -\pi$

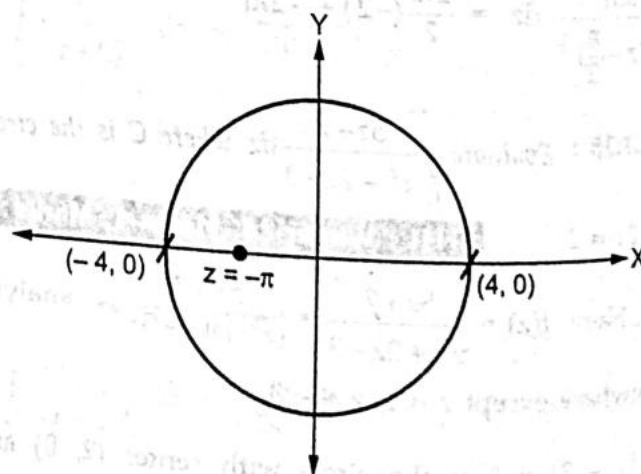


Fig. 6.7.20

Now, circle $|z| = 4$ is given with center $z = 0$ and radius 4.

But $z = -\pi$ is inside of the circle $|z| = 4$

$\therefore$ By Cauchy Integral formula

$$\oint_C \frac{\cos z}{z + \pi} dz = 2\pi i f(z_0)$$

Where, $f(z) = \cos z$ and $z_0 = -\pi$

$$\therefore f(z_0) = f(-\pi) = \cos(-\pi) = -1$$

$$\therefore \oint_C \frac{\cos z}{z + \pi} dz = 2\pi i (-1) = -2\pi i$$

Ex. 6.7.19 : Evaluate $\int_C \frac{z + e^z}{(z-1)^3} dz$, where C is $|z| = 2$

Using Cauchy's integral formula.

GTU : Winter-22, CVPDE, Marks 4

Sol. :

Here, $f(z) = \frac{z + e^z}{(z-1)^3}$ is not analytic at $z = 1$.

$C : |z| = 2$ is circle with center $z = 0$ and radius is 2.

Here, $z = 1$ is inside of the circle $|z| = 2$

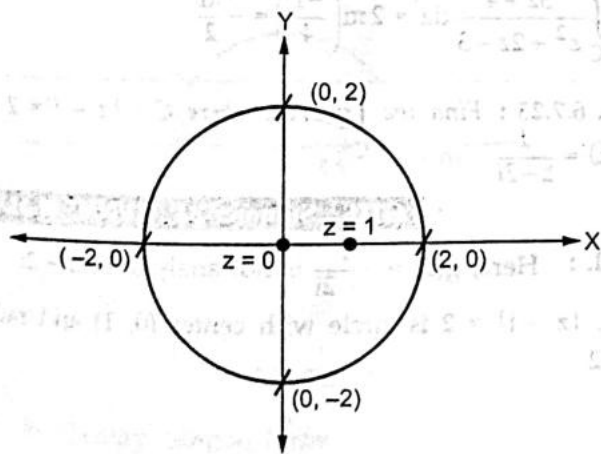


Fig. 6.7.21

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{z + e^z}{(z-1)^3} dz = \frac{2\pi i}{(3-1)!} f''(z_0)$$

Where, $f(z) = z + e^z$ and $z_0 = 1$

$$\therefore f'(z) = 1 + e^z$$

$$f''(z) = e^z \Rightarrow f''(1) = e^1 = e$$

$$\therefore \oint_C \frac{z + e^z}{(z-1)^3} dz = \frac{2\pi i}{(2)} \times (e)$$

$$= \pi i e$$

Ex. 6.7.20 : Evaluate i) $\oint_C \frac{e^z}{(z-1)(z-3)} dz$, $C : |z| = 2$ and

ii) $\oint_C e^z dz$, $C : |z| = 3$.

GTU : Winter-21, CVPDE, Marks 7

Sol. : i) $f(z) = \frac{e^z}{(z-1)(z-3)}$ is not analytic at $z = 1$ and $z = 3$.

$C : |z| = 2$ Circle with center $z = 0$ and radius is 2.

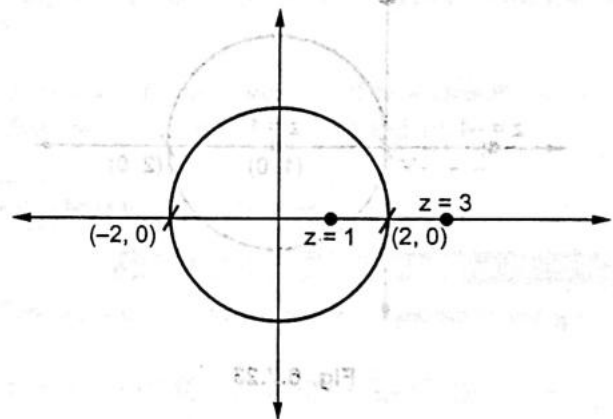


Fig. 6.7.22

Here, $z = 1$ is inside and $z = 3$ is outside of the region $|z| = 2$

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{e^z}{(z-1)(z-3)} dz = 2\pi i f(z_0)$$

Where $f(z) = \frac{e^z}{z-3}$ and $z_0 = 1$

$$\therefore f(z_0) = f(1) = \frac{e^1}{1-3} = -\frac{e}{2}$$

$$\therefore \oint_C \frac{e^z}{(z-1)(z-3)} dz = 2\pi i \left(-\frac{e}{2}\right) = -\pi i e$$

ii) $\oint_C e^z dz$, $C : |z| = 3$.

Here, $f(z) = e^z$ is analytic everywhere

$\therefore$ By Cauchy Goursat theorem

$$\oint_C e^z dz = 0$$

Ex. 6.7.21 : Evaluate $\int_C \frac{z^2+1}{z^2-1} dz$ where C is $|z-1| = 1$.

GTU : Summer-23, CVPDE, Marks 4

Sol. : Here, $f(z) = \frac{z^2+1}{z^2-1} = \frac{z^2+1}{(z-1)(z+1)}$ is not analytic at $z = 1$ and $z = -1$.

Curve $C : |z-1| = 1$ is a circle with center $(1, 0)$ and radius is 1

$\therefore z = 1$ is inside but $z = -1$ is outside of the region $|z-1| = 1$

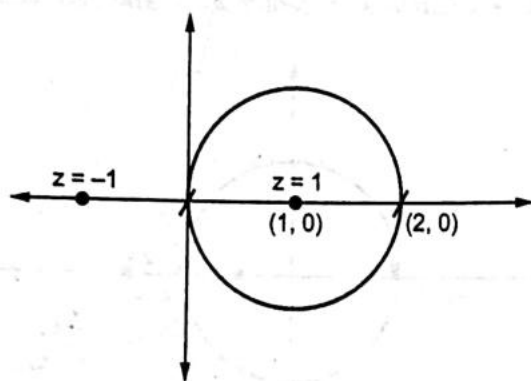


Fig. 6.7.23

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{z^2+1}{(z-1)(z+1)} dz = 2\pi i f(z_0)$$

Where $f(z) = \frac{z^2+1}{(z+1)}$ and $z_0 = 1$

$$\therefore f(z_0) = f(1) = \frac{2}{2} = 1$$

$$\therefore \oint_C \frac{z^2+1}{z^2-1} dz = 2\pi i (1) = 2\pi i$$

Ex. 6.7.22 : Evaluate $\oint_C \frac{3z+4}{z^2+2z-3} dz$, where C is $|z-i| = 2$.

GTU : Summer-22, CVPDE, Marks 4

Sol. : Here, $f(z) = \frac{3z+4}{z^2+2z-3} = \frac{3z+4}{(z-3)(z+1)}$ is not analytic at $z = 3$ and $z = -1$.

$C : |z-i| = 2$ is the circle with radius 2 and center $(0, 0)$

Now, $z = -1$ is inside but $z = 3$ outside of the region $|z-i| = 2$

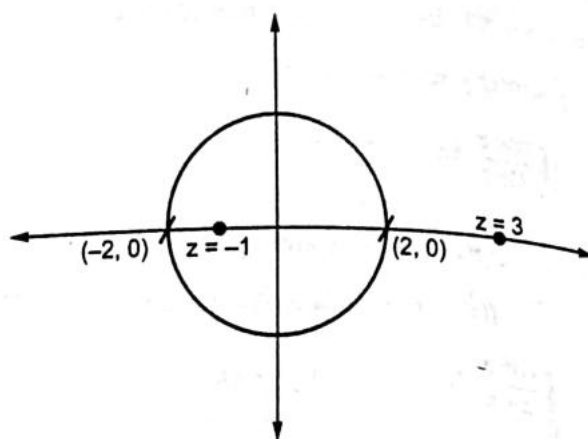


Fig. 6.7.24

$\therefore$ By Cauchy integral formula

$$\oint_C \frac{3z+4}{z^2+2z-3} dz = 2\pi i f(z_0)$$

Where $f(z) = \frac{3z+4}{(z-3)}$ and $z_0 = -1$

$$\therefore f(z_0) = f(-1) = \frac{3(-1)+4}{(-1-3)} = \frac{-1}{4}$$

$$\therefore \oint_C \frac{3z+4}{z^2+2z-3} dz = 2\pi i \left(\frac{-1}{4} \right) = -\frac{\pi i}{2}$$

Ex. 6.7.23 : Find the $\int_C g(z) dz$ where $C : |z-i| = 2$ and $g(z) = \frac{1}{z-2i}$

GTU : Summer-21, CVNM, Marks 3

Sol. : Here, $g(z) = \frac{1}{z-2i}$ is not analytic at $z = 2i$

$C : |z-i| = 2$ is circle with center $(0, 1)$ and radius is 2

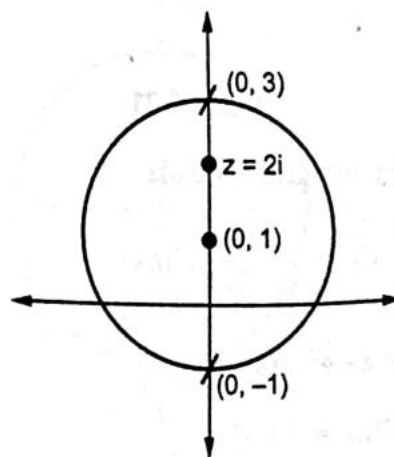


Fig. 6.7.25

But $z = 2i$ is inside of the circle $|z - i| = 2$

∴ By Cauchy integral formula

$$\oint_C \frac{1}{z-2i} dz = 2\pi i f(z_0)$$

Where $f(z) = 1$, and $z_0 = 2i$

$$f(z_0) = f(2i) = 1$$

$$\therefore \oint_C \frac{1}{z+2i} dz = 2\pi i$$

Ex. 6.7.24 : Find the $\int_C g(z) dz$ where $C : |z| = 2$ and $g(z) = \frac{1}{z^2 - 5z + 4}$.

GTU : Summer-21, CVNM, Marks 4

Sol. : Here, $g(z) = \frac{1}{z^2 - 5z + 4} = \frac{1}{(z-4)(z-1)}$ is not

analytic at $z = 4$ and $z = 1$.

$C : |z| = 2$ is the circle with center $z = 0$ and radius is 2

Here, $z = 1$ is inside but $z = 4$ is outside of the circle $|z| = 2$.

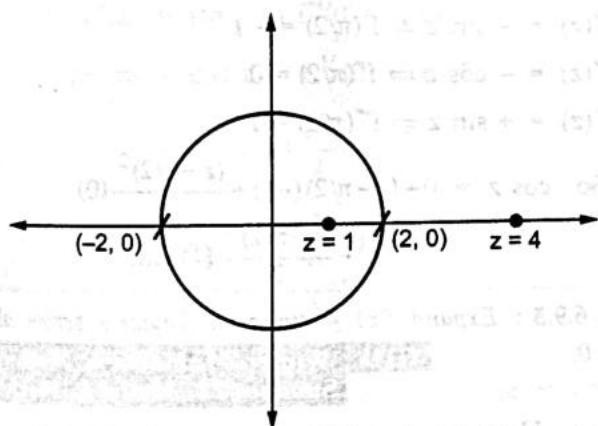


Fig. 6.7.26

∴ By Cauchy integral formula

$$\oint_C \frac{1}{z^2 - 5z + 4} dz = 2\pi i f(z_0)$$

Where, $f(z) = \frac{1}{z-4}$ and $z_0 = 1$

$$\therefore f(z_0) = f(1) = \frac{1}{1-4} = -\frac{1}{3}$$

$$\therefore \oint_C \frac{1}{z^2 - 5z + 4} dz = 2\pi i \left(-\frac{1}{3}\right) = -\frac{2}{3}\pi i$$

6.8 Taylor's Series

• Taylor's series :

If $f(z)$ is analytic function inside and on a simple closed curve C in the z -plane, the higher order derivative of $f(z)$ is also exist inside C then

$$\begin{aligned} f(z) &= f(z_0) + \frac{(z-z_0)}{1!} f'(z_0) + \frac{(z-z_0)^2}{2!} f''(z_0) \\ &\quad + \frac{(z-z_0)^3}{3!} f'''(z_0) + \dots + \frac{(z-z_0)^n}{n!} f^{(n)}(z_0) + \dots \\ &= \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{n!} f^{(n)}(z_0) \end{aligned}$$

is known as Taylor's series expansion of $f(z)$ about $z = z_0$.

The radius of convergent of this series is $|z - z_0| < R$, a disk centered on $z = z_0$ and radius R .

6.9 Maclaurin's Series

GTU : Summer-21,22, Winter-21,22,23,24

When $z_0 = 0$ in Taylor's series expansion we get,

$$\begin{aligned} f(z) &= f(0) + z f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots + \frac{z^n}{n!} f^{(n)}(0) \\ &= \sum_{n=0}^{\infty} \frac{z^n}{n!} f^{(n)}(0) \end{aligned}$$

is said to be a Maclaurin's series expansion of $f(z)$ about $z = 0$.

• Binomial series for any positive integer m is

$$\frac{1}{(1-z)^m} = \sum_{n=0}^{\infty} \frac{(m+n-1)!}{(m-1)! n!} z^n ; |z| < 1$$

• Geometric series :

$$\text{i) } \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n ; |z| < 1$$

$$= 1 + z + z^2 + z^3 + \dots$$

$$\text{ii) } \frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n ; |z| < 1$$

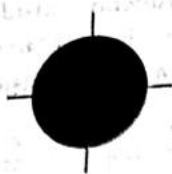
$$= 1 - z + z^2 - z^3 + \dots$$

Note : Closed disk with boundaries.

Mathematics - 2

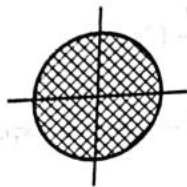
i) $r = |z| = a$

circle with center 0 and radius a.



ii) $r = |z| \leq a$

circle with center 0 and radius a.



iii) $r = |z| \geq a$

circle with center 0 and radius a.

**Note : Open disk without boundaries.**

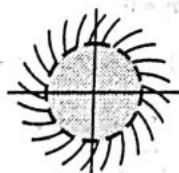
i) $r = |z| < a$

circle with center 0 and radius a.



ii) $r = |z| > a$

circle with center 0 and radius a.

**Ex. 6.9.1 :** Find series expansion of $f(z) = e^{3z}$ in power at z.or Expand e^{3z} using Maclaurin's series.or Expand e^{3z} at $z = 0$.**Sol. :**

$$f(z) = f(0) + zf'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots$$

Now

$$f(z) = e^{3z} \Rightarrow f(0) = e^0 = 1$$

$$f'(z) = 3e^{3z} \Rightarrow f'(0) = 3e^0 = 3$$

$$f''(z) = 9e^{3z} \Rightarrow f''(0) = 9e^0 = 9$$

$$f'''(z) = 27e^{3z} \Rightarrow f'''(0) = 27e^0 = 27$$

$$\begin{aligned} \text{So } f(z) &= 1 + z(3) + \frac{z^2}{2!}(9) + \frac{z^3}{3!}(27) + \dots \\ &= 1 + 3z + \frac{9}{2}z^2 + \frac{9}{2}z^3 + \dots \end{aligned}$$

Ex. 6.9.2 : Find expansion of $f(z) = \cos z$ in power of $(z - \pi/2)$.or Using Taylor's series expand $\cos z$ at $z = \pi/2$.**Sol. :**

$$\begin{aligned} f(z) &= f(\pi/2) + (z - \pi/2)f'(\pi/2) \\ &\quad + \frac{(z - \pi/2)^2}{2!} f''(\pi/2) + \frac{(z - \pi/2)^3}{3!} f'''(\pi/2) + \dots \end{aligned}$$

$$f(z) = \cos z \Rightarrow f(\pi/2) = \cos \pi/2 = 0$$

$$f'(z) = -\sin z \Rightarrow f'(\pi/2) = -1$$

$$f''(z) = -\cos z \Rightarrow f''(\pi/2) = 0$$

$$f'''(z) = +\sin z \Rightarrow f'''(\pi/2) = 1$$

$$\begin{aligned} \text{So } \cos z &= 0 + (z - \pi/2)(-1) + \frac{(z - \pi/2)^2}{2!}(0) \\ &\quad + \frac{(z - \pi/2)^3}{3!}(1) + \dots \end{aligned}$$

Ex. 6.9.3 : Expand $f(z) = \sin z$ in Taylor's series about $z = 0$.**GTU : Winter-24, Maths-4, Marks 7.**
Summer-22, Maths-4, Marks 4**Sol. :** Here, we use Taylor's series about $z = 0$.

$$\begin{aligned} \therefore f(z) &= f(0) + z \cdot f'(0) + \frac{z^2}{2!} f''(0) \\ &\quad + \frac{z^3}{3!} f'''(0) + \frac{z^4}{4!} f^{IV}(0) + \dots \end{aligned}$$

$$\text{Now, } f(z) = \sin z \Rightarrow f(0) = \sin 0 = 0$$

$$f'(z) = \cos z \Rightarrow f'(0) = \cos 0 = 1$$

$$f''(z) = -\sin z \Rightarrow f''(0) = -\sin 0 = 0$$

$$f'''(z) = -\cos z \Rightarrow f'''(0) = -\cos 0 = -1$$

$$f^{IV}(z) = \sin z \Rightarrow f^{IV}(0) = \sin 0 = 0$$

$$f^V(z) = \cos z \Rightarrow f^V(0) = \cos 0 = 1$$

$$\sin z \approx 0 + z \cdot (1) + \frac{z^2}{2!}(0) + \frac{z^3}{3!}(-1)$$

$$+ \frac{z^4}{4!}(0) + \frac{z^5}{5!}(-1) + \dots$$

$$\sin z \approx z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots$$

Ex. 6.9.4 : Expand $f(z) = \cos z$ as a Taylor's series about $z = 0$.

GTU : Winter-23, CVPDE, Marks 4

Sol. : Here, $f(z) = \cos z$ is given and point $z = 0$ is also given.

$$f(z) = f(0) + z f'(0) + \frac{z^2}{2!} f''(0)$$

$$+ \frac{z^3}{3!} f'''(0) + \frac{z^4}{4!} f^{IV}(0) + \dots$$

Now,

$$f(z) = \cos z \Rightarrow f(0) = \cos 0 = 1$$

$$f'(z) = -\sin z \Rightarrow f'(0) = -\sin 0 = 0$$

$$f''(z) = -\cos z \Rightarrow f''(0) = -\cos 0 = -1$$

$$f'''(z) = \sin z \Rightarrow f'''(0) = \sin 0 = 0$$

$$f^{IV}(z) = \cos z \Rightarrow f^{IV}(0) = \cos 0 = 1$$

$$\cos z \approx 1 + z(0) + \frac{z^2}{2!}(-1) +$$

$$\frac{z^3}{3!}(0) + \frac{z^4}{4!}(1) + \dots$$

$$\cos z \approx 1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \dots$$

Ex. 6.9.5 : Expand $\ln(1+z)$ as a Taylor's series about $z = 0$ for $|z| < 1$

GTU : Winter-22, CVPDE, Marks 3

Sol. $f(z) = \ln(1+z)$ and $z = 0$ is given

$$f(z) = f(0) + z f'(0) + \frac{z^2}{2!} f''(0)$$

$$+ \frac{z^3}{3!} f'''(0) + \frac{z^4}{4!} f^{IV}(0) + \dots$$

$$f(z) = \ln(1+z) \Rightarrow f(0) = 0$$

$$f'(z) = \frac{1}{1+z} \Rightarrow f'(0) = 1$$

$$f''(z) = -\frac{1}{(1+z)^2} \Rightarrow f''(0) = -1$$

$$f'''(z) = \frac{2}{(1+z)^3} \Rightarrow f'''(0) = 2$$

$$f^{IV}(z) = \frac{-6}{(1+z)^4} \Rightarrow f^{IV}(0) = -6$$

$$\therefore \ln(1+z) = 0 + z(1) + \frac{z^2}{2!}(-1)$$

$$= \frac{z^3}{3!}(2) + \frac{z^4}{4!}(-6) + \dots$$

$$\therefore \ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots$$

Ex. 6.9.6 : Find the Maclaurin series for

$f(z) = \sin^2 z$. Also find R.O.C.

Sol. :

$$f(z) = f(0) + z f'(0) + \frac{z^2}{2!} f''(0) + \frac{z^3}{3!} f'''(0) + \dots$$

$$f(z) = \sin^2 z \Rightarrow f(0) = 0$$

$$f'(z) = 2 \sin z \cos z \Rightarrow f'(z) = \sin 2z$$

$$\Rightarrow f'(0) = 0$$

$$f''(z) = 2 \cos 2z \Rightarrow f''(0) = 2$$

$$f'''(z) = -4 \sin 2z \Rightarrow f'''(0) = 0$$

$$f^{IV}(z) = -8 \cos 2z \Rightarrow f^{IV}(0) = -8$$

$$f^V(z) = 16 \sin 2z \Rightarrow f^V(0) = 0$$

$$f^{VI}(z) = 32 \cos 2z \Rightarrow f^{VI}(0) = 32$$

$$\text{So, } f(z) = \frac{z^2}{2!}(2) + \frac{z^4}{4!}(-8) + \frac{z^6}{6!}(32) + \dots$$

$$= z^2 - \frac{z^4}{3} + \frac{2z^6}{45} + \dots$$

$$\text{Here } f(z) = \sin^2 z \quad |z| < \infty$$

$$\text{So, } f(z) = \sin^2 z \quad |z| < \infty$$

So, radius of convergence is ∞ .

Ex. 6.9.7 : Expand $f(z) = \sin z$, at $z = \pi/4$.

GTU : Winter-21, CVPDE Marks 4

Sol. :

$$f(z) = f(\pi/4) + (z - \pi/4) f'(\pi/4)$$

$$+ \frac{(z - \pi/4)^2}{2!} f''(\pi/4) + \frac{(z - \pi/4)^3}{3!} f'''(\pi/4) + \dots$$

$$f(z) = \sin z \Rightarrow f(\pi/4) = \frac{1}{\sqrt{2}}$$

$$f'(z) = \cos z \Rightarrow f'(\pi/4) = \cos \pi/4 = \frac{1}{\sqrt{2}}$$

$$f''(z) = -\sin z \Rightarrow f''(\pi/4) = -\frac{1}{\sqrt{2}}$$

$$f'''(z) = -\cos z \Rightarrow f'''(\pi/4) = -\frac{1}{\sqrt{2}}$$

$$\sin z = \frac{1}{\sqrt{2}} + (z - \pi/4) \frac{1}{\sqrt{2}} + \frac{(z - \pi/4)^2}{2!} (-1/\sqrt{2}) + \frac{(z - \pi/4)^3}{3!} - \left(\frac{1}{\sqrt{2}}\right) + \dots$$

Ex. 6.9.8 : Expand $\frac{1}{4-3z}$ at $z = 1 + i$

Sol. : $f(z) = f(1+i) + (z-1-i)f'(1+i) + \frac{(z-(1+i))^2}{2!} f''(1+i) + \dots$

$$= \frac{1}{4-3z} \Rightarrow f(1+i) = \frac{1}{4-3(1+i)}$$

$$= \frac{1}{4-3-3i} = \frac{1}{1-3i}$$

$$f'(z) = -\frac{1}{(4-3z)^2} (-3) = \frac{3}{(4-3z)^2} = \frac{3}{(1-3i)^2}$$

$$f''(z) = -\frac{2}{(4-3z)^3} (-3) = \frac{6}{(4-3z)^3} = \frac{6}{(1-3i)^3}$$

$$f(z) = \frac{1}{1-3i} + (z-(1+i)) \frac{3}{(1-3i)^2} + (z-(1+i))^2 \frac{6}{(1-3i)^3} + \dots$$

• Standard expansion :

i) $\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \quad |z| < \infty$

ii) $\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \quad |z| < \infty$

iii) $e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \quad |z| < \infty$

iv) $\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots \quad |z| < 1$

v) $\frac{1}{1+z} = 1 - z + z^2 - z^3 + \dots \quad |z| < 1$

Ex. 6.9.9 : Expand $f(z) = \log\left(\frac{2+z}{2-z}\right)$ using Taylor's series about $z = 0$.

or Expand $\log\left(\frac{2+z}{2-z}\right)$ about $z = 0$.

Sol. : $f(z) = \log\left(\frac{2+z}{2-z}\right)$

$$= \log(2+z) - \log(2-z)$$

$$f'(z) = \frac{1}{2+z} - \frac{1}{2-z} = \frac{1}{2(1+z/2)} - \frac{1}{2(1-z/2)}$$

$$= \frac{1}{2} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots \right) - \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right)$$

$$= \frac{1}{2} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots - 1 - \frac{z}{2} - \frac{z^2}{4} - \frac{z^3}{8} - \dots \right)$$

$$= \frac{1}{2} \left(-\frac{z}{2} - \frac{z}{2} - \frac{z^2}{4} - \frac{z^2}{4} - \frac{z^3}{8} - \frac{z^3}{8} - \dots \right)$$

$$f'(z) = \frac{1}{2} \left(-z - \frac{z^3}{4} - \dots \right)$$

then $f(z) = \frac{1}{2} \left(-\frac{z^2}{2} - \frac{z^4}{16} - \dots \right) = -\frac{z^2}{4} - \frac{z^4}{32} - \dots$

Ex. 6.9.10 : Prove that

$$\tan^{-1} z = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \dots \quad |z| < 1$$

or Expand $\tan^{-1} z$ in the power of z .

or Expand $\tan^{-1} z$ about $z = 0$.

Sol. : $f(z) = \tan^{-1} z$

$$f'(z) = \frac{1}{1+z^2} = (1-z^2+z^4-z^6+z^8+\dots)$$

Taking integration

then $f(z) = z - \frac{z^3}{3} + \frac{z^5}{5} - \frac{z^7}{7} + \frac{z^9}{9} - \dots$

Ex. 6.9.11 : Expand $f(z) = z^3 e^{2/z}$ about $z = \infty$.

Sol. :

$$z \rightarrow \infty \Rightarrow \frac{1}{z} \rightarrow 0$$

Now $u = \frac{1}{z} \rightarrow 0$

$$f(z) = \frac{1}{u^3} e^{2u}$$

$$= \frac{1}{u^3} \left(1 + (2u) + \frac{(2u)^2}{2!} + \frac{(2u)^3}{3!} + \dots \right)$$

$$= \frac{1}{u^3} \left(1 + 2u + 2u^2 + \frac{4u^3}{3} + \dots \right)$$

$$= z^3 \left(1 + \frac{2}{z} + 2 \frac{1}{z^2} + \frac{4}{3} \frac{1}{z^3} + \dots \right)$$

$$f(z) = \left(z^3 + 2z^2 + 2z + \frac{4}{3} + \dots \right)$$

Ex. 6.9.12 : Expand $\frac{z+2}{1-z^2}$ about $z = 0$.

Sol. : $\frac{z}{1-z^2} + \frac{2}{1-z^2}$

$$= z(1-z^2)^{-1} + 2(1-z^2)^{-1}$$

$$= z(1+z^2+z^4+\dots) + 2(1+z^2+z^4+\dots)$$

$$= z + z^3 + z^5 + \dots + 2 + 2z^2 + 2z^4 + \dots$$

$$= 2 + z + 2z^2 + z^3 + 2z^4 + z^5 + \dots$$

Ex. 6.9.13 : Find the Taylor's series of

$$f(z) = \frac{2z^2+1}{z(z-1)} \text{ about i) } z = 0 \text{ ii) } z = 1$$

Sol. :

$$\frac{2z^2+1}{z(z-1)} = \frac{A}{z} + \frac{B}{z-1}$$

$$2z^2+1 = A(z-1) + Bz$$

$$z = 0 \Rightarrow 1 = A(-1) \Rightarrow A = -1$$

$$z = 1 \Rightarrow 3 = B$$

i) About $z = 0$

$$\frac{2z^2+1}{z(z-1)} = -\frac{1}{z} + \frac{3}{z-1} = -\frac{1}{z} + \frac{3}{-1-z}$$

$$= -\frac{1}{z} - \frac{3}{1+z}$$

$$= -\frac{1}{z} - 3(1+z+z^2+z^3+\dots)$$

$$= -\frac{1}{z} - 3 - 3z - 3z^2 - 3z^3 + \dots$$

ii) About $z = 1 \Rightarrow (z-1) = u \Rightarrow z = 1+u$

$$\frac{2z^2+1}{z(z-1)} = \frac{2(1+u)^2+1}{(1+u) \cdot u} = \frac{2(1+2u+u^2)+1}{u(u+1)}$$

$$= \frac{2u^2+4u+3}{u(u+1)}$$

$$\frac{2u^2+4u+3}{u(u+1)} = \frac{A}{u} + \frac{B}{u+1}$$

$$2u^2+4u+3 = A(u+1) + Bu$$

$$u = 0$$

$$3 = A$$

$$u = -1$$

$$2 - 4 + 3 = -B$$

$$\Rightarrow -1 = B$$

$$\frac{2u^2+4u+3}{u(u+1)} = \frac{3}{u} - \frac{1}{u+1}$$

$$= \frac{3}{u} - (1+u)^{-1}$$

$$= \frac{3}{u} - (1-u+u^2-u^3+\dots)$$

$$= \frac{3}{u} - 1 + u - u^2 + u^3 + \dots$$

$$= \frac{3}{(z-1)} - 1 + (z-1) - (z-1)^2 + (z-1)^3 + \dots$$

Ex. 6.9.14 : Expand $f(z) = \frac{z-1}{z+1}$ about

i) $z = 0$ ii) $z = 1$

Sol. :

i) $z = 0$

$$f(z) = \frac{z-1}{z+1} = \frac{z}{1+z} - \frac{1}{1+z} = z(1+z)^{-1} - 1(1+z)^{-1}$$

$$= z(1-z+z^2-z^3+\dots) - (1-z+z^2-z^3+\dots)$$

$$= z - z^2 + z^3 - z^4 + \dots - 1 + z - z^2 + z^3 + \dots$$

$$= -1 + 2z - 2z^2 + 2z^3 - 2z^4 + \dots$$

ii) About $z = 1$

$$\text{So } z - 1 = 0$$

We can take $u = z - 1 \quad z = 1 + u$

$$f(z) = \frac{z-1}{z+1} = \frac{1+u-1}{1+u+1} = \frac{u}{u+2} = \frac{u}{2(1+u/2)}$$

$$= \frac{u}{2} \left(1 - \frac{u}{2} + \frac{u^2}{4} - \frac{u^3}{8} + \dots + \dots \right)$$

$$= \frac{u}{2} - \frac{u^2}{4} + \frac{u^3}{8} - \frac{u^4}{16} + \dots$$

$$= \frac{(z-1)}{2} - \frac{(z-1)^2}{4} + \frac{(z-1)^3}{8} - \frac{(z-1)^4}{16} + \dots$$

Ex. 6.9.15 : Expand $\frac{2z^2+9z+5}{z^3+z^2-8z-12}$ about $z=1$

Sol. : $\frac{2z^2+9z+5}{(z-3)(z+2)^2}$

Using partial fraction

$$\frac{2z^2+9z+5}{(z-3)(z+2)^2} = \frac{A}{z-3} + \frac{B}{z+2} + \frac{C}{(z+2)^2}$$

$$2z^2+9z+5 = A(z+2)^2 + B(z+2)(z-3) + C(z-3)$$

Now $z=3 \quad 50 = 25A \Rightarrow A = \frac{1}{2}$

$z=-2 \quad -5 = -5C \Rightarrow C = 1$

$z=0 \quad 5 = 4A - 6B - 3C$

$$5 = \frac{4}{2} - 6B - 3$$

$$\Rightarrow 6B = 2 - 3 - 5 \Rightarrow 6B = -6 \quad B = -1$$

$$\frac{2z^2+9z+5}{z^3+z^2-8z-12} = \frac{1}{2} \frac{1}{z-3} - \frac{1}{z+2} + \frac{1}{(z+2)^2}$$

Now about $z=1 \quad z-1=0$

So, $u = z-1 \Rightarrow z = 1+u$

$$= \frac{1}{2} \frac{1}{1+u-3} - \frac{1}{u+1+2} + \frac{1}{(u+1+2)^2}$$

$$= \frac{1/2}{u-2} - \frac{1}{u+3} + \frac{1}{(u+3)^2}$$

$$= \frac{1/2}{-2\left(1-\frac{u}{2}\right)} - \frac{1}{3\left(1+\frac{u}{3}\right)} + \frac{1}{9\left(1+\frac{u}{3}\right)^2}$$

$$= -\frac{1}{4} \left(1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right)$$

$$- \frac{1}{3} \left(1 - \frac{u}{3} + \frac{u^2}{9} - \frac{u^3}{27} + \dots \right) + \frac{1}{9} \left(1 - \frac{2u}{3} + \frac{6u^2}{9} + \dots \right)$$

$$\left(\because \frac{1}{(1+z)^m} = \sum_{n=0}^{\infty} \frac{(m+n-1)!}{(m-1)! n!} (-1)^n z^n \right)$$

$$= -\frac{1}{4} - \frac{u}{8} - \frac{u^2}{16} - \frac{u^3}{32} + \dots - \frac{1}{3} + \frac{u}{9} - \frac{u^2}{27} + \frac{u^3}{81} + \dots$$

$$+ \frac{1}{9} - \frac{2u}{27} + \frac{1}{27} u^2 + \dots$$

$$= -\frac{17}{36} - \frac{19}{216} u - \frac{1}{16} u^2 + \dots$$

$$= -\frac{17}{36} - \frac{19}{216} (z-1) - \frac{1}{16} (z-1)^2 + \dots$$

Ex. 6.9.16 : Find the Maclaurin's series for

$$f(z) = \frac{1}{z+3i}$$

Sol. : $f(z) = \frac{1}{z+3i} = \frac{1}{3i} \frac{1}{(1+z/3i)}$

$$= \frac{1}{3i} \left(1 - \frac{z}{3i} + \frac{z^2}{(3i)^2} - \frac{z^3}{(3i)^3} + \dots \right)$$

$$= \frac{1}{3i} \left(1 - \frac{z}{3i} + \frac{z^2}{-9} + \frac{z^3}{27i} + \dots \right)$$

$$= \left(\frac{1}{3i} + \frac{z}{9} - \frac{z^2}{27i} - \frac{z^3}{81} + \dots \right)$$

Ex. 6.9.17 : Show that $\frac{1+2z}{z^2+z^3} = \frac{1}{z^2} + \frac{1}{z} - 1 + z - z^2 + \dots$

$$0 < |z| < 1$$

GTU : Summer-22, CVNM, Marks 3

Sol. : $\frac{1+2z}{z^2(1+z)} = \frac{1}{z^2(1+z)} + \frac{2}{z(1+z)}$

$$= \frac{1}{z^2} (1 - z + z^2 - z^3 + z^4)$$

$$+ \frac{2}{z} (1 - z + z^2 - z^3 + z^4)$$

$$= \left(\frac{1}{z^2} - \frac{1}{2} + 1 - z + z^2 \dots \right)$$

$$+ \left(\frac{2}{z} - 2 + 2z - 2z^2 + \dots \right)$$

$$= \frac{1}{z^2} + \frac{1}{z} - 1 + z - z^2 + \dots$$

Ex. 6.9.18 : Show that when $z \neq 0$

$$\frac{e^z}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \frac{z}{3!} + \frac{z^2}{4!} + \dots$$

GTU : Summer-21, CVNM, Marks 3

Sol. : $\frac{e^z}{z^2}$ is given

We know that

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \dots$$

$$\therefore \frac{e^z}{z^2} = \frac{1}{z^2} \left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \dots \right)$$

$$= \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \frac{z}{3!} + \frac{z^2}{4!} + \dots$$

Laurent's Series

6.10 GTU : Summer-10,11,12,19,21,22,23,24,
Winter-10,11,12,21,22,23,24

If a function f fails to be analytic at a point z_0 , one cannot apply Taylor's theorem at that point. In that case however to find a series for $f(z)$ in the power at $(z - z_0)$. So we now present the theory of such representation and we begin with Laurent's theorem.

Theorem :

Suppose that a function f is analytic throughout an annular domain $R_1 < |z - z_0| < R_2$ centered at z_0 and let C denote any positively oriented simple closed contour around z_0 and lying in that domain.

Then

$$f(z) = \sum_{n=0}^{\infty} C_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n} \quad (R_1 < |z - z_0| < R_2)$$

↑

Principal part of Laurent's series.

$$\text{Where } C_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 0, 1, 2, \dots)$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{-n+1}} dz \quad (n = 1, 2, \dots)$$

As for example the Laurent's series at

$$e^{1/z} = 1 + \frac{1}{1!z} + \frac{1}{2!}\left(\frac{1}{z}\right)^2 + \frac{1}{3!}\left(\frac{1}{z}\right)^3 + \dots$$

$$0 < |z| < \infty$$

Ex. 6.10.1 : Find the Laurent's series of the function

$$f(z) = z^2 \sin\left(\frac{1}{z^2}\right) \text{ in the domain } 0 < |z| < \infty$$

$$\text{Sol. : } f(z) = z^2 \sin\left(\frac{1}{z^2}\right)$$

$$= z^2 \left(\frac{1}{z^2} - \frac{1}{3!} \left(\frac{1}{z^2}\right)^3 + \frac{1}{5!} \left(\frac{1}{z^2}\right)^5 + \dots \right)$$

$$\text{Using } \left(\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} + \dots \right)$$

$$= z^2 \left(\frac{1}{z^2} - \frac{1}{3!z^6} + \frac{1}{5!} \frac{1}{z^{10}} + \dots \right)$$

$$= \left(1 - \frac{1}{3!z^4} + \frac{1}{5!} \frac{1}{z^8} + \dots \right)$$

$$= 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{1}{z^{4n}}$$

Ex. 6.10.2 : Derive the Laurent's series representation

$$\frac{e^z}{(1+z)^2} = \frac{1}{e} \left[\sum_{n=0}^{\infty} \frac{(z+1)^n}{(n+2)!} + \frac{1}{z+1} + \frac{1}{(z+1)^2} \right]$$

$$0 < |z+1| < \infty$$

Sol. : Now we can take

$$(z+1) = u$$

$$z = u - 1$$

$$\text{So, } \frac{e^z}{(z+1)^2} = \frac{e^{u-1}}{u^2} = \frac{e^u e^{-1}}{u^2} = \frac{1}{e u^2} (e^u)$$

$$= \frac{1}{u^2} e \left(1 + u + \frac{u^2}{2!} + \frac{u^3}{3!} + \dots \right)$$

$$= \frac{1}{e} \left(\frac{1}{u^2} + \frac{1}{u} + \frac{1}{2!} + \frac{u}{3!} + \dots \right)$$

$$= \frac{1}{e} \left(\frac{1}{(z+1)^2} + \frac{1}{(z+1)} + \frac{1}{2!} + \frac{(z+1)}{3!} + \dots \right)$$

$$= \frac{1}{e} \left(\frac{1}{(z+1)^2} + \frac{1}{(z+1)} + \sum_{n=0}^{\infty} \frac{(z+1)^n}{(n+2)!} \right)$$

Ex. 6.10.3 : Give two Laurent's series expansion in the power of z for the function.

$f(z) = \frac{1}{z^2(1-z)}$ and specify the region in which those expansions are valid

GTU : Summer-21,23, Maths-4, Marks 7
Winter-24, CVPDE, Marks 7

Sol. :

$$f(z) = \frac{1}{z^2(1-z)} \text{ is not analytic at point } z = 0 \text{ and}$$

$$z = 1. (\because z^2(1-z) = 0 \Rightarrow z = 0 \text{ and } z = 1)$$

So valid region of Laurent's series is

$$(\because -\infty < |z| < 0 \text{ and } |z| < 1 \text{ and } |z| > 1)$$

$$9 < |z| < 1$$

$$1 < |z| < \infty$$

i) $f(z) = \frac{1}{z^2(1-z)}$ for $0 < |z| < 1$

$$\begin{aligned} f(z) &= \frac{1}{z^2(1-z)} = \frac{1}{z^2} (1-z)^{-1} \\ &= \frac{1}{z^2} (1+z+z^2+z^3+z^4+\dots) \\ &= \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \dots \\ &= \sum_{n=0}^{\infty} z^n + \frac{1}{z} + \frac{1}{z^2} \end{aligned}$$

ii) For $|z| > 1$

$$\Rightarrow \frac{1}{|z|} < 1$$

$$\begin{aligned} \text{So } f(z) &= \frac{1}{z^2(1-z)} = \frac{1}{z^2} \frac{1}{(-z)(1-1/z)} \\ &= -\frac{1}{z^3} \frac{1}{(1-1/z)} = -\frac{1}{z^3} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) \\ &= \left(-\frac{1}{z^3} - \frac{1}{z^4} - \frac{1}{z^5} - \frac{1}{z^6} + \dots \right) \\ &= -\sum_{n=3}^{\infty} \frac{1}{z^n} \end{aligned}$$

Ex. 6.10.4 : Show that when $0 < |z-1| < 2$

$$\frac{z}{(z-1)(z-3)} = -3 \sum_{n=0}^{\infty} \frac{(z-1)^n}{2^{n+2}} - \frac{1}{2(z-1)}$$

Sol. : Here given that $0 < |z-1| < 2$.

Now first we use partial fraction.

$$\frac{z}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$$

$$z = A(z-3) + B(z-1)$$

$$z = 1$$

$$1 = -2A \Rightarrow A = -\frac{1}{2}$$

$$z = 3 \quad 3 = 2B \Rightarrow B = \frac{3}{2}$$

$$\frac{z}{(z-1)(z-3)} = -\frac{1}{2(z-1)} + \frac{3}{2(z-3)}$$

$$= -\frac{1}{2(z-1)} + \frac{3}{2} \frac{1}{z-3}$$

Now let $0 < |z-1| < 2$

$$z-1 = u \Rightarrow z = 1+u$$

$$\text{So } 0 < |u| < 2$$

$$\text{So } \left| \frac{u}{2} \right| < 1$$

$$= -\frac{1}{2u} + \frac{3}{2} \frac{1}{1+u-3} = -\frac{1}{2u} + \frac{3}{2} \frac{1}{u-2}$$

$$= -\frac{1}{2u} + \frac{3}{2} \frac{1}{(-2)(1-u/2)}$$

$$= -\frac{1}{2u} - \frac{3}{4} \left(1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right)$$

$$= -\frac{1}{2u} - \frac{3}{4} - \frac{3}{8}u - \frac{3}{16}u^2 - \frac{3}{32}u^3$$

$$= -\frac{1}{2(z-1)} - \frac{3}{4} - \frac{3}{8}(z-1) - \frac{3}{16}(z-1)^2 - \frac{3}{32}(z-1)^3 + \dots$$

$$= -\frac{1}{2(z-1)} - 3 \sum_{n=0}^{\infty} \frac{(z-1)^n}{2^{n+2}}$$

Ex. 6.10.5 : Write the two Laurent's series in the power of z that represent the function.

$f(z) = \frac{1}{z(1-z)}$ in certain domain and specify those domains.

Sol. : Valid region

$$(0 < |z| < 1$$

$$1 < |z| < \infty)$$

i) $f(z) = \frac{1}{z(1-z)} \quad |z| < 1$

$$= \frac{1}{z} (1+z+z^2+z^3+\dots) = \left(\frac{1}{z} + 1 + z + z^2 + \dots \right)$$

$$= \frac{1}{z} + \sum_{n=0}^{\infty} z^n$$

ii) $1 < |z| \Rightarrow \frac{1}{|z|} < 1$

$$\text{So, } f(z) = \frac{1}{z(-z)(1-1/z)}$$

$$= -\frac{1}{z^2} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right)$$

$$= \left(-\frac{1}{z^2} - \frac{1}{z^3} - \frac{1}{z^4} - \frac{1}{z^5} - \dots \right)$$

$$= -\sum_{n=2}^{\infty} \frac{1}{z^n}$$

Mathematics - 2

Ex. 6.10.6 : Find the Laurent's series for the function $f(z) = \frac{1}{z(1-z)}$ in the region.

i) $|z+1| < 1$ ii) $1 < |z+1| < 2$

iii) $|z+1| > 2$

GTU : Winter-23, CVPDE, Marks 7

Sol. : $z+1 = u$

$$z = u - 1$$

$$f(z) = \frac{1}{z(1-z)} = \frac{1}{u-1(1-u+1)}$$

$$= \frac{1}{(u-1)(2-u)}$$

Using partial fraction

$$\frac{1}{(u-1)(2-u)} = \frac{A}{u-1} + \frac{B}{2-u}$$

$$1 = A(2-u) + B(u-1)$$

$$1 = 2B \Rightarrow B = \frac{1}{2}$$

$$1 = A(1) \Rightarrow A = 1$$

$$\frac{1}{(u-1)(2-u)} = \frac{1}{u-1} + \frac{1}{2} \cdot \frac{1}{2-u}$$

i) $|z+1| < 1$

$\Rightarrow |u| < 1$

$$= \frac{1}{-(1-u)} + \frac{1}{4} \cdot \frac{1}{\left(1-\frac{u}{2}\right)}$$

$$= -(1+u+u^2+u^3+\dots) + \frac{1}{4} \left(1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right)$$

$$= -1-u-u^2-u^3+\dots + \frac{1}{4} + \frac{u}{8} + \frac{u^2}{16} + \frac{u^3}{32} + \dots$$

$$= -\frac{3}{4} - \frac{7}{8}u - \frac{15}{16}u^2 - \frac{31}{32}u^3 + \dots$$

$$= -\frac{3}{4} - \frac{7}{8}(z+1) - \frac{15}{16}(z+1)^2 - \frac{31}{32}(z+1)^3 + \dots$$

ii) $1 < |z+1| < 2$

$\Rightarrow 1 < |u| < 2$

$$1 < |u| \text{ and } |u| < 2$$

$$\frac{1}{|u|} < 1 \text{ and } \left| \frac{u}{2} \right| < 1$$

$$= \frac{1}{u-1} + \frac{1}{2} \cdot \frac{1}{2-u} = \frac{1}{u(1-1/u)} + \frac{1}{4(1-u/2)}$$

$$= \frac{1}{u} \left(1 + \frac{1}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \dots \right)$$

$$+ \frac{1}{4} \left(1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right)$$

$$= \frac{1}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \dots + \frac{1}{4} + \frac{1}{8}u + \frac{u^2}{16} + \dots$$

$$= \dots + \frac{1}{u^3} + \frac{1}{u^2} + \frac{1}{u} + \frac{1}{4} + \frac{1}{8}u + \frac{u^2}{16} + \dots$$

$$= \dots + \frac{1}{(z+1)^3} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)} + \frac{1}{4}$$

$$+ \frac{1}{8}(z+1) + \frac{(z+1)^2}{16} + \dots$$

iii) $|z+1| > 2$

$$|u| > 2$$

$$2 < |u|$$

$$\frac{2}{|u|} < 1$$

So, $\frac{1}{|u|} < \frac{2}{|u|} < 1$

$$\therefore = \frac{1}{u-1} + \frac{1}{2} \cdot \frac{1}{2-u}$$

$$= \frac{1}{u\left(1-\frac{1}{u}\right)} + \frac{1}{2(-u)\left(1-\frac{2}{u}\right)}$$

$$= \frac{1}{4} \left(1 + \frac{1}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \dots \right)$$

$$- \frac{1}{2u} \left(1 + \frac{2}{u} + \frac{4}{u^2} + \frac{8}{u^3} + \dots \right)$$

$$= \frac{1}{u} + \frac{1}{u^2} + \frac{1}{u^3} + \frac{1}{u^4} + \dots$$

$$- \frac{1}{2u} - \frac{1}{u^2} - \frac{2}{u^3} - \frac{4}{u^4} + \dots$$

$$= \frac{1}{2u} - \frac{1}{u^3} - \frac{3}{u^4} + \dots$$

$$= \frac{1}{2(z+1)} - \frac{1}{(z+1)^3} - \frac{3}{(z+1)^4} + \dots$$

Ex. 6.10.7 : Find the Taylor series and Laurent's series for the following function.

a) $f(z) = \frac{z^2 - 1}{(z+2)(z+3)}$ in the region

i) $|z| < 2$ ii) $2 < |z| < 3$ iii) $|z| > 3$

b) $f(z) = \frac{-2z+3}{z^2-3z+2}$ with center 0

Summer-10

GTU : Summer-22, Maths-4, Marks 7

c) $f(z) = \frac{2z-3i}{z^2-3iz-2}$ valid for

i) $0 < |z| < 1$ ii) $|z| > 2$

d) $f(z) = \frac{1}{(z^2+1)(z^2+2)}$ in the power of z when

i) $|z| < 1$ ii) $1 < |z| < \sqrt{2}$ iii) $|z| > \sqrt{2}$

e) $f(z) = \frac{z}{(z-1)(z-3)}$

i) $|z| < 1$ ii) $1 < |z| < 3$

iii) $|z| > 3$ iv) $0 < z-1 < 2$

GTU : Winter-12

f) $f(z) = e^{3/z}$ about $z = 0$

g) $f(z) = \frac{1}{(z-1)^2(z-3)}$

i) $0 < |z| < 1$ ii) $1 < |z| < 3$ iii) $3 < |z| < \infty$

h) $f(z) = \frac{1-e^z}{z}$ about $z = 0$

Summer-10

GTU : Summer-23, CVPDE, Marks 3

i) $f(z) = -\frac{1}{(z-1)(z-2)}$ in the region

i) $|z| < 1$ ii) $1 < |z| < 2$ iii) $2 < |z| < \infty$

Winter-10

**GTU : Winter-22, CVNM, Marks 7,
Winter-23, Maths-4, Marks 7**

j) $f(z) = \frac{1}{(z+1)(z+3)}$ $1 < |z| < 3$

Winter-11

**GTU : Winter-21,24, CVNM, Marks 7,
Winter-22, Maths-4, Marks 3**

k) $f(z) = \frac{7z-2}{(z+1)z(z-2)}$ $1 < |z+1| < 3$

Summer-11

**GTU : Winter-24, Maths-4, Marks 7
Summer-24, CVPDE Marks 7**

l) $f(z) = \frac{1}{(z+2)(z+4)}$

GTU : Summer-21,22, Maths-4, Marks 7

i) $|z| < 2$ ii) $2 < |z| < 4$ iii) $|z| > 4$

Summer-12

GTU : Summer-22, CVPDE, Marks 4

m) $f(z) = \frac{z}{(z-1)(z-4)}$ in the valid region

$|z+3| < 4$

GTU : Summer-12

Sol. :

a) $f(z) = \frac{z^2-1}{(z+2)(z+3)}$

Using a partial fraction

$$\frac{z^2-1}{(z+2)(z+3)} = \frac{A}{z+2} + \frac{B}{z+3}$$

$$z^2-1 = A(z+3) + B(z+2)$$

$$z = -2$$

$$3 = A \Rightarrow A = 3$$

$$z = -3$$

$$8 = -B \Rightarrow B = -8$$

$$\frac{z^2-1}{(z+2)(z+3)} = \frac{3}{z+2} - \frac{8}{z+3}$$

Now i) $|z| < 2$ ii) $2 < |z| < 3$ iii) $|z| > 3$

i) $|z| < 2 \Rightarrow \left| \frac{z}{2} \right| < 1$

So $\frac{3}{z+2} - \frac{8}{z+3} = \frac{3}{2(1+z/2)} - \frac{8}{3(1+z/3)}$

$$= \frac{3}{2} \left(1 - \frac{z}{2} + \frac{z^2}{4} - \frac{z^3}{8} + \dots \right) - \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right)$$

$$= \frac{3}{2} - \frac{3}{4}z + \frac{3}{8}z^2 - \frac{3z^3}{16} + \dots - \frac{8}{3} + \frac{8z}{9} - \frac{8z^2}{27} - \frac{8z^3}{81} + \dots$$

$$= -\frac{7}{6} + \frac{5}{36}z - \frac{17}{216}z^2 + \dots$$

ii) $2 < |z| < 3$

$$2 < |z| \text{ and } |z| < 3$$

$$\frac{2}{|z|} < 1 \text{ and } \left| \frac{2}{3} \right| < 1$$

So, $\frac{3}{z+2} - \frac{8}{z+3}$

$$= \frac{3}{z(1+2/z)} - \frac{8}{3(1+z/3)}$$

$$= \frac{3}{z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \dots \right) - \frac{8}{3} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right)$$

$$= \frac{3}{z} - \frac{6}{z^2} + \frac{12}{z^3} - \frac{24}{z^4} + \dots - \frac{8}{3} + \frac{8z}{9} - \frac{8}{27} z^2 + \dots$$

$$= -\frac{12}{z^3} - \frac{6}{z^2} + \frac{3}{z} - \frac{8}{3} + \frac{8}{9} z - \frac{8z^2}{27} + \dots$$

iii)

$$|z| > 3$$

$$3 < |z| \Rightarrow \left| \frac{3}{z} \right| < 1$$

$$\Rightarrow \left| \frac{2}{z} \right| < \left| \frac{3}{z} \right| < 1$$

$$\frac{3}{z+2} - \frac{8}{z+3} = \frac{3}{z \left(1 + \frac{2}{z} \right)} - \frac{8}{z \left(1 + \frac{3}{z} \right)}$$

$$= \frac{3}{z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \dots \right) - \frac{8}{z} \left(1 - \frac{3}{z} + \frac{9}{z^2} - \frac{27}{z^3} + \dots \right)$$

$$= \frac{3}{z} - \frac{6}{z^2} + \frac{12}{z^3} - \frac{24}{z^4} + \dots - \frac{8}{z} + \frac{24}{z^2} - \frac{72}{z^3} + \dots = -\frac{5}{z} + \frac{18}{z^2} - \frac{60}{z^3} + \dots$$

b)

$$f(z) = \frac{-2z+3}{z^2-3z+2}$$

Center $z = 0$

$$f(z) = \frac{-2z+3}{(z-2)(z-1)}$$

Using partial fraction

$$\frac{-2z+3}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$-2z+3 = A(z-2) + B(z-1)$$

$$z = 1$$

$$1 = -A \Rightarrow A = -1$$

$$z = 2$$

$$-1 = B \Rightarrow B = -1$$

So

$$\frac{-2z+3}{(z-1)(z-2)} = -\frac{1}{(z-1)} - \frac{1}{(z-2)}$$

$$\begin{aligned}
 &= \frac{1}{(1-z)} + \frac{1}{2(1-z/2)} \\
 &= (1+z+z^2+z^3+\dots) + \frac{1}{2} \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots \right) \\
 &= \frac{3}{2} + \frac{5z}{4} + \frac{9}{8}z^2 + \frac{17}{16}z^3 + \dots
 \end{aligned}$$

c) $f(z) = \frac{2z-3i}{z^2-3iz-2}$ i) $0 < |z| < 1$ ii) $|z| > 2$

$$= \frac{2z-3i}{z^2-3iz-2i^2}$$

$$\Rightarrow \frac{2z-3i}{(z-2i)(z-i)} = \frac{A}{z-2i} + \frac{B}{z-i}$$

$$\Rightarrow 2z-3i = A(z-i) + B(z-2i)$$

$$z = i$$

$$\Rightarrow 2i-3i = B(i-2i)$$

$$\Rightarrow -i = B(-i) \Rightarrow B = 1$$

$$z = 2i$$

$$\Rightarrow 4i-3i = Ai \Rightarrow A = 1$$

$$\therefore \frac{2z-3i}{(z-2i)(z-i)} = \frac{1}{z-2i} + \frac{1}{z-i}$$

i) $|z| < 1$

$$\begin{aligned}
 \therefore \frac{1}{z-2i} + \frac{1}{z-i} &= \frac{1}{-2i(1-z/2i)} + \frac{1}{-i(1-z/i)} \\
 &= \frac{-1}{2i} \left(1 + \frac{z}{2i} + \frac{z^2}{4i^2} + \frac{z^3}{8i^3} + \dots \right) - \frac{1}{i} \left(1 + \frac{z}{i} + \frac{z^2}{i^2} + \frac{z^3}{i^3} + \dots \right) \\
 &= -\frac{1}{2i} + \frac{z}{4} + \frac{z^3}{8i} + \dots - \frac{1}{i} + \frac{z}{1} + \frac{z^2}{i} + \dots \\
 &= \left(-\frac{1}{2} - 1 \right) i + \left(\frac{1}{4} + 1 \right) z + \left(\frac{1}{8i} + \frac{1}{i} \right) z^3 + \dots \\
 &= -\frac{3}{2}i + \frac{5}{4}z + \frac{9}{8i}z^3 + \dots
 \end{aligned}$$

ii) $|z| > 2 \Rightarrow 2 < |z|$

$$\Rightarrow \left| \frac{2}{z} \right| < 1 \Rightarrow \left| \frac{1}{z} \right| < \left| \frac{2}{z} \right| < 1$$

$$\frac{1}{z-2i} + \frac{1}{z-i} = \frac{1}{z(1-2i/z)} + \frac{1}{z(1-i/z)}$$

$$= \frac{1}{2} \left(1 + \frac{2i}{z} + \frac{4i^2}{z^2} + \dots \right) + \frac{1}{2} \left(1 + \frac{i}{z} + \frac{i^2}{z^2} + \dots \right)$$

$$= \frac{1}{z} + \frac{2i}{z^2} + \frac{4i^2}{z^3} + \dots + \frac{1}{z} + \frac{i}{z^2} + \frac{i^3}{z^3} + \dots$$

d)

$$f(z) = \frac{1}{(z^2 + 1)(z^2 + 2)}$$

$$i) |z| < 1 \quad ii) 1 < |z| < \sqrt{2} \quad iii) |z| > \sqrt{2}$$

$$\Rightarrow \frac{1}{(z^2 + 1)(z^2 + 2)} = \frac{1}{(z+i)(z-i)(z+\sqrt{2}i)(z-\sqrt{2}i)} \cdot \frac{1}{(z+i)(z-i)(z+\sqrt{2}i)(z-\sqrt{2}i)}$$

$$= \frac{A}{z+i} + \frac{B}{z-i} + \frac{C}{z+\sqrt{2}i} + \frac{D}{z-\sqrt{2}i}$$

 $\therefore$

$$1 = A(z-i)(z+\sqrt{2}i)(z-\sqrt{2}i) + B(z+i)(z+\sqrt{2}i)(z-\sqrt{2}i) \\ + C(z+i)(z-i)(z-\sqrt{2}i) + D(z+i)(z-i)(z+\sqrt{2}i)$$

$$z = i$$

$$1 = B(2i)(i + \sqrt{2}i)(i - \sqrt{2}i)$$

$$1 = B(2i) i^2 (1 + \sqrt{2})(1 - \sqrt{2})$$

$$1 = B(2) i^3 (1 - 2)$$

$$B = \frac{1}{2i}$$

$$z = -i$$

$$1 = A(-2i)(-i + \sqrt{2}i)(-i - \sqrt{2}i)$$

$$1 = A(-2i) i^2 (-1 + \sqrt{2})(-1 - \sqrt{2})$$

$$1 = A(2i)((-1)^2 - (\sqrt{2})^2)$$

$$1 = A(2i)(-1)$$

$$A = -\frac{1}{2i}$$

$$z = -\sqrt{2}i$$

$$1 = C(-\sqrt{2}i + i)(-\sqrt{2}i - i)(-\sqrt{2}i - \sqrt{2}i)$$

$$1 = C i(-\sqrt{2} + 1) i(-\sqrt{2} - 1) - \sqrt{2}i(2)$$

$$1 = C 2\sqrt{2}i(1)$$

$$C = \frac{1}{2\sqrt{2}i}$$

$$z = \sqrt{2}i$$

$$1 = D(\sqrt{2}i + i)(\sqrt{2}i - i)(\sqrt{2}i + \sqrt{2}i)$$

$$1 = D i(\sqrt{2} + 1)i(\sqrt{2} - 1)\sqrt{2}i(2)$$

$$1 = D(-2\sqrt{2}i)$$

$$D = -\frac{1}{2\sqrt{2}i}$$

So

$$\frac{1}{(z^2 + 1)(z^2 + 2)} = -\frac{1}{2i} \frac{1}{z+i} + \frac{1}{2i} \frac{1}{z-i} + \frac{1}{2\sqrt{2}i} \frac{1}{z+\sqrt{2}i} - \frac{1}{2\sqrt{2}i} \frac{1}{z-\sqrt{2}i}$$

i) $|z| < 1$

$$\begin{aligned} &= \frac{-1}{2i} \frac{1}{i(1+z/i)} + \frac{-1}{2i(-i)1(1-z/i)} + \frac{1}{(2\sqrt{2}i)(\sqrt{2}i)} \frac{1}{(1+z/\sqrt{2}i)} - \frac{-1}{2\sqrt{2}i(-\sqrt{2}i)} \frac{1}{(1-z/\sqrt{2}i)} \\ &= +\frac{1}{2} \left(1 - \frac{z}{i} + \frac{z^2}{i^2} - \frac{z^3}{i^3} + \dots \right) + \frac{1}{2} \left(1 + \frac{z}{i} + \frac{z^2}{i^2} + \frac{z^3}{i^3} + \dots \right) \\ &\quad - \frac{1}{4} \left(1 - \frac{z}{\sqrt{2}i} + \frac{z^2}{2i^2} - \frac{z^3}{2\sqrt{2}i^3} + \dots \right) - \frac{1}{4} \left(1 + \frac{z}{\sqrt{2}i} + \frac{z^2}{2i^2} - \frac{z^3}{2\sqrt{2}i^3} + \dots \right) \end{aligned}$$

ii) $1 < |z| < \sqrt{2}$

$$1 < |z| \text{ and } |z| < \sqrt{2}$$

$$\frac{1}{|z|} < 1 \text{ and } \left| \frac{z}{\sqrt{2}} \right| < 1$$

$$\begin{aligned} \therefore -\frac{1}{2i(z+i)} + \frac{1}{2i(z-i)} + \frac{1}{2\sqrt{2}i} \frac{1}{z+\sqrt{2}i} - \frac{1}{2\sqrt{2}i} \frac{1}{z-\sqrt{2}i} &= -\frac{1}{2iz(1+i/z)} + \frac{1}{2iz(1-i/z)} + \frac{1}{2\sqrt{2}i(\sqrt{2}i)} + \frac{1}{\left(1 + \frac{z}{\sqrt{2}i}\right)} \\ &\quad - \frac{1}{2\sqrt{2}i(-\sqrt{2}i)} \frac{1}{(1-z/\sqrt{2}i)} \\ &= -\frac{1}{2iz} \left(1 - \frac{i}{z} + \frac{i^2}{z^2} - \frac{i^3}{z^3} + \dots \right) + \frac{1}{2iz} \left(1 + \frac{i}{z} + \frac{i^2}{z^2} + \frac{i^3}{z^3} + \dots \right) \\ &\quad + \frac{1}{4i^2} \left(1 - \frac{z}{\sqrt{2}i} + \frac{z^2}{2i^2} + \dots \right) + \frac{1}{4i^2} \left(1 + \frac{z}{\sqrt{2}i} + \frac{z^2}{2i^2} + \dots \right) \end{aligned}$$

iii) $|z| > \sqrt{2} \Rightarrow \sqrt{2} < |z|$

$$\frac{1}{|z|} < \frac{\sqrt{2}}{|z|} < 1$$

$$\therefore -\frac{1}{2iz\left(1 + \frac{i}{z}\right)} + \frac{1}{2iz\left(1 - \frac{i}{z}\right)} + \frac{1}{2\sqrt{2}iz\left(1 + \frac{\sqrt{2}i}{z}\right)} - \frac{1}{2\sqrt{2}iz\left(1 - \frac{\sqrt{2}i}{z}\right)}$$

$$= -\frac{1}{2iz} \left(1 - \frac{i}{z} + \frac{i^2}{z^2} - \frac{i^3}{z^3} + \dots \right) + \frac{1}{2iz} \left(1 + \frac{i}{z} + \frac{i^2}{z^2} + \frac{i^3}{z^3} + \dots \right)$$

$$+ \frac{1}{2\sqrt{2}iz} \left(1 - \frac{\sqrt{2}i}{z} + \frac{(\sqrt{2}i)^2}{z^2} + \dots \right) - \frac{1}{2\sqrt{2}iz} \left(1 + \frac{\sqrt{2}i}{z} + \frac{(\sqrt{2}i)^2}{z^2} + \dots \right)$$

$$f(z) = \frac{z}{(z-1)(z-3)}$$

$$\frac{z}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$$

$$z = A(z-3) + B(z-1)$$

$$z = 1$$

$$1 = A(-2) \Rightarrow A = -\frac{1}{2}$$

$$z = 3$$

$$3 = 2B \Rightarrow B = \frac{3}{2}$$

$$= -\frac{1}{2(z-1)} + \frac{3}{2(z-3)}$$

$$i) |z| < 1$$

$$= +\frac{1}{2(1-z)} + \frac{3}{2(-3)(1-z/3)} = \frac{1}{2}(1+z+z^2+z^3+\dots) - \frac{1}{2}\left(1+\frac{z}{3}+\frac{z^2}{9}+\frac{z^3}{27}+\dots\right)$$

$$= \frac{1}{2} + \frac{z}{2} + \frac{z^2}{2} + \frac{z^3}{2} + \dots - \frac{1}{2} + \frac{z}{6} - \frac{z^2}{18} + \frac{z^3}{54} - \dots$$

$$= \frac{1}{3}z + \frac{4z^2}{9} + \frac{13}{27}z^3 + \dots$$

$$ii) 1 < |z| < 3$$

$$1 < |z| \text{ and } |z| < 3$$

$$\left| \frac{1}{z} \right| < 1 \text{ and } \left| \frac{z}{3} \right| < 1$$

$$= -\frac{1}{2(z-1)} + \frac{3}{2(z-3)} = -\frac{1}{2z(1-1/z)} - \frac{1}{2(1-z/3)}$$

$$= -\frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) - \frac{1}{2} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \dots \right)$$

$$= -\frac{1}{2z} - \frac{1}{2z^2} - \frac{1}{2z^3} - \frac{1}{2z^4} + \dots - \frac{1}{2} - \frac{z}{6} - \frac{z^2}{18} - \frac{z^3}{54} + \dots$$

$$= -\frac{1}{2z} - \frac{1}{2z^2} - \frac{1}{2z^3} - \frac{1}{2z^4} - \frac{1}{2} - \frac{z}{6} - \frac{z^2}{18} - \frac{z^3}{54} + \dots$$

$$\text{iii) } |z| > 3 \Rightarrow 3 < |z|$$

$$\left| \frac{1}{z} \right| < \left| \frac{3}{z} \right| < 1$$

$$= -\frac{1}{2(z-1)} + \frac{3}{2(z-3)}$$

$$= -\frac{1}{2z(1-1/z)} + \frac{3}{2z(1-3/z)}$$

$$= -\frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) + \frac{3}{2z} \left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \dots \right)$$

$$= -\frac{1}{2z} - \frac{1}{2z^2} - \frac{1}{2z^3} - \frac{1}{2z^4} + \dots + \frac{3}{2z} + \frac{9}{2z^2} + \frac{27}{2z^3} + \dots$$

$$= \frac{1}{z} + \frac{4}{z^2} + \frac{13}{z^3} + \dots$$

$$\text{iv) } 0 < |z-1| < 2$$

$$\text{Now } z-1 = u$$

$$z = 1+u$$

$$\therefore 0 < |u| < 2$$

$$\therefore -\frac{1}{2(z-1)} + \frac{3}{2(z-3)} = -\frac{1}{2(1+u-1)} + \frac{3}{2(1+u-3)}$$

$$= -\frac{1}{2u} + \frac{3}{2(u-2)}$$

$$\therefore |u| < 2$$

$$\Rightarrow \left| \frac{u}{2} \right| < 1$$

$$= -\frac{1}{2u} + \frac{3}{2(-2)} - \frac{1}{(1-u/2)}$$

$$= -\frac{1}{2u} - \frac{3}{4} \left(1 + \frac{u}{2} + \frac{u^2}{4} + \frac{u^3}{8} + \dots \right)$$

$$= -\frac{1}{2u} - \frac{3}{4} - \frac{3}{8}u - \frac{3}{16}u^2 - \frac{3}{32}u^3 + \dots$$

$$= -\frac{1}{2(z-1)} - \frac{3}{4} - \frac{3}{8}(z-1) - \frac{3}{16}(z-1)^2 - \frac{3}{32}(z-1)^3 + \dots$$

$$\text{f) } f(z) = e^{3/z} \text{ about } z = 0$$

$$\therefore e^{3/z} = 1 + \left(\frac{3}{z} \right) + \frac{1}{2!} \left(\frac{3}{z} \right)^2 + \frac{1}{3!} \left(\frac{3}{z} \right)^3 + \dots$$

$$= 1 + \frac{3}{z} + \frac{9}{4z^2} + \frac{1}{3!} \frac{27}{z^3} + \dots = 1 + \frac{3}{z} + \frac{9}{4z^2} + \frac{9}{2z^3} + \dots$$

$$f(z) = \frac{1}{(z-1)^2(z-3)}$$

$$\frac{1}{(z-1)^2(z-3)} = \frac{A}{z-3} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

$$1 = A(z-1)^2 + B(z-1)(z-3) + C(z-3)$$

$$z=1, 1 = C(-2) \Rightarrow C = -\frac{1}{2}$$

$$z=3, 1 = A(4) \Rightarrow A = \frac{1}{4}$$

$$z=0,$$

$$1 = A + 3B - 3C$$

$$1 = \frac{1}{4} + 3B - 3\left(-\frac{1}{2}\right)$$

$$1 - \frac{1}{4} - \frac{3}{2} = 3B$$

$$-\frac{3}{4} = 3B \Rightarrow B = -\frac{1}{4}$$

$$\frac{1}{(z-1)^2(z-3)} = \frac{1}{4(z-3)} - \frac{1}{4(z-1)} - \frac{1}{2(z-1)^2}$$

$$i) |z| < 1$$

$$\therefore \frac{1}{4(z-3)} - \frac{1}{4(z-1)} - \frac{1}{2(z-1)^2} = \frac{1}{-12(1-z/3)} + \frac{1}{4(1-z)} - \frac{1}{2(1-z)^2}$$

$$= -\frac{1}{12} \left(1 + \frac{z}{3} + \frac{z^2}{9} + \frac{z^3}{27} + \dots \right) + \frac{1}{4} (1 + z + z^2 + z^3 + \dots)$$

$$- \frac{1}{2} (1 + 2z + 3z^2 + 4z^3 + \dots)$$

$$= -\frac{1}{3} - \frac{7}{9}z - \frac{34z^2}{27} + \dots$$

$$ii) 1 < |z| < 3$$

$$1 < |z| \text{ and } |z| < 3$$

$$\left| \frac{1}{z} \right| < 1 \text{ and } \left| \frac{z}{3} \right| < 1$$

$$\therefore \frac{1}{4(z-3)} - \frac{1}{4(z-1)} - \frac{1}{2(z-1)^2} = \frac{1}{4z(1-3/z)} + \frac{1}{4(1-z)} - \frac{1}{2(1-z)^2}$$

$$= \frac{1}{4z} \left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \dots \right) + \frac{1}{4} (1 + z + z^2 + z^3 + \dots) - \frac{1}{2} (1 + 2z + 3z^2 + 4z^3 + \dots)$$

$$\begin{aligned}
 &= \frac{1}{4z} + \frac{3}{4z^2} + \frac{9}{4z^3} + \frac{27}{4z^4} + \dots + \frac{1}{4} + \frac{z}{4} + \frac{z^2}{4} + \frac{z^3}{4} + \dots - \frac{1}{2} - z - \frac{3}{2}z^2 - 2z^3 + \dots \\
 &= \dots - \frac{9}{4z^3} + \frac{3}{4z^2} + \frac{1}{4z} - \frac{1}{4} - \frac{3z}{4} - \frac{5}{4}z^2 + \dots
 \end{aligned}$$

iii)

$$3 < |z| \Rightarrow \left| \frac{3}{z} \right| < 1$$

 $\therefore$

$$\Rightarrow \left| \frac{1}{z} \right| < \left| \frac{3}{z} \right| < 1$$

$$\begin{aligned}
 &\frac{1}{4(z-3)} - \frac{1}{4(z-1)} - \frac{1}{2(z-1)^2} \\
 &= \frac{1}{4z(1-3/z)} - \frac{1}{4z(1-1/z)} - \frac{1}{2z^2(1-1/z)^2} \\
 &= \frac{1}{4z} \left(1 - \frac{3}{z} \right)^{-1} - \frac{1}{4z} \left(1 - \frac{1}{z} \right)^{-1} - \frac{1}{2z^2} \left(1 - \frac{1}{z} \right)^{-2} \\
 &= \frac{1}{4z} \left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \dots \right) - \frac{1}{4z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) - \frac{1}{2z^2} \left(1 + \frac{2}{z} + \frac{3}{z^2} + \frac{4}{z^3} + \dots \right) \\
 &= \frac{1}{4z} + \frac{3}{4z^2} + \frac{9}{4z^3} + \frac{27}{4z^4} + \dots - \frac{1}{4z} - \frac{1}{4z^2} - \frac{1}{4z^3} - \frac{1}{4z^4} + \dots - \frac{1}{2z^2} - \frac{1}{z^3} - \frac{3}{2z^4} - \frac{2}{z^5} + \dots \\
 &= \frac{1}{z^3} + \frac{5}{z^4} + \dots
 \end{aligned}$$

h) $f(z) = \frac{1-e^z}{z}$ about $z=0$

$$\begin{aligned}
 &= \frac{1}{z} \left(1 - \left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \right) \right) = \frac{1}{z} \left(1 - 1 - z - \frac{z^2}{2!} - \frac{z^3}{3!} + \dots \right) \\
 &= \frac{1}{z} \left(-z - \frac{z^2}{2!} - \frac{z^3}{3!} - \dots \right)
 \end{aligned}$$

$$f(z) = \left(-1 - \frac{z}{2!} - \frac{z^2}{3!} - \dots \right)$$

i) $f(z) = -\frac{1}{(z-1)(z-2)}$ in the region

i) $|z| < 1$ ii) $1 < |z| < 2$ iii) $2 < |z| < \infty$

$$f(z) = -\frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$\Rightarrow -1 = A(z-2) + B(z-1)$$

$$\Rightarrow \text{Now } z = 1$$

$$-1 = A(-1) \Rightarrow A = 1$$

$$z = 2$$

$$-1 = B(1) \Rightarrow B = -1$$

$$\frac{1}{(z-1)(z-2)} = \frac{1}{z-1} - \frac{1}{z-2}$$

$$i) |z| < 1$$

$$= \frac{1}{z-1} - \frac{1}{z-2} = -\frac{1}{1-z} + \frac{1}{2(1-z/2)}$$

$$= -(1-z)^{-1} + \frac{1}{2}(1-z/2)^{-1}$$

$$= -(1+z+z^2+\dots) + \frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\dots\right)$$

$$ii) 1 < |z| < 2$$

$$1 < |z| \text{ and } |z| < 2$$

$$\left|\frac{1}{z}\right| < 1 \text{ and } \left|\frac{z}{2}\right| < 1$$

$$\frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z(1-1/z)} + \frac{1}{2(1-z/2)}$$

$$= \frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\dots\right) + \frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\dots\right)$$

$$iii) 2 < |z| < \infty$$

$$\Rightarrow 2 < |z| \Rightarrow \left|\frac{2}{z}\right| < 1 \Rightarrow \left|\frac{1}{z}\right| < \left|\frac{2}{z}\right| < 1$$

$$\frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z(1-1/z)} - \frac{1}{z(1-2/z)}$$

$$= \frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\dots\right) - \frac{1}{z}\left(1+\frac{2}{z}+\frac{4}{z^2}+\dots\right)$$

$$j) f(z) = \frac{1}{(z+1)(z+3)} \quad 1 < |z| < 3$$

$$\frac{1}{(z+1)(z+3)} = \frac{A}{z+1} + \frac{B}{z+3}$$

$$\Rightarrow 1 = A(z+3) + B(z+1)$$

$$z = -1$$

$$1 = A(+2) \Rightarrow A = +\frac{1}{2}$$

$$z = -3$$

$$1 = B(-2) \Rightarrow B = -\frac{1}{2}$$

$$\frac{1}{(z+1)(z+3)} = \frac{1}{2(z+1)} - \frac{1}{2(z+3)}$$

$$i) 1 < |z| < 3$$

$$1 < |z| \text{ and } |z| < 3$$

$$\left| \frac{1}{z} \right| < 1 \text{ and } \left| \frac{z}{3} \right| < 1$$

$$\begin{aligned} \frac{1}{2(z+1)} - \frac{1}{2(z+3)} &= \frac{1}{2z(1+1/z)} - \frac{1}{6(1+z/3)} \\ &= \frac{1}{2z} \left(1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots \right) - \frac{1}{6} \left(1 - \frac{z}{3} + \frac{z^2}{9} - \frac{z^3}{27} + \dots \right) \end{aligned}$$

$$k) f(z) = \frac{7z-2}{(z+1)z(z-2)} \quad 1 < |z+1| < 3$$

$$\Rightarrow f(z) = \frac{7z-2}{z(z+1)(z-2)}$$

$$\frac{7z-2}{z(z+1)(z-2)} = \frac{A}{z} + \frac{B}{(z+1)} + \frac{C}{z-2}$$

$$\therefore 7z-2 = A(z+1)(z-2) + Bz(z-2) + Cz(z+1)$$

$$i) z = 0 \Rightarrow -2 = -2A \Rightarrow A = 1$$

$$ii) z = -1 \Rightarrow -9 = 3B \Rightarrow B = -3$$

$$iii) z = 2 \Rightarrow 12 = 6C \Rightarrow C = 2$$

$$\frac{7z-2}{z(z+1)(z-2)} = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

$$\text{Now } 1 < |z+1| < 3$$

$$u = z+1 \Rightarrow z = u-1$$

$$= \frac{1}{u-1} - \frac{3}{u} + \frac{2}{u-1-2} = \frac{1}{u-1} - \frac{3}{u} + \frac{2}{u-3}$$

$$\text{Here } 1 < |u| < 3$$

$$\therefore 1 < |u| \text{ and } |u| < 3$$

$$\left| \frac{1}{u} \right| < 1 \text{ and } \left| \frac{u}{3} \right| < 1$$

$$= \frac{1}{u(1-1/u)} - \frac{3}{u} + \frac{2}{-3(1-u/3)}$$

$$= \frac{1}{u} \left(1 + \frac{1}{u} + \frac{1}{u^2} + \dots \right) - \frac{3}{u} - \frac{2}{3} \left(1 + \frac{u}{3} + \frac{u^2}{9} + \dots \right)$$

$$= \frac{1}{(z+1)} \left(1 + \frac{1}{(z+1)} + \frac{1}{(z+1)^2} + \dots \right) - \frac{3}{(z+1)} - \frac{2}{3} \left(1 + \frac{(z+1)}{3} + \frac{(z+1)^2}{9} + \dots \right)$$

1) $f(z) = \frac{1}{(z+2)(z+4)}$

i) $|z| < 2$ ii) $2 < |z| < 4$ iii) $|z| > 4$

$$\frac{1}{(z+2)(z+4)} = \frac{A}{z+2} + \frac{B}{z+4}$$

$$1 = A(z+4) + B(z+2)$$

$$z = -2 \Rightarrow 1 = 2A \Rightarrow A = \frac{1}{2}$$

$$z = -4 \Rightarrow 1 = -2B \Rightarrow B = -\frac{1}{2}$$

$$\frac{1}{(z+2)(z+4)} = \frac{1}{2(z+2)} - \frac{1}{2(z+4)}$$

i) $|z| < 2 \Rightarrow \text{and } \left| \frac{z}{2} \right| < 1$

$$\frac{1}{2(z+2)} - \frac{1}{2(z+4)} = \frac{1}{4(1+z/2)} - \frac{1}{8(1+z/4)}$$

$$= \frac{1}{4} \left(1 - \frac{z}{2} + \frac{z^2}{4} + \dots \right) - \frac{1}{8} \left(1 - \frac{z}{4} + \frac{z^2}{16} + \dots \right)$$

ii) $2 < |z| < 4$

$$2 < |z| \text{ and } |z| < 4$$

$$\left| \frac{2}{z} \right| < 1 \text{ and } \left| \frac{z}{4} \right| < 1$$

$$\frac{1}{2(z+2)} - \frac{1}{2(z+4)} = \frac{1}{2z(1+2/z)} - \frac{1}{8(1+z/4)}$$

$$= \frac{1}{2z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \dots \right) - \frac{1}{8} \left(1 - \frac{z}{4} + \frac{z^2}{16} - \dots \right)$$

iii) $|z| > 4 \Rightarrow 4 < |z|$

$$\Rightarrow \left| \frac{4}{z} \right| < 1$$

$$\Rightarrow \left| \frac{2}{z} \right| < \left| \frac{4}{z} \right| < 1$$

$$\begin{aligned}
 \therefore \quad & \frac{1}{2(z+2)} - \frac{1}{2(z+4)} \\
 &= \frac{1}{2z(1+2/z)} - \frac{1}{2z(1+4/z)} \\
 &= \frac{1}{2z} \left(1 - \frac{2}{z} + \frac{4}{z^2} - \frac{8}{z^3} + \dots \right) - \frac{1}{2z} \left(1 - \frac{4}{z} + \frac{16}{z^2} + \dots \right)
 \end{aligned}$$

$$m) \quad f(z) = \frac{z}{(z-1)(z-4)} \quad (|z+3| < 4)$$

$$\frac{z}{(z-1)(z-4)} = \frac{A}{z-1} + \frac{B}{z-4}$$

$$z = A(z-4) + B(z-1)$$

$$z=1 \Rightarrow 1 = A(-3) \Rightarrow A = -\frac{1}{3}$$

$$z=4 \Rightarrow 4 = 3B \Rightarrow B = \frac{4}{3}$$

$$\frac{z}{(z-1)(z-4)} = \frac{-1}{3(z-1)} + \frac{4}{3(z-4)}$$

$\Rightarrow$ Now take

$$z+3 = u$$

$$z = u-3$$

$$= -\frac{1}{3(u-3-1)} + \frac{4}{3(u-3-4)} = -\frac{1}{3(u-4)} + \frac{4}{3(u-7)}$$

$$|z+3| < 4$$

$$|u| < 4$$

$$\left| \frac{u}{4} \right| < 1$$

$$\Rightarrow -\frac{1}{3(u-4)} + \frac{4}{3(u-7)}$$

$$= +\frac{1}{12\left(1-\frac{u}{4}\right)} + \frac{4}{3(-7)\left(1-\frac{u}{7}\right)} = \frac{1}{12\left(1-\frac{u}{4}\right)} - \frac{4}{21\left(1-\frac{u}{7}\right)}$$

$$= \frac{1}{12}\left(1-\frac{u}{4}\right)^{-1} - \frac{4}{21}\left(1-\frac{u}{7}\right)^{-1} = \frac{1}{12}\left(1+\frac{u}{4}+\frac{u^2}{16}+\dots\right) - \frac{4}{21}\left(1+\frac{u}{7}+\frac{u^2}{49}+\dots\right)$$

$$= \frac{1}{12}\left(1+\frac{(z+3)}{4}+\frac{(z+3)^2}{16}+\dots\right) - \frac{4}{21}\left(1+\frac{(z+3)}{7}+\frac{(z+3)^2}{49}+\dots\right)$$

Ex. 6.10.8 : Expand $\frac{1}{z(z^2 - 3z + 2)}$ about $z = 0$ for the region i) $1 < |z| < 2$ ii) $|z| > 2$.

GTU : Winter-21, Summer-22, Maths-4, Marks 7

Sol. : The singular point of the function

$$f(z) = \frac{1}{z(z^2 - 3z + 2)} = \frac{1}{z(z-1)(z-2)} \text{ is}$$

$$z = 0, z = 1 \text{ and } z = 2$$

Using partial fraction, we write

$$\frac{1}{z(z-1)(z-2)} = \frac{1}{2z} - \frac{1}{z-1} + \frac{1}{2(z-2)}$$

Now our region

$$\text{i) } 1 < |z| < 2, \text{ we write } \left(\frac{1}{|z|} < 1 \text{ and } \frac{|z|}{2} < 1 \right)$$

$$\begin{aligned} \therefore f(z) &= \frac{1}{2z} - \frac{1}{2(1-1/z)} - \frac{1}{4(1-z/2)} \\ &= \frac{1}{2z} - \frac{1}{z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \dots \right) - \frac{1}{4} \left(1 + z/2 + z^2/4 + z^3/8 + \dots \right) \\ &= \left(-\frac{1}{2z} - \frac{1}{z^2} - \frac{1}{z^3} + \dots \right) - \frac{1}{4} \left(1 + z/2 + z^2/4 + z^3/8 + \dots \right) \end{aligned}$$

$$\text{ii) } |z| > 2 \Rightarrow 2 < |z| \Rightarrow \frac{2}{|z|} < 1$$

$$\begin{aligned} \therefore f(z) &= \frac{1}{2z} - \frac{1}{z(1-1/z)} + \frac{1}{2z(1-2/z)} \\ &= \frac{1}{2z} - \frac{1}{z} \left(1 + 1/z + 1/z^2 + 1/z^3 + \dots \right) + \frac{1}{2z} \left(1 + 2/z + 4/z^2 + 8/z^3 + \dots \right) \\ &= (2-1) \cdot \frac{1}{z^3} + (2^2-1) \frac{1}{z^4} + (2^3-1) \frac{1}{z^5} + \dots \end{aligned}$$

Ex. 6.10.9 : Expand $f(z) = \frac{1}{(z+1)(z-2)}$ in Laurent's series in the region

$$\text{i) } |z| < 1 \quad \text{ii) } 1 < |z| < 2 \quad \text{iii) } |z| > 2$$

GTU : Summer-19, Maths-4, Marks 7,
Winter-22, CVPDE, Marks 7

Sol. : Here, $f(z) = \frac{1}{(z+1)(z-2)}$ is given

Now, First we apply partial fraction

$$\frac{1}{(z+1)(z-2)} = \frac{A}{z+1} + \frac{B}{z-2}$$

$$1 = A(z-2) + B(z+1)$$

$$z = 2 \Rightarrow 1 = 3 \cdot B \Rightarrow B = 1/3$$

$$z = -1 \Rightarrow 1 = A(-3) \Rightarrow A = -1/3$$

$$\therefore \frac{1}{(z+1)(z-2)} = -\frac{1}{3} \cdot \frac{1}{z+1} + \frac{1}{3} \cdot \frac{1}{z-2}$$

$$\text{i) } |z| < 1$$

$$\begin{aligned} \therefore \frac{1}{(z+1)(z-2)} &= -\frac{1}{3} \cdot \frac{1}{1+z} - \frac{1}{6} \cdot \frac{1}{1-z/2} \\ &= -\frac{1}{3}(1-z+z^2-z^3+\dots) - \frac{1}{6}(1+z/2+z^2/4+z^3/8+\dots) \\ &= \left(-\frac{1}{3} + \frac{z}{3} - \frac{z^2}{3} + \frac{z^3}{3} - \dots\right) + \left(-\frac{1}{6} - \frac{z}{12} - \frac{z^2}{24} - \frac{z^3}{48} - \dots\right) \\ &= -\frac{1}{12} + \frac{z}{4} - \frac{27}{72}z^2 + \dots \end{aligned}$$

$$\text{ii) } 1 < |z| < 2 \Rightarrow 1 < |z| \text{ and } |z| < 2$$

$$\Rightarrow \frac{1}{|z|} < 1 \text{ and } \frac{|z|}{2} < 1$$

$$\begin{aligned} \therefore \frac{1}{(z+1)(z-2)} &= -\frac{1}{3} \cdot \frac{1}{z+1} + \frac{1}{3} \cdot \frac{1}{z-2} \\ &= -\frac{1}{3} \frac{1}{z(1+1/z)} - \frac{1}{6} \frac{1}{(1-z/2)} \\ &= -\frac{1}{3z}(1-1/z+1/z^2-1/z^3+\dots) - \frac{1}{6}(1+z/2+z^2/4+z^3/8+\dots) \\ &= \left(-\frac{1}{3z} + \frac{1}{3z^2} - \frac{1}{3z^3} + \frac{1}{3z^4} - \dots\right) + \left(-\frac{1}{6} - \frac{z}{12} - \frac{z^2}{24} - \frac{z^3}{48} - \dots\right) \\ &= -\frac{1}{3z^4} - \frac{1}{3z^3} + \frac{1}{3z^2} - \frac{1}{3z} - \frac{1}{6} - \frac{z}{12} - \frac{z^2}{24} - \frac{z^3}{48} - \dots \end{aligned}$$

$$\text{iii) } |z| > 2 \Rightarrow z < |z| \Rightarrow \frac{2}{|z|} < 1$$

$$\begin{aligned} \therefore \frac{1}{(z+1)(z-2)} &= -\frac{1}{3} \cdot \frac{1}{1+z} + \frac{1}{3} \cdot \frac{1}{z-2} \\ &= -\frac{1}{3} \cdot \frac{1}{z(1+1/z)} + \frac{1}{3z(1-2/z)} \\ &= -\frac{1}{3z}(1-1/z+1/z^2-1/z^3+\dots) + \frac{1}{3z}(1+2/z+4/z^2+8/z^3+\dots) \\ &= \left(-\frac{1}{3z} + \frac{1}{3z^2} - \frac{1}{3z^3} + \frac{1}{3z^4} - \dots\right) + \left(\frac{1}{3z} + \frac{2}{3z^2} + \frac{4}{3z^3} + \frac{8}{3z^4} + \dots\right) \\ &= \frac{1}{z^2} + \frac{1}{z^3} + \frac{3}{z^4} + \dots \end{aligned}$$

Ex. 6.10.10 : Find the Laurent's series that represent $f(z) = \frac{1}{(z-2)(z-3)}$ in the region $2 < |z| < 3$.

GTU : Summer-22, CVPDE, Marks 4, Summer-21, CVNM, Marks 4

Sol. : Here, $f(z) = \frac{1}{(z-2)(z-3)}$ is given Using partial

fraction,

$$\frac{1}{(z-2)(z-3)} = \frac{A}{z-2} + \frac{B}{z-3}$$

$$1 = A(z-3) + B(z-2)$$

$$z=2 \Rightarrow 1 = -A \Rightarrow A = -1$$

$$z=3 \Rightarrow 1 = B \Rightarrow B = 1$$

$$\therefore \frac{1}{(z-2)(z-3)} = \frac{-1}{z-2} + \frac{1}{z-3}$$

$$i) 2 < |z| < 3$$

$$\Rightarrow 2 < |z| \text{ and } |z| < 3$$

$$\Rightarrow \frac{2}{|z|} < 1 \text{ and } \frac{|z|}{3} < 1$$

$$\begin{aligned} \frac{1}{(z-2)(z-3)} &= -\frac{1}{z(1-2/z)} + \frac{1}{(-3)(1-z/3)} \\ &= -\frac{1}{z} \left(1 - 2/z + 4/z^2 + 8/z^3 + \dots \right) \\ &\quad - \frac{1}{3} \left(1 + z/3 + z^2/9 + z^3/27 + \dots \right) \end{aligned}$$

6.11 Different Types of Singularities

GTU : Summer-21,23, Marks 3

The functions which are analytic at all points of a bounded region except at a finite number of points (such points are called singular points).

i) Isolated singularity :

A point z_0 is called an isolated singularity of a function $f(z)$ if $f(z)$ has a singularity at z_0 but single valued and analytic in the annular $0 < |z - z_0| < R$ (i.e. analytic in some neighbourhood of z_0).

Otherwise it is non-isolated singular points.

As for example

$$f(z) = \frac{z^2 - 1}{z(z^2 + 1)(z - 2)}$$

Here $z = 0$, $z = \pm i$ and $z = 2$ is a isolated singularities.

$$f(z) = \frac{\tan z}{z^2} \text{ has a singularity at } z = 0$$

But $f(z) = \frac{\tan z}{z}$ has no singular points.

$$\text{Because } f(z) = \frac{\tan z}{z} = \frac{1}{2} \left(z + \frac{z^3}{3} + \dots \right) = \left(1 + \frac{z^2}{3} + \dots \right)$$

Note : 1 $f(z) = \log z$ has a singularity at $z = 0$ but this is non isolated singularity because at $z = 0$ its neighbourhood at zero points on the negative real axis ; where $\log z$ is not analytic.

Note : 2 : $f(z) = |z|^2 = x^2 + y^2$ is nowhere analytic. Thus any point in the z -plane is a non-isolated singular points.

Types of singularities :

1. Poles :

If principal part of Laurent's series has finite number of terms.

$$\text{i.e. } f(z) = \sum_{n=0}^{\infty} C_n(z-z_0)^n + \frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \dots + \frac{b_n}{(z-z_0)^n}$$

Then the singularity $z = z_0$ is said to be pole of order n , if $b_1 \neq 0$ and $b_2 = b_3 = \dots = b_n = 0$, then

$$f(z) = \sum_{n=0}^{\infty} C_n(z-z_0)^n + \frac{b_1}{z-z_0}$$

Then singularity $z = z_0$ is said to be pole of order 1 or a simple pole.

2. Removable singularity :

If in the Laurent's series expansion the principal part is zero.

$$\text{i.e. } f(z) = \sum_{n=0}^{\infty} C_n(z-z_0)^n + 0$$

then the singularity $z = z_0$ is said to be removable singularity.

3. Essential singularity :

If in the Laurent's series expansion, the principal part contains an infinite number of terms, then the singularity $z = z_0$ is said to be an essential singularity.

4. Singularities of Infinity :

Let $z = \frac{1}{\omega}$ in $f(z)$ then $f(z) = f(1/\omega)$

The nature of singularities at $z = \infty$ is defined to be the same as that of $f(\omega)$ at $\omega=0$.

Ex. 6.11.1 : Determine the location and order of zeroes of

i) $(z^2 + 1)(e^z - 1)$ ii) $(z^4 - z^2 - 6)^3$

Sol. :

i) Here $f(z) = (z^2 + 1)(e^z - 1)$

$\therefore z^2 + 1 = 0 \Rightarrow z = \pm i$

$(e^z - 1) = 0 \Rightarrow z = 0$

So $z = \pm i$ and $z = 0$ are zeroes of a function.

All zeroes of order is 1.

ii) $(z^4 - z^2 - 6)^3$

$f(z) = (z^4 - z^2 - 6)^3$

$\therefore z^4 - z^2 - 6 = 0$

$(z^2 - 3)(z^2 + 2) = 0$

$z^2 = 3$ and $z^2 = -2$

$z = \pm\sqrt{3}$ $z = \pm\sqrt{2}i$

So $z = \pm\sqrt{3}$ and $z = \pm\sqrt{2}i$ are zeroes of function and All zeroes of order is 3.

Ex. 6.11.2 : Determine and classify the singularities of following functions.

i) $\frac{1}{z-z^3}$ ii) $\frac{z^4}{1+z^2}$

Sol. :

i) $f(z) = \frac{1}{z-z^3} = \frac{1}{z(1-z^2)}$

$\therefore z = 0$ and $z = \pm 1$ are isolated singular points. All points are pole of order 1.

ii) $f(z) = \frac{z^4}{1+z^2}$

$\therefore 1+z^2 = 0$

$\Rightarrow z^2 + 1 = 0$

$\Rightarrow z^2 = -1$

$\Rightarrow z = \pm i$

$z = \pm i$ is a isolated singularities.

$\therefore z = \pm i$ is a pole of order 1.

Ex. 6.11.3 : Find the nature and location of singularities of the following function.

i) $f(z) = \frac{z^2+1}{(z-1)^4}$ ii) $f(z) = \frac{e^z}{z(z-1)^4}$

iii) $f(z) = \sin\left(\frac{1}{1+z}\right)$ iv) $f(z) = \frac{\tan z}{z}$

v) $f(z) = \frac{\cot z}{(z-1)^3}$

Sol. : i) $f(z) = \frac{z^2+1}{(z-1)^4}$

Here $f(z)$ is analytic every where except $z = 1$.

$\therefore z = 1$ is a pole of order 4.

ii) $f(z) = \frac{e^z}{z(z-1)^4}$

Here $f(z)$ is analytic every where except $z = 0$ and $z = 1$.

$\therefore z = 0$ is a simple pole and $z = 1$ is a pole of order 4.

iii) $f(z) = \sin\left(\frac{1}{1+z}\right)$

Now $1+z = u$

$\sin\left(\frac{1}{1+z}\right) = \sin\left(\frac{1}{u}\right) = \frac{1}{u} - \frac{1}{3!u^3} + \frac{1}{5!u^5} - \dots$

$= \frac{1}{(z+1)} - \frac{1}{3!(z+1)^3} + \frac{1}{5!(z+1)^5} - \dots$

$\therefore z = -1$ is a essential singularities.

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$$\text{iv) } f(z) = \frac{\tan z}{z}$$

$$f(z) = \frac{\sin z}{\cos z \cdot z}$$

$$z = 0 \text{ and } \cos z = 0$$

$$z = 0, \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots \text{ are simple pole.}$$

$$\text{v) } f(z) = \frac{\cot z}{(z-1)^3}$$

$$f(z) = \frac{\cot z}{(z-1)^3} = \frac{\cos z}{\sin z(z-1)^3}$$

$$\sin z = 0 \text{ and } (z-1)^3 = 0$$

$$z = \pm n\pi \text{ and } z = 1 \text{ is a pole}$$

$$z = 1 \text{ is a pole of order 3}$$

$$\text{and } z = \pm\pi, \pm 2\pi, \pm 3\pi, \dots \text{ are simple pole.}$$

Ex. 6.11.4 : Discuss the singularity of the function $\frac{1-\sin z}{z^3}$.

$$\text{Sol: } f(z) = \frac{1-\sin z}{z^3} = \frac{1}{z^3}(1-\sin z)$$

$$= \frac{1}{z^3} \left(1 - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right) \right)$$

$$= \frac{1}{z^3} \left(1 - z + \frac{z^3}{3!} - \frac{z^5}{5!} + \frac{z^7}{7!} + \dots \right)$$

$$= \frac{1}{z^3} \left(1 - z + \frac{z^3}{3!} - \frac{z^5}{5!} + \frac{z^7}{7!} + \dots \right)$$

$$f(z) = \left(\frac{1}{z^3} - \frac{1}{z^2} + \frac{1}{3!} - \frac{z^2}{5!} + \frac{z^4}{7!} + \dots \right)$$

$z = 0$ is a pole at order 3.

Ex. 6.11.5 : Find the singularity of $f(z) = \frac{e^z + e^{-z}}{2z^4}$.

GTU : Summer-23, Maths-4, Marks 3

Sol: Here, $f(z) = \frac{e^z + e^{-z}}{2z^4}$ is not analytic at

$$z = 0.$$

$\therefore f(z)$ has pole at $z = 0$ of order 4.

Ex. 6.11.6 : Classify the singularities of

$$f(z) = \frac{1 - \cos z}{z^2}$$

GTU : Summer-21, Maths-4, Marks 3

Sol: Here, $f(z) = \frac{1 - \cos z}{z^2}$ is not analytic at

$$z = 0.$$

$\therefore f(z)$ has pole at $z = 0$ of order 2.

6.12 Cauchy Residue Theorem

**GTU : Summer-10,11,12,19,21,22,23,24,
Winter-10,11,12,21,22,24, June-10**

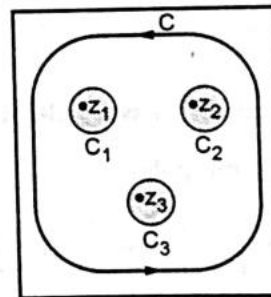
Theorem : State and prove Cauchy residue theorem.

State : If $f(z)$ is analytic in a closed curve C except at a finite number at singular points within C , then,

$$\int_C f(z) dz = 2\pi i \text{ (Sum of residue at the singular points)}$$

Proof : Let $z_1, z_2, \dots, z_n$ are singularities of $f(z)$ inside C .

Let us construct $C_1, C_2, \dots, C_n$ small circle with centre $z_1, z_2, \dots, z_n$ respectively.



By Cauchy integral formula,

$$\begin{aligned} \int_C f(z) dz &= \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \dots + \int_{C_n} f(z) dz \\ &= 2\pi i \text{Res}(f(z_1)) + 2\pi i \text{Res}(f(z_2)) \\ &\quad + \dots + 2\pi i \text{Res}(f(z_n)) \\ &= 2\pi i \text{ (sum of residue)} \end{aligned}$$

6.12.1 Techniques to Find Residue

i) If $f(z)$ has a simple pole at $z = z_0$ then

$$\text{Res}(f(z_0)) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

ii) If $f(z)$ has a multiple pole of order n (means order is more than one) at $z = z_0$ then

$$\text{Res } f(z_0) = \frac{1}{(n-1)!} \lim_{z \rightarrow z_0} \left[\frac{d^{n-1}}{dz^{n-1}} (z - z_0)^n f(z) \right]$$

iii) If $f(z) = \frac{P(z)}{Q(z)}$ has a simple pole at $z = z_0$ then

$$\text{Res } f(z_0) = \frac{P(z_0)}{Q'(z_0)}$$

Ex. 6.12.1 : Determine the pole of the function and find the residue at each pole.

i) $f(z) = \frac{z^2 + 1}{(z-1)(z+1)}$ ii) $f(z) = \frac{z^2}{(z-1)^2(z+1)}$

GTU : Summer-24, CVPDE, Marks 4

iii) $f(z) = \frac{1-e^z}{z^4}$ iv) $f(z) = \cot z$

v) $f(z) = \frac{\tan z}{z}$

Sol. :

i) $f(z) = \frac{z^2 + 1}{(z-1)(z+1)}$

$\therefore$ Here $z = 1$ and $z = -1$ is a pole of order 1.

$\therefore$ Both are simple poles.

i) $\text{Res } f(z) = \lim_{z \rightarrow 1} \frac{(z-1)(z^2+1)}{(z-1)(z+1)} = \frac{2}{2} = 1$

ii) $\text{Res } f(z) = \lim_{z \rightarrow -1} \frac{(z+1)(z^2+1)}{(z-1)(z+1)} = \frac{2}{-2} = -1$

ii) $f(z) = \frac{z^2}{(z-1)^2(z+1)}$

$\Rightarrow$ Here $z = 1$ is a pole of order 2 and $z = -1$ is a simple pole.

At $z = -1$

$$\text{Res } f(z) = \lim_{z \rightarrow -1} \frac{(z+1)z^2}{(z-1)^2(z+1)} = \frac{1}{4}$$

At $z = 1$, here $n = 2$.

$$\begin{aligned} \text{Res } f(z)_{z=1} &= \frac{1}{(2-1)!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \frac{(z-1)^2 z^2}{(z-1)^2 (z+1)} \right] \\ &= \frac{1}{1!} \lim_{z \rightarrow 1} \left[\frac{2z(z+1) - z^2(1)}{(z+1)^2} \right] \\ &= \frac{2(1+1) - 1}{4} = \frac{3}{4} \end{aligned}$$

iii) $f(z) = \frac{1-e^z}{z^4}$

Here $z = 0$ is a pole of order 4

$$\begin{aligned} \text{Res } f(z)_{z=0} &= \frac{1}{3!} \lim_{z \rightarrow 0} \left[\frac{d^3}{dz^3} z^4 \frac{(1-e^z)}{z^4} \right] \\ &= \frac{1}{6} \lim_{z \rightarrow 0} \left[\frac{d^3}{dz^3} (1-e^z) \right] \\ &= \frac{1}{6} \lim_{z \rightarrow 0} \left[\frac{d^2}{dz^2} (-e^z) \right] = \frac{1}{6} \lim_{z \rightarrow 0} [-e^z] \\ &= \frac{1}{6}(-1) = -\frac{1}{6} \end{aligned}$$

iv) $f(z) = \cot z$

$$f(z) = \frac{\cos z}{\sin z}$$

$$\sin z = 0 \Rightarrow z = \pm n\pi$$

$\therefore z = 0, \pm\pi, \pm2\pi, \pm3\pi$ are simple poles.

$$\text{Res } f(z)_{z=0} = \frac{P(z_0)}{Q'(z_0)} = \frac{\cos 0}{\cos 0} = 1$$

$$\text{Res } f(z)_{z=\pm n\pi} = \frac{\cos(\pm n\pi)}{\cos(\pm n\pi)} = 1$$

v) $f(z) = \frac{\tan z}{z} = \frac{\sin z}{\cos z \cdot z}$

$\therefore z = 0, \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}$ are simple poles.

$$\begin{aligned} \text{Res } f(z)_{z=0} &= \frac{\sin z}{\cos z - z \sin z} \\ &= \frac{\sin 0}{\cos 0 - 0 \sin 0} = 0 \end{aligned}$$

$$\text{Res } f(z)_{z=\pm n\pi/2} = \frac{\sin z}{-\sin z \cdot z + \cos z} = \frac{\sin 0}{0 + \cos 0} = 1$$

Ex. 6.12.2 : Find the residue at each pole of

$$f(z) = \frac{z \cdot e^{iz}}{z^2 + 9}$$

GTU : Summer-19, Maths-4, Marks 4

Sol. :

Here, $f(z) = \frac{z \cdot e^{iz}}{z^2 + 9} = \frac{z \cdot e^{iz}}{(z+3i)(z-3i)}$ is not analytic at

$$z = -3i \text{ and } z = 3i$$

$\therefore f(z)$ has a pole at $z = -3i$ and $z = 3i$ of order 1

$\therefore z = -3i$ and $z = 3i$ both are simple pole.

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow -3i} \frac{(z+3i) \cdot z e^{iz}}{(z+3i)(z-3i)}$$

$$= \frac{(-3i) e^{i(-3i)}}{(-6i)} = \frac{1}{2} e^3$$

$$\text{Res } f(z) = \lim_{z \rightarrow 3i} \frac{(z-3i) \cdot z e^{iz}}{(z+3i)(z-3i)}$$

$$= \frac{3i e^{i(3i)}}{6i} = \frac{1}{2} e^{-3}$$

Ex. 6.12.3 : Obtain the residue of

$$f(z) = \frac{z-3}{(z+1)(z+2)} \text{ at it's poles.}$$

GTU : Winter-24, CVPDE, Marks 3

Sol. : Here, $f(z) = \frac{z-3}{(z+1)(z+2)}$ is not analytic at $z = -1$

and $z = -2$.

Both $z = -1$ and $z = -2$ are simple pole's

$$\text{Res } f(z) = \lim_{z \rightarrow -1} \frac{(z+1)(z-3)}{(z+1)(z+2)} = \frac{-4}{1} = -4$$

$$\text{Res } f(z) = \lim_{z \rightarrow -2} \frac{(z+2)(z-3)}{(z+1)(z+2)} = \frac{-5}{-1} = 5$$

Ex. 6.12.4 : Find the residue of the function

$$f(z) = \frac{z}{(z+1)^2(z^2-4)} \text{ at it's poles.}$$

GTU : Summer-22, CVPDE, Marks 7

Sol. : Here, $f(z) = \frac{z}{(z+1)^2(z-2)(z+2)}$ is not analytic at

$z = -1$, $z = 2$ and $z = -2$.

$\therefore z = -1$ is a pole of order 2 and $z = -2$,

$z = 2$ is a simple pole.

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow 2} \frac{(z-2) z}{(z+1)^2(z-2)(z+2)}$$

$$= \frac{2}{9 \times 4} = \frac{1}{18}$$

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow -2} \frac{(z+2) z}{(z+1)^2(z-2)(z+2)}$$

$$= \frac{-2}{(-1)^2(-4)} = \frac{-2}{-4} = \frac{1}{2}$$

$$\text{Res } f(z) = \frac{1}{1!} \lim_{z \rightarrow -1} \left[\frac{d}{dz} \frac{(z+1)^2 z}{(z+1)^2(z^2-4)} \right]$$

$$= \lim_{z \rightarrow -1} \left[\frac{d}{dz} \frac{z}{(z^2-4)} \right]$$

$$= \lim_{z \rightarrow -1} \left[\frac{(z^2-4) - z(2z)}{(z^2-4)^2} \right]$$

$$= \frac{(1-4) - (-1)(-2)}{(1-4)^2}$$

$$= \frac{-3-2}{(-3)^2} = \frac{-5}{9}$$

Ex. 6.12.5 : Using Cauchy residue theorem evaluate

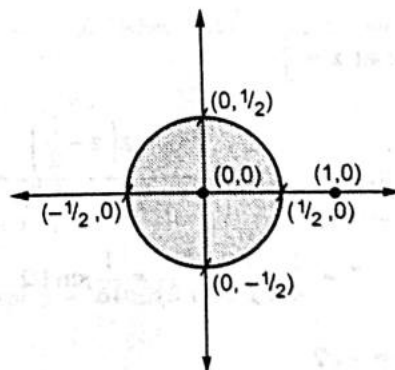
$$\oint_C \frac{e^z}{z(1-z)^3} dz \text{ where } C : \text{ is } |z| = \frac{1}{2}.$$

Sol. : Here $\frac{e^z}{z(1-z)^3}$ is analytic every where except

$z = 0$ and $z = 1$.

$\therefore z = 0$ is a pole of order 1 (simple pole) and $z = 1$ is a pole of order 3.

$C : |z| = \frac{1}{2}$ is a circle with center $(0, 0)$ and radius $\frac{1}{2}$.



Here $z = 0 \Rightarrow x + iy = 0 \Rightarrow x = 0, y = 0$

$\therefore z = 0$ or $(x, y) = (0, 0)$ points is inside in $|z| = \frac{1}{2}$.

But $z = 1 \Rightarrow x + iy = 1 \Rightarrow x = 1, y = 0$.

$\therefore z = 1$ or $(x, y) = (1, 0)$ is outside $|z| = \frac{1}{2}$

$$\text{Res}_{z=0} f(z) = \lim_{z \rightarrow 0} \frac{(z-0)e^z}{z(1-z)^3} = \frac{e^0}{1} = 1$$

Using Cauchy residue theorem

$$\oint_C \frac{e^z}{z(1-z)^3} dz = 2\pi i (\text{Sum of Residue})$$

$$= 2\pi i (1) = 2\pi i$$

Ex. 6.12.6 : Using residue theorem evaluate

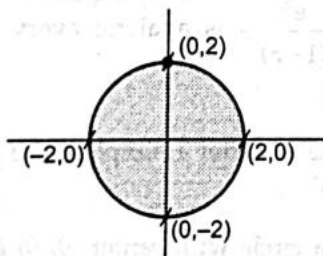
$$\oint_C \frac{z^2 \sin z}{4z^2 - 1} dz \quad C : |z| = 2 \quad \text{GTU : Summer-10, Marks 4}$$

Sol. : $\frac{z^2 \sin z}{4z^2 - 1}$ is analytic at every point except

$4z^2 - 1 = 0 \Rightarrow z^2 = \frac{1}{4} \Rightarrow z = \pm \frac{1}{2}$ both are simple poles.

$$f(z) = \frac{z^2 \sin z}{4(z-1/2)(z+1/2)}$$

Here $z = \frac{1}{2}$ and $z = -\frac{1}{2}$ both inside in $|z| = 2$



Now residue at $z = \frac{1}{2}$

$$\text{Res}_{z=1/2} f(z) = \lim_{z \rightarrow 1/2} \frac{z^2 \sin z \left(z - \frac{1}{2}\right)}{4(z-1/2)(z+1/2)} = 0$$

$$= \frac{1}{4} \frac{\sin 1/2}{4(1/2+1/2)} = \frac{1}{16} \sin 1/2$$

Residue at $z = -1/2$

$$\text{Res}_{z=-1/2} f(z) = \lim_{z \rightarrow -1/2} \frac{z^2 \sin z (z+1/2)}{4(z-1/2)(z+1/2)}$$

$$= \lim_{z \rightarrow -1/2} \frac{z^2 \sin z}{4(z-1/2)}$$

$$= \frac{1}{4} \frac{\sin(-1/2)}{4(-1/2-1/2)} = \frac{1}{16} \frac{-\sin(1/2)}{-1}$$

$$= \frac{\sin(1/2)}{16} = 2\pi i (\text{sum of residue})$$

$$\oint_C \frac{z^2 \sin z}{2z^2 - 1} dz = 2\pi i \left(\frac{\sin 1/2}{16} + \frac{\sin 1/2}{16} \right) \sin\left(\frac{1}{2}\right) i$$

$$= 2\pi i \left(\frac{2 \sin 1/2}{16} \right) = \frac{\pi}{4} \sin \frac{1}{2}$$

$$= \frac{\pi i}{4} \sin\left(\frac{1}{2}\right)$$

Ex. 6.12.7 : Determine the poles of the function

$f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and Residue at each pole. Hence evaluate $\oint_C f(z) dz$, where $C : |z| = 3$.

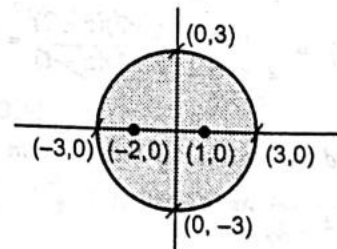
GTU : Winter-24, Summer-24, Maths 4, Marks 7, Summer-23, CVPDE, Marks 7

Sol. :

$f(z) = \frac{z^2}{(z-1)^2(z+2)}$ is analytic any where except

$z = 1$ and $z = -2$.

$z = 1$ is a pole of order 2 and $z = -2$ is a pole of order 1 (simple pole).



$z = 1 \Rightarrow x + iy = 1 \Rightarrow x = 1$ and $y = 0$

$\Rightarrow (x, y) = (1, 0)$

$z = -2 \Rightarrow x + iy = -2$

$\Rightarrow x = -2$ and $y = 0$

$\Rightarrow (x, y) = (-2, 0)$

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Here $z = 1$ and $z = -2$ both inside in $|z| = 3$.

$$\text{Res } f(z) = \lim_{z \rightarrow -2} \frac{z^2(z+2)}{(z-1)^2(z+2)}$$

$$= \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2} = \frac{4}{9}$$

$$\text{Res } f(z) = \frac{1}{(2-1)!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} (f(z)(z-1)^2) \right]$$

$$= 1 \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{z^2(z-1)^2}{(z-1)^2(z+2)} \right) \right]$$

$$= \lim_{z \rightarrow 1} \left[\frac{2z(z+2) - z^2}{(z+2)^2} \right] = \frac{2(1+2) - 1}{(3)^2} = \frac{5}{9}$$

$$\oint_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i (\text{Sum of residue})$$

$$= 2\pi i \left(\frac{4}{9} + \frac{5}{9} \right) = 2\pi i$$

Ex. 6.12.8 : Use Cauchy's residue theorem to evaluate the integral of each of these functions around the circle $|z| = 3$ in the positive sense

a) $\frac{e^{-z}}{(z-1)^2}$ b) $\frac{z+1}{z^2-2z}$

Sol. : a) $\frac{e^{-z}}{(z-1)^2}$

Here $\frac{e^{-z}}{(z-1)^2}$ is a analytic any where except $z = 1$.

$\therefore z = 1$ is a pole of order 2 and $z = 1$ is inside in $|z| = 3$.

$$\text{Res } f(z) = \frac{1}{1!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{e^{-z}(z-1)^2}{(z-1)^2} \right) \right]$$

$$= \lim_{z \rightarrow 1} [-e^{-z}] = -e^{-1}$$

$$= -e^{-1}$$

$$\oint_C \frac{e^{-z}}{(z-1)^2} dz = 2\pi i (\text{sum of residue})$$

$$= 2\pi i (-e^{-1}) = -\frac{2\pi i}{e}$$

b) $\frac{z+1}{z^2-2z}$

Here $f(z) = \frac{z+1}{z(z-2)}$ is analytic any where except $z = 0$ and $z = 2$.

$\therefore z = 0$ and $z = 2$ is simple poles.

Here $z = 0$ and $z = 2$ both inside in $|z| = 3$.

$$\text{Res } f(z) = \lim_{z \rightarrow 0} \frac{(z+1)(z-0)}{z(z-2)} = -\frac{1}{2}$$

$$\text{Res } f(z) = \lim_{z \rightarrow 2} \frac{(z+1)(z-2)}{z(z-2)} = \frac{3}{2}$$

$$\oint_C \frac{z+1}{z^2-2z} dz = 2\pi i (-1/2 + 3/2) = 2\pi i$$

Ex. 6.12.9 : Show that the singular point of the function $f(z) = \frac{(1-\cosh z)}{z^3}$ is a pole. Determine the order m of that pole and the corresponding residue.

GTU : Winter-12, Marks 3

Sol. : Here $f(z) = \frac{1-\cosh z}{z^3}$ is analytic any where

except $z = 0$.

So $z = 0$ is a singularities.

Now Here $z = 0$ is also a pole because

$f(z) = \frac{1-\cosh z}{z^3}$ is a analytic any where except $z = 0$.

Now here pole of order $f(z)$ is $m = 3$.

$$\text{Res } f(z) = \frac{1}{2!} \lim_{z \rightarrow 0} \left[\frac{d^2}{dz^2} \left(\frac{1-\cosh z}{z^3} z^3 \right) \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow 0} \left[\frac{d}{dz} (0 - \sinh z) \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow 0} [-\cosh z] = -\frac{1}{2}$$

Ex. 6.12.10 : Find the value of the integral

$$\oint_C \frac{3z^2+2}{(z-1)(z^2+9)} dz \text{ taken counter clockwise around the}$$

circle $C : |z-2| = 2$

Sol. :

Here $f(z) = \frac{3z^2+2}{(z-1)(z^2+9)}$ is analytic any

where except $z = 1$ and $z = \pm 3i$.

All are simple poles.

$$C : |z - 2| = 2$$

Circle with radius 2 and centre (2, 0)

Here $z = 1 \Rightarrow (x, y) = (1, 0)$

$$z = 3i \Rightarrow (x, y) = (0, 3)$$

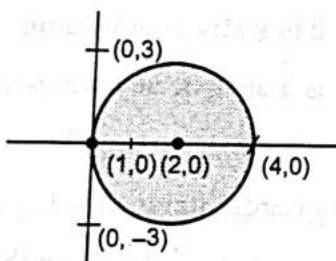
$$z = -3i \Rightarrow (x, y) = (0, -3)$$

Here $z = \pm 3i$ is outside

But $z = 1$ is inside in $|z - 2| = 2$

$$\begin{aligned} \text{Res}_{z=1} f(z) &= \lim_{z \rightarrow 1} \frac{3z^2 + 2(z-1)}{(z-1)(z^2+9)} = \frac{3+2}{1+9} = \frac{5}{10} \\ &= \frac{1}{2} \end{aligned}$$

$$\int_C \frac{3z^2 + 2}{(z-1)(z^2+9)} dz = 2\pi i \left(\frac{1}{2} \right) = \pi i$$



Ex. 6.12.11 : Evaluate $\oint_C \frac{dz}{z^2 + 1}$, where C is

$|z + i| = 1$, Counter clockwise.

GTU : Summer-10, Marks 4

Sol : $f(z) = \frac{1}{z^2 + 1}$ is analytic any where except

$$z^2 + 1 = 0 \Rightarrow z^2 = -1 \Rightarrow z = \pm i$$

Both are simple pole.

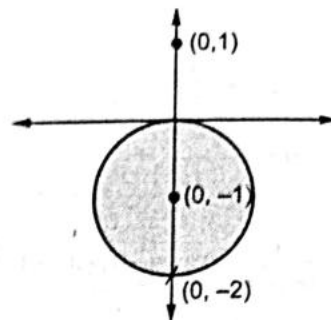
$$C : |z + i| = 1 \Rightarrow \text{Centre } (0, -1) \text{ and radius } 1$$

$$|x + iy + i| = 1$$

$$|x + i(y+1)| = 1$$

$$\sqrt{x^2 + (y+1)^2} = 1$$

Circle with centre (0, -1) and radius 1.



$$z = i \Rightarrow (x, y) = (0, 1)$$

$$z = -i \Rightarrow (x, y) = (0, -1)$$

$$z = i \text{ outside and}$$

$$z = -i \text{ inside in } |z + i| = 1$$

$$\text{Res}_{z=-i} f(z) = \lim_{z \rightarrow -i} \frac{1(z+i)}{(z+i)(z-i)} = \frac{1}{-i-i} = \frac{-1}{2i}$$

$$\oint_C \frac{1}{z^2 + i} dz = 2\pi i \left(\frac{-1}{2i} \right) = -\pi$$

Ex. 6.12.12 : Using residue theorem, evaluate

$$\oint_C \frac{e^z}{z(1-z)^3} dz \text{ where}$$

$$C : a) |z| = \frac{1}{2} \quad b) |z - 1| = \frac{1}{2}$$

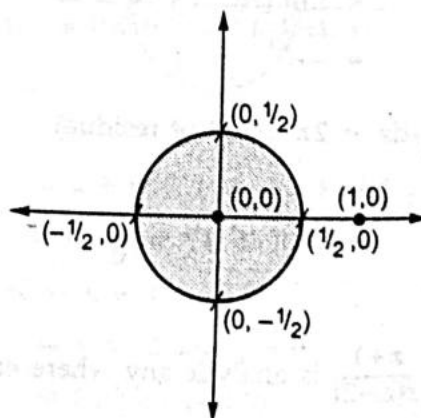
GTU : Summer-10, Marks 7

Sol. : $\frac{e^z}{z(1-z)^3}$ is analytic any where except $z = 0$ and $z = 1$.

$z = 0$ is a simple pole and $z = 1$ is a pole of order 3.

$$a) C : |z| = \frac{1}{2}$$

Centre (0, 0) and radius = $\frac{1}{2}$



$$z=0 \Rightarrow (x, y) = (0, 0)$$

$$z=1 \Rightarrow (x, y) = (1, 0)$$

Here $z=0$ is inside in $|z| = \frac{1}{2}$

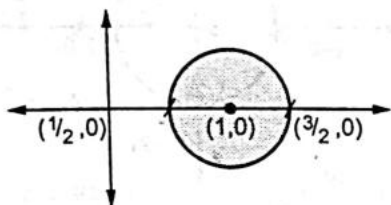
$$\text{Res } f(z) = \lim_{z \rightarrow 0} \frac{e^z}{z(1-z)^3} \cdot z = \frac{e^0}{(1-0)^3} = 1$$

$$\oint_C \frac{e^z}{z(1-z)^3} dz = 2\pi i(1) = 2\pi i$$

$C: |z| = \frac{1}{2}$

$$b) C: |z-1| = \frac{1}{2}$$

Circle with centre $(1, 0)$ and radius is $\frac{1}{2}$.



$$z=0 \Rightarrow (x, y) = (0, 0)$$

$$z=1 \Rightarrow (x, y) = (1, 0)$$

$$\text{Res } f(z) = \frac{1}{2!} \lim_{z \rightarrow 1} \left[\frac{d^2}{dz^2} \frac{e^z (z-1)^3}{z(1-z)^3} \right]$$

$$= \frac{1}{2!} \lim_{z \rightarrow 1} \left[-\frac{d^2}{dz^2} \frac{e^z}{z(z-1)^3} \cdot (z-1)^3 \right]$$

$$\text{Res } f(z) = \frac{1}{2!} \lim_{z \rightarrow 1} \left[\frac{d^2}{dz^2} \frac{e^z}{z(1-z)^3} (z-1)^3 \right]$$

$$= -\frac{1}{2!} \lim_{z \rightarrow 1} \left[\frac{d^2}{dz^2} \left(\frac{e^z}{z} \right) \right]$$

$$= -\frac{1}{2!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{e^z z - e^z}{z^2} \right) \right]$$

$$= -\frac{1}{2} \lim_{z \rightarrow 1} \left[\frac{(e^z + ze^z - e^z)z^2 - (e^z z - e^z) \cdot 2z}{z^4} \right]$$

$$= -\frac{1}{2} \lim_{z \rightarrow 1} \left[\frac{z^3 e^z - 2z^3 e^z + 2z e^z}{z^4} \right]$$

$$= -\frac{1}{2} \left[\frac{1e - 2e + 2e}{1} \right]$$

$$= -\frac{e}{2}$$

$$\oint_C \frac{e^z}{z(1-z)^3} dz = 2\pi i \left(-\frac{e}{2} \right) = -\pi i e$$

Ex. 6.12.13 : Evaluate $\oint_C \frac{e^{2z}}{(z+1)^4} dz$ $C: |z| = 3$

Winter-10

GTU : Winter-24, CVPDE, Marks 4

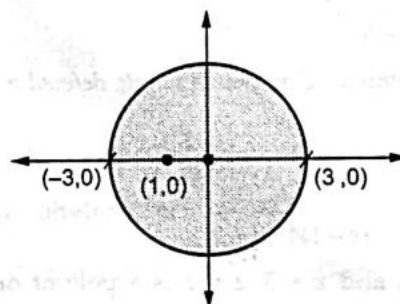
Sol. : $\frac{e^{2z}}{(z+1)^4}$ is analytic any where except $z = -1$

$\therefore z = -1$ is a pole of order 4.

$$C: |z| = 3$$

$$z = -1 \Rightarrow (-1, 0)$$

$z = -1$ is inside in $|z| = 3$



$$\text{Res } f(z) = \frac{1}{3!} \lim_{z \rightarrow -1} \left[\frac{d^3}{dz^3} \left(\frac{e^{2z}}{(z+1)^4} (z+1)^4 \right) \right]$$

$$= \frac{1}{3!} \lim_{z \rightarrow -1} \left[\frac{d^2}{dz^2} (2e^{2z}) \right]$$

$$= \frac{1}{3!} \lim_{z \rightarrow -1} \left[\frac{d}{dz} (4e^{2z}) \right]$$

$$= \frac{1}{3!} \lim_{z \rightarrow -1} [8e^{2z}] = \frac{8}{3!} e^{2(-1)}$$

$$= \frac{8^4 e^{-2}}{3 \times 2}$$

$$\oint_C \frac{e^{2z}}{(z+1)^4} dz = \frac{2\pi i \times 4e^{-2}}{3} = \frac{8\pi i}{3e^2} = \frac{8\pi i}{3e^2}$$

Ex. 6.12.14 : Evaluate $\oint_C \frac{5z-2}{z(z-1)} dz$ $C : |z| = 2$

Winter-10

GTU : Summer-23, Maths 4, Marks 4,
Summer-21, Maths 4, Marks 7

Sol. : $f(z) = \frac{5z-2}{z(z-1)}$ is analytic any where except $z = 0$

and $z = 1$.

Both are Simple pole

Here $z = 0$ and $z = 1$ both inside in $|z| = 2$.

$$\text{Res}_{z=0} f(z) = \lim_{z \rightarrow 0} \frac{5z-2}{z(z-1)} (z-0) = \frac{-2}{(-1)} = 2$$

$$\text{Res}_{z=1} f(z) = \lim_{z \rightarrow 1} \frac{5z-2}{z(z-1)} (z-1) = \frac{5-2}{1} = 3$$

$$\oint_C \frac{5z-2}{z(z-1)} dz = 2\pi i (2+3) = 10\pi i$$

Ex. 6.12.15 : Use Cauchy residue theorem

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz$$

$C : a)$ The contour C is the rectangle defined $x = 0, x = 4, y = 1, y = 1$.

$b)$ The contour C is the circle $|z| = 2$

Sol. : $f(z) = \frac{1}{(z-1)^2(z-3)}$ is analytic any where

except $z = 1$ and $z = 3$. $z = 1$ is a pole of order 2 and $z = 3$ is a Simple pole.

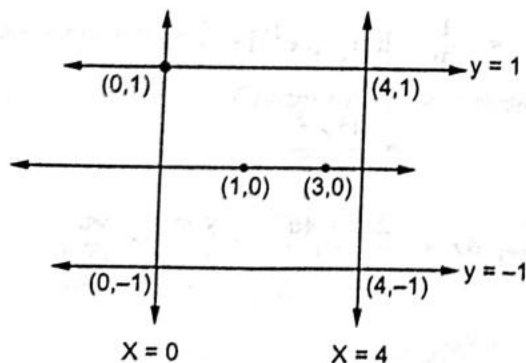
$a)$ $C : x = 0, x = 4, y = -1, y = 1$.

$$z = 1 \Rightarrow (x, y) = (1, 0)$$

$$z = 3 \Rightarrow (x, y) = (3, 0)$$

$z = 1$ and $z = 3$ both inside.

$$\text{Res}_{z=1} f(z) = \frac{1}{1!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{1}{(z-1)^2(z-3)} (z-1)^2 \right) \right]$$



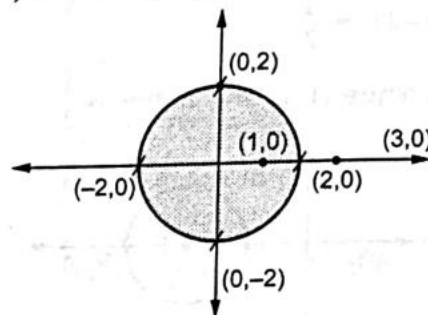
$$= \lim_{z \rightarrow 1} \left[\frac{-1}{(z-3)^2} \right] = -\frac{1}{4}$$

$$\text{Res}_{z=3} f(z) = \lim_{z \rightarrow 3} \frac{1}{(z-1)^2(z-3)} (z-3) = \frac{1}{4}$$

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i \left(\frac{1}{4} - \frac{1}{4} \right) = 0$$

$b) C : |z| = 2$

Center $(0,0)$ and radius $= 2$



$$z = 1 \Rightarrow (1, 0) = (x, y)$$

$$z = 3 \Rightarrow (3, 0) = (x, y)$$

$z = 1$ is inside and $z = 3$ is outside.

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i \left(-\frac{1}{4} \right) = -\frac{\pi i}{2}$$

Ex. 6.12.16 : Evaluate $\oint_C \frac{\sin 3z}{z + \pi/2} dz$ $C : |z| = 5$.

GTU : Winter-11, Marks 4

Sol. : Here $\frac{\sin 3z}{z + \pi/2}$ is analytic any where except $z = -\pi/2$.

$z = -\frac{\pi}{2}$ is a simple pole.

$z = -\frac{\pi}{2}$ is inside in $|z| = 5$

$$\begin{aligned} \text{Res}_{z=-\pi/2} f(z) &= \lim_{z \rightarrow -\pi/2} \frac{\sin 3z}{\left(z + \frac{\pi}{2}\right)} \left(z + \frac{\pi}{2}\right) \\ &= \sin 3 \left(-\frac{\pi}{2}\right) = -\sin 3 \frac{\pi}{2} \\ &= -(-1) = 1 \end{aligned}$$

$$\oint_C \frac{\sin 3z}{z + \pi/2} dz = 2\pi i (1) = 2\pi i$$

Ex. 6.12.17 : Evaluate $\oint_C \frac{dz}{\sinh 2z}$ $C : |z| = 2$.

Sol. : $\frac{1}{\sinh 2z}$ is analytic any where except

$z = 0$ because $\sinh 2z = 0$

$$\Rightarrow \frac{e^{2z} - e^{-2z}}{2} = 0 \Rightarrow e^{2z} - e^{-2z} = 0$$

$$\Rightarrow z = 0$$

$z = 0$ is a simple pole.

$$\text{Res } f(z)_{z=0} = \frac{P(z)}{Q'(z_0)} = \frac{1}{2 \cosh 2z} = \frac{1}{2 \cos(0)} = \frac{1}{2} = \frac{1}{2}$$

where $P(z) = 1$ and $Q(z) = \sinh 2z$.

$$\oint_C \frac{dz}{\sinh 2z} = 2\pi i \left(\frac{1}{2} \right) = 2\pi i \cdot \frac{1}{2} = \pi i$$

Ex. 6.12.18 : Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)} dz$ $C : |z| = 3$.

Sol. : Here $\frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)}$ is a analytic any where

except $z = 1$ and $z = 2$.

$\therefore z = 1$ is a pole of order 2 and $z = 2$ is a simple pole.

Both are inside in $|z| = 3$.

$$\begin{aligned} \text{Res } f(z)_{z=2} &= \lim_{z \rightarrow 2} \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)} (z-2) \\ &= \frac{\sin 4\pi + \cos 4\pi}{1} = 1 \end{aligned}$$

$$\begin{aligned} \text{Res } f(z)_{z=1} &= \frac{1}{1!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} (z-1)^2 \right) \right] \\ &= \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)} \right) \right] \\ &= \lim_{z \rightarrow 1} \left[\frac{\cos(\pi z^2) \cdot 2\pi z - \sin(\pi z^2)}{(z-2)^2} \right] \\ &= \lim_{z \rightarrow 1} \left[\frac{\cos(\pi z^2) \cdot 2\pi z - \sin(\pi z^2) + \cos(\pi z^2) \cdot 1}{(z-2)^2} \right] \\ &= \frac{[(\cos \pi) \cdot (2\pi) - \sin(\pi) (2\pi) (-1) - (\sin \pi + \cos \pi)]}{1} \\ &= + \frac{2\pi+1}{1} (2\pi+1) \end{aligned}$$

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)} dz = 2\pi i$$

$$= 2\pi i (2\pi+1+1)$$

$$= 2\pi i (2\pi+2) = 4\pi i (\pi+1)$$

Ex. 6.12.19 : $\oint_C \frac{\cos \pi z^2}{(z-2)(z-1)} dz$ $C : |z| = 3$

Summer-11

GTU : Summer-23, Maths 4, Marks 7

Sol. : $f(z) = \frac{\cos \pi z^2}{(z-2)(z-1)}$ is analytic any where except $z = 1$ and $z = 2$.

$\therefore z = 1$ and $z = 2$ both are simple pole and both are inside in $|z| = 3$

$$\text{Res } f(z)_{z=1} = \lim_{z \rightarrow 1} \frac{\cos \pi z^2}{(z-1)(z-2)} (z-1) = \frac{\cos \pi}{(1-2)} = 1$$

$$\text{Res } f(z)_{z=2} = \lim_{z \rightarrow 2} \frac{\cos \pi z^2}{(z-1)(z-2)} (z-2) = \frac{\cos 4\pi}{(2-1)} = 1$$

$$\oint_C \frac{\cos \pi z^2}{(z-2)(z-1)} dz = 2\pi i (1+1) = 4\pi i$$

Ex. 6.12.20 : Evaluate $\oint_C \frac{dz}{(z^2+1)^2}$ Where

$$C : |z+i| = 1$$

GTU : Summer-12, Marks 4

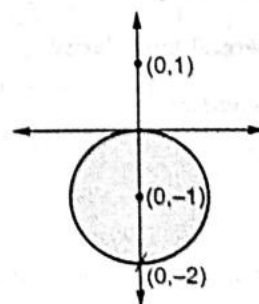
Sol. : Here $f(z) = \frac{1}{(z+i)^2 (z-i)^2}$ is analytic any where

except $z = i$ and $z = -i$.

Both are pole at order 2

$$C : |z+i| = 1$$

Circle with centre $(0, -1)$ and radius 1.



$$z = i \Rightarrow (x, y) = (0, 1)$$

$$z = -i \Rightarrow (x, y) = (0, -1)$$

$\therefore z = -i$ is inside and $z = i$ is outside in $|z + i| = 1$

$$\begin{aligned} \text{Res } f(z)_{z=-1} &= \frac{1}{1!} \lim_{z \rightarrow -i} \left[\frac{d}{dz} \left(\frac{1}{(z+i)^2(z-i)^2} \right) \right] \\ &= \lim_{z \rightarrow -i} \left[\frac{-2}{(z-i)^3} \right] = \frac{-2}{(-i-i)^3} \\ &= \frac{-2}{(-2i)^3} = \frac{-2}{-8i^3} = \frac{-1}{+4i} \end{aligned}$$

$$\oint_C \frac{dz}{(z^2+1)^2} = 2\pi i \left(-\frac{1}{4i} \right) = \frac{-\pi}{2}$$

Ex. 6.12.21 : Classify the poles of $f(z) = \frac{1}{z^2 - z^6}$.

Summer-12, Winter-22. CVPDE Marks 3

Sol. :

$$\begin{aligned} f(z) &= \frac{1}{z^2(1-z^4)} = \frac{1}{z^2(1-z^2)(1+z^2)} \\ &= \frac{1}{z^2(1-z)(1+z)(z+i)(z-i)} \end{aligned}$$

$\therefore z = 0, z = 1, z = -1$, and $z = \pm i$ is a pole

$z = 0$ is a pole of order 2.

$z = \pm 1$ and $z = \pm i$ is a simple pole.

Ex. 6.12.22 : Find the residue at $z = 0$ of the function

a) $z \cos\left(\frac{1}{z}\right)$ b) $\frac{z - \sin z}{z}$ c) $\frac{\cot z}{z}$

Sol. :

$$\begin{aligned} \text{a) } f(z) &= z \cos\left(\frac{1}{z}\right) = z \left(1 - \frac{1}{2!} \left(\frac{1}{z^2}\right) + \frac{1}{4!} \left(\frac{1}{z^4}\right) + \dots \right) \\ &= \left(z - \frac{1}{2} \frac{1}{z} + \frac{1}{4!} \frac{1}{z^3} + \dots \right) \end{aligned}$$

Here $z = 0$ is an essential singularity.

$\therefore \text{Res } f(z)_{z=0}$ is coefficient at $\frac{1}{z}$

$$\therefore \text{Res } f(z)_{z=0} = -\frac{1}{2}$$

b) $f(z) = \frac{z - \sin z}{z}$

Here $z = 0$ is a simple pole.

$$\therefore \text{Res } f(z)_{z=0} = \lim_{z \rightarrow 0} \frac{z - \sin z}{z} (z-0) = 0 - 0 = 0$$

c) $f(z) = \frac{\cot z}{z} = \frac{\cos z}{\sin z}$

Here $z = 0$ is a pole at order -2 and $z = \pm \pi, \pm 2\pi, \dots$ is a simple pole.

$$\begin{aligned} \text{Res } f(z)_{z=0} &= \frac{1}{1!} \lim_{z \rightarrow 0} \left[\frac{d}{dz} \left(\frac{\cot z}{z} (z-0)^2 \right) \right] \\ &= \lim_{z \rightarrow 0} \left[\frac{d}{dz} (z \cot z) \right] \\ &= \lim_{z \rightarrow 0} [z(-\cos z) + \cot z] \\ &= \lim_{z \rightarrow 0} \left[\frac{-z}{\sin^2 z} + \frac{\cos z}{\sin z} \right] \\ &= \lim_{z \rightarrow 0} \left[\frac{-z + \cos z \sin z}{\sin^2 z} \right] \\ &= \lim_{z \rightarrow 0} \left[\frac{-z + \frac{1}{2} \sin 2z}{\sin^2 z} \right] \end{aligned}$$

using L- hospital rule

$$\begin{aligned} &= \lim_{z \rightarrow 0} \left[\frac{-1 + \frac{1}{2} \cos 2z \cdot 2}{\sin^2 z} \right] = \lim_{z \rightarrow 0} \left[\frac{\cos 2z}{\sin 2z} \right] \\ &= \lim_{z \rightarrow 0} \left[\frac{-2 \sin 2z}{2 \cos 2z} \right] \end{aligned}$$

$$\text{Res } f(z)_{z=0} = 0$$

Ex. 6.12.23 : Use Cauchy residue theorem to evaluate the integral around the circle $|z| = 3$ in the positive sense.

$$f(z) = z^2 \exp\left(\frac{1}{z}\right)$$

Sol. :

$$\begin{aligned} f(z) &= z^2 e^{\left(\frac{1}{z}\right)} \\ &= z^2 \left(1 + \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \frac{1}{3!} \frac{1}{z^3} + \dots \right) \\ &= \left(z^2 + z + \frac{1}{2} + \frac{1}{z} + \frac{1}{3!} + \dots \right) \end{aligned}$$

$z = 0$ is an essential singularity

$z = 0$ is inside in $|z| = 3$

Res $f(z)$ is a co-efficient at $\frac{1}{z} z = 0$

$$\therefore \text{Res}_{z=0} f(z) = \frac{1}{3!} = \frac{1}{6}$$

$$\therefore \oint_C z^2 e^{\frac{1}{z}} dz = 2\pi i \left(\frac{1}{6} \right) = \frac{\pi i}{3}$$

Ex. 6.12.24 : Evaluate $\oint_C \frac{z^5}{1-z^3} dz$; $C : |z| = 2$

Sol. :
$$f(z) = \frac{z^5}{1-z^3} = \frac{-z^5}{z^3-1}$$

$$= \frac{-z^5}{(z-1)(z^2+z+1)}$$

$z-1=0 \Rightarrow z=1$ is a simple pole.

$$z^2+z+1=0 \Rightarrow a=1, b=1, c=1$$

$$\Rightarrow \Delta = b^2 - 4ac$$

$$\Rightarrow \Delta = 1 - 4 = -3$$

$$z_1, z_2 = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1}{2} \pm \frac{\sqrt{3}}{2} i$$

Here $z=1, z = \frac{-1}{2} \pm \frac{\sqrt{3}}{2} i$ are simple pole.

All are inside in $|z| = 2$.

$$\begin{aligned} \text{Res}_{z=1} f(z) &= \lim_{z \rightarrow 1} \frac{z^5 (z-1)}{(z-1) \left(z + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \left(z + \frac{1}{2} + \frac{\sqrt{3}}{2} i \right)} \\ &= \frac{-1}{\left(\frac{3}{2} - \frac{\sqrt{3}}{2} i \right) \left(\frac{3}{2} + \frac{\sqrt{3}}{2} i \right)} \\ &= \frac{-1}{\left(\frac{9}{4} + \frac{3}{4} \right)} = \frac{-1}{12^3/4} = \frac{-1}{3} \end{aligned}$$

$$\begin{aligned} \text{Res}_{z = \frac{-1}{2} + \frac{\sqrt{3}}{2} i} f(z) &= \lim_{z \rightarrow \frac{-1}{2} + \frac{\sqrt{3}}{2} i} \frac{z^5 \left(z + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right)}{(z-1) \left(z + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \left(z + \frac{1}{2} + \frac{\sqrt{3}}{2} i \right)} \end{aligned}$$

$$= \frac{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)^5}{\left(-\frac{3}{2} - \frac{\sqrt{3}}{2} i \right) \left(2 - \frac{\sqrt{3}}{2} i \right)}$$

$$\begin{aligned} \text{Res}_{z = \frac{-1}{2} - \frac{\sqrt{3}}{2} i} f(z) &= \lim_{z \rightarrow \frac{-1}{2} - \frac{\sqrt{3}}{2} i} \frac{z^5 \left(z + \frac{1}{2} + \frac{\sqrt{3}}{2} i \right)}{\left(z + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right) \left(z + \frac{1}{2} + \frac{\sqrt{3}}{2} i \right) (z-1)} \end{aligned}$$

$$= \frac{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)^5}{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i - 1 \right) \left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i + \frac{1}{2} - \frac{\sqrt{3}}{2} i \right)}$$

$$= \frac{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)^5}{\left(-\frac{3}{2} - \frac{\sqrt{3}}{2} i \right) (-\sqrt{3} i)}$$

$$\therefore \oint_C \frac{z^5}{1-z^3} dz = 2\pi i \quad (\text{Sum of residue} = 2\pi i)$$

$$= 2\pi i$$

$$\left(-\frac{1}{3} + \frac{\left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right)^5}{\left(-\frac{3}{2} + \frac{\sqrt{3}}{2} i \right) (-\sqrt{3} i)} + \frac{\left(-\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)^5}{\left(-\frac{3}{2} - \frac{\sqrt{3}}{2} i \right) (-\sqrt{3} i)} \right)$$

After the calculation we get $= -2\pi i$

Ex. 6.12.25 : Using Residue theorem, evaluate $\int_C \frac{e^z + z}{z^3 - 2} dz$,

Where $C : |z| = \pi/2$.

GTU : Winter-21, Summer-22, Maths-4, Marks 7

Sol. : Here,

$f(z) = \frac{e^z + z}{z^3 - 2} = \frac{e^z + z}{z(z^2 - 1)} = \frac{e^z + z}{z(z-1)(z+1)}$ is not analytic at $z = 0, z = 1$ and $z = -1$

Now $C : |z| = \pi/2$ is a circle with center $(0, 0)$ and radius is $\pi/2$.

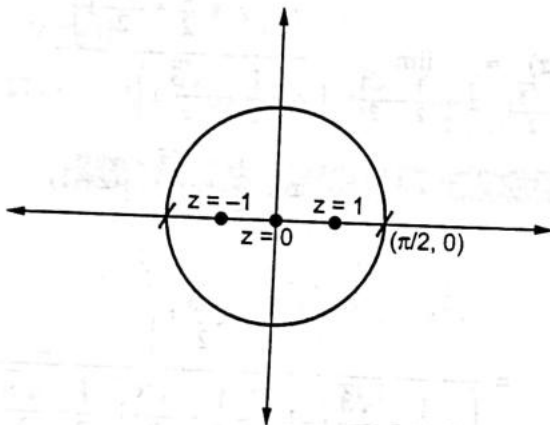


Fig. 6.21.1

Here, All points $z = 0, z = 1$ and $z = -1$ is inside of the circle $|z| = \pi/2$.

$\therefore$ First we find residue at each point $z = 0, z = 1$ and $z = -1$.

All points are simple pole,

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow 0} \frac{(e^z + z)(z-1)(z+1)}{z(z-1)(z+1)} = \frac{1}{-1} = -1$$

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow 1} \frac{(e^z + z)(z-1)(z+1)}{z(z-1)(z+1)} = \frac{(e^1 + 1)}{(2)} = \frac{e+1}{2}$$

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow -1} \frac{(e^z + z)(z-1)(z+1)}{z(z-1)(z+1)} = \frac{e^{-1} - 1}{(-1)(-2)} = \frac{e^{-1} - 1}{2}$$

Now, By residue theorem

$$\begin{aligned} \oint_C \frac{e^z + z}{z^3 - 2} dz &= 2\pi i (\text{Sum of residue}) \\ &= 2\pi i \left(-1 + \left(\frac{e+1}{2} \right) + \left(\frac{e^{-1} - 1}{2} \right) \right) \\ &= 2\pi i \left(\frac{-2 + e + 1 + e^{-1} - 1}{2} \right) \\ &= \pi i \left(e + \frac{1}{e} - 2 \right) \end{aligned}$$

Ex. 6.12.26 : Evaluate $\oint_C \frac{2z-1}{z(z+1)(z-3)} dz$, where C is the circle $|z| = 2$. **GTU : Summer-19, Maths-4, Marks 4**

Sol. : Here, $f(z) = \frac{2z-1}{z(z+1)(z-3)}$ is not analytic at $z = 0, -1$ and $z = 3$

Region : $|z| = 2$ is circle with center $(0, 0)$ and radius is 2.

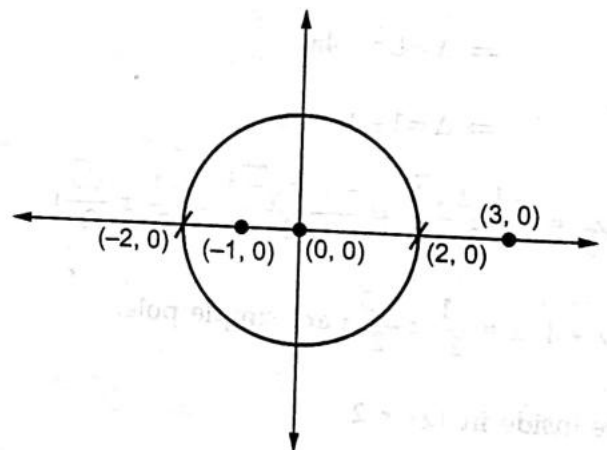


Fig. 6.12.2

$\therefore z = 0$ and $z = -1$ is inside but $z = 3$ out side of the circle $|z| = 2$.

Here, $z = 0$ and $z = -1$ both are simple pole.

$$\therefore \text{Res } f(z) = \lim_{z \rightarrow 0} \frac{(z-1)(2z-1)}{z(z-1)(z-3)} = \frac{(-1)}{(1)(-3)} = \frac{1}{3}$$

$$\begin{aligned} \text{Res } f(z) &= \lim_{z \rightarrow -1} \frac{(z-1)(2z-1)}{z(z-1)(z-3)} \\ &= \frac{(-3)}{(-1)(-4)} = \frac{-3}{4} \end{aligned}$$

By Cauchy residue theorem.

$$\oint_C f(z) dz = 2\pi i \text{ (Sum of residue)}$$

$$\begin{aligned} \therefore \oint_C \frac{2z-1}{z(z+1)(z-3)} dz &= 2\pi i \text{ (Sum of residue)} \\ &= 2\pi i \left(\frac{1}{3} - \frac{3}{4} \right) = 2\pi i \left(\frac{-4-9}{12} \right) \\ &= 2\pi i \left(\frac{-13}{12} \right) \\ &= -\frac{13\pi i}{6} \end{aligned}$$

Ex. 6.12.27 : Using Cauchy residue theorem, compute

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz \text{ where } C \text{ is } |z| = 4.$$

GTU : Winter-22, CVPDE, Marks 4

Sol. : Here, $f(z) = \frac{1}{(z-1)^2(z-3)}$ is not analytic at

$z=1$ and $z=3$.

$\therefore z=1$ is a pole of order 2 and

$z=3$ is a simple pole.

Both points $z=1$ and $z=3$ is inside the circle $|z|=4$

$\therefore$ First we find residue at $z=1$ and $z=3$

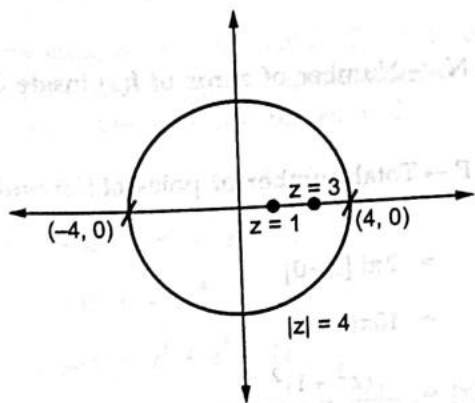


Fig. 6.12.3

$$\text{i) Res } f(z) = \lim_{z \rightarrow 3} \frac{(z-3) \cdot 1}{z(z-1)^2(z-3)} = \frac{1}{4}$$

$$\text{ii) Res } f(z) = \frac{1}{1!} \lim_{z \rightarrow 1} \left[\frac{d}{dz} \left(\frac{(z-1)^2}{(z-1)^2(z-3)} \right) \right]$$

$$= \lim_{z \rightarrow 1} \left(\frac{-1}{(z-3)^2} \right) = \frac{-1}{4}$$

$\therefore$ By Residue theorem

$$\oint_C \frac{1}{(z-1)^2(z-3)} dz = 2\pi i \left(\frac{1}{4} - \frac{1}{4} \right) = 0$$

Ex. 6.12.28 : Evaluate using Cauchy residue theorem

$$\oint_C \frac{2z+6}{z^2+4} dz \text{ where } C : |z-i|=2$$

GTU : Summer-24, CVPDE, Marks 7, Summer-21, CVNM, Marks 7

Sol. : Here, $f(z) = \frac{2z+6}{z^2+4} = \frac{2z+6}{(z+2i)(z-2i)}$ is analytic

every where except $z=-2i$ and $z=2i$

Both $z=2i$ and $z=-2i$ are simple pole

$C : |z-i|=2$ is a circle with center $(0, 1)$ and radius 2

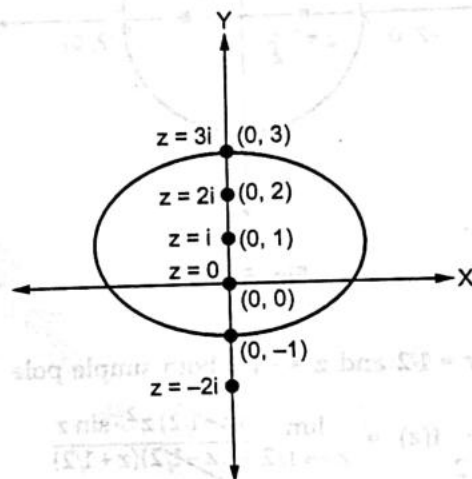


Fig. 6.12.4

$\therefore z=2i$ is inside but $z=-2i$ outside of the circle $|z-i|=2$

$$\begin{aligned} \therefore \text{Res } f(z)_{z=2i} &= \lim_{z \rightarrow 2i} \frac{2z+6}{(z+2i)(z-2i)} \cdot (z-2i) \\ &= \frac{2(2i)+6}{4i} = \frac{4i+6}{4i} \end{aligned}$$

$\therefore$ By Cauchy residue theorem,

$$\begin{aligned} \oint_C \frac{2z+6}{z^2+4} dz &= 2\pi i \left(\frac{4i+6}{4i} \right) \\ &= \frac{\pi}{2} (4i+6) \end{aligned}$$

Ex. 6.12.29 : Evaluate $\oint_C \frac{z^2 \sin z}{4z^2 - 1} dz$ using residue theorem

where $C : |z| = 2$

GTU : Summer-23, CVNM, Marks 7

Sol. : Here $f(z) = \frac{z^2 \sin z}{4z^2 - 1} = \frac{z^2 \sin z}{4(z-1/2)(z+1/2)}$ is not analytic at $z = 1/2$ and $z = -1/2$

$C : |z| = 2$ is the circle with center $(0, 0)$ and radius 2.

$\therefore$ Both $z = 1/2$ and $z = -1/2$ are inside of the circle $|z| = 2$

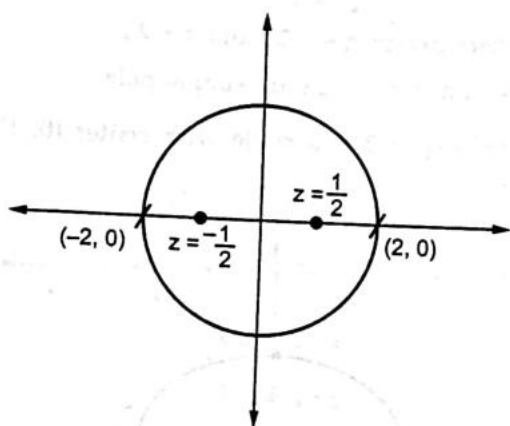


Fig. 6.12.5

$\therefore$ Here, $z = 1/2$ and $z = -1/2$ both simple pole

$$\begin{aligned} \therefore \text{Res } f(z)_{z=1/2} &= \lim_{z \rightarrow 1/2} \frac{(z-1/2) z^2 \cdot \sin z}{4(z-1/2)(z+1/2)} \\ &= \frac{(1/2)^2 \cdot \sin 1/2}{4(1/2+1/2)} = \frac{1}{4} \cdot \frac{1}{4} \sin 1/2 \\ &= \frac{1}{16} \sin 1/2 \end{aligned}$$

$$\begin{aligned} \text{Res } f(z)_{z=-1/2} &= \lim_{z \rightarrow -1/2} \frac{(z+1/2) z^2 \cdot \sin z}{4(z+1/2)(z-1/2)} \\ &= \frac{(-1/2)^2 \cdot \sin(-1/2)}{4(-1/2-1/2)} \\ &= \frac{-1}{4} \cdot \frac{1}{4} (-\sin 1/2) \end{aligned}$$

$\therefore$ By residue theorem $= +\frac{1}{16} \sin 1/2$

$$\begin{aligned} \oint_C \frac{z^2 \sin z}{4z^2 - 1} dz &= 2\pi i \left(\frac{1}{16} \sin 1/2 + \frac{1}{16} \sin 1/2 \right) \\ &= \frac{\pi i}{4} \sin 1/2 \end{aligned}$$

6.13 Rouché's Theorem

Let $f(z)$ and $g(z)$ be analytic within and on a simple closed curve C bounding a region R . Further $|f(z)| > |g(z)|$ on C and f and g are non-vanishing on C . Then the total number of zeros inside C of $f(z) + g(z)$ is equal to total number of zeroes of $f(z)$.

Note : 1) The counting of the total number of zeroes of an analytic function in a given domain can be appreciably simplified by Rouché's theorem.

2) If $\omega = 0$, origin lies inside C .

Ex. 6.13.1 : Evaluate $\int_C \frac{f'(z)}{f(z)} dz$, where C is a simple closed curve, for the function $f(z) = z^5 - 3iz^2 + 2z - 1 + i$

Sol. : i) By fundamental theorem of algebra $f(z)$ has 5 zeroes. But $f(z)$ has no poles.

$\therefore$ By argument principal,

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i [N - P]$$

$$\sum_{r=1}^n n_r = N \rightarrow \text{Number of zeros of } f(z) \text{ inside } C.$$

$$\sum_{r=1}^i P_r = P \rightarrow \text{Total number of poles of } f(z) \text{ inside } C.$$

$$= 2\pi i [5 - 0]$$

$$= 10\pi i$$

$$\text{ii) } f(z) = \frac{(z^2 + 1)^2}{(z^2 + 2z + 1)^3}$$

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i [N - P]$$

Here $z^2 + 1 = 0 \Rightarrow z = \pm i$ is a zeroes and each of order is 2.

$\therefore$ (Multiplicity of $z = i$ is 2 and $z = -i$ is 2)

$$\text{Pole of } f(z) = (z + 1)^6$$

$$z = -1 \text{ is a pole at order } 6$$

$$= 2\pi i [2 \times 2 - 1 \times 6]$$

$$= 2\pi i [4 - 6]$$

$$= -4\pi i$$

Ex. 6.13.2: Apply Rouché's theorem to determine the number of roots (zeroes) of $f(z)$ that lie within the circle region for the following.

$$\text{i) } P(z) = z^9 - 2z^6 + z^2 - 8z - 2 \quad C: |z| = 1$$

$$\text{ii) } P(z) = z^6 - 5z^4 + z^3 - 2z \quad C: |z| = 1$$

$$\text{iii) } P(z) = e^z - 4z^n + 1 \quad C: |z| = 1$$

$$\text{Sol: i) } P(z) = z^9 - 2z^6 + z^2 - 8z - 2 \quad C: |z| = 1$$

$$f(z) = -8z, \quad g(z) = z^9 - 2z^6 + z^2 - 2$$

$$C: |z| = 1 \Rightarrow |f(z)| = |-8z| = 8|z| = 8$$

$$C: |z| = 1 \Rightarrow |g(z)| = |z^9 - 2z^6 + z^2 - 2|$$

$$\leq |z^9| + 2|z^6| + |z^2| + |z|$$

$$= 1 + 2 + 1 + 2 = 6$$

$$|f| = 8 \text{ and } |g| = 6$$

Thus $|f| = 8 > 6 = |g|$ on C .

f and g are analytic on and within C . So by Rouché's theorem f and $p = f + g$ has same number of roots within C . But f has only one zeroes in C .

Hence $P(z)$ has one zero inside C . $|z| = 1$.

$$\text{ii) } P(z) = z^6 - 5z^4 + z^3 - 2z$$

$$f(z) = -5z^4$$

$$g(z) = z^6 + z^3 - 2z$$

$$C: |z| = 1 \therefore |f| = |-5z^4| = 5|z^4| = 5$$

$$g(z) = |z^6 + z^3 - 2z|$$

$$\leq |z^6| + |z^3| + 2|z|$$

$$\leq 1 + 1 + 2$$

$$\leq 4$$

Here $f = 5 \geq 4 = g$.

Here f has 4 zeroes inside the circle $|z| = 1$

$\therefore P(z)$ has 4 zeroes inside the circle $|z| = 1$

$$\text{iii) } P(z) = e^z - 4z^n + 1$$

$$\therefore f(z) = -4z^n, \quad g(z) = e^z + 1$$

$$|z| = 1$$

$$\therefore |f| = |-4z^n| = 4|z|^n = 4$$

$$\therefore |g| = |e^z + 1| \leq |e^z| + |1|$$

$$\leq |e^z| + 1 \leq e + 1$$

$$\therefore 4 = |f| > |g| = e + 1$$

$\therefore f(z)$ has n zeroes within C .

$\therefore P(z)$ has n zeroes within C .

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